Supply Chain Management Expert Field Handbook

A practical field reference from first principles to advanced execution



OPERATE	ANALYZE	TRANSFORM
Plan • source • make • store • deliver	Excel • SQL • Power BI • data platforms	SAP • Ariba • QuickBooks • greenfield

Sri Sai Lakshmi Ganesh Murukutla

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A field handbook for real work

Supply-chain problems rarely stay inside one department. A late delivery may begin with a planning assumption, become a purchasing or warehouse exception, and finally appear as an invoice dispute or customer complaint. I wrote this handbook for that moment: when someone needs the next safe question, a practical method and a way to prove the answer.

Use the book with one real transaction at a time. Start at [FIND-01], follow the shortest relevant route, replace only the marked example values, and reconcile the result before changing a live process. The examples are intentionally connected so that a new reader can learn the language while an experienced practitioner can move directly to a checklist, formula, query or control.

— Sri Sai Lakshmi Ganesh Murukutla

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Use and responsibility

- The sample company, people, identifiers, prices and transactions are fictional teaching data.
- Test formulas, queries, configurations and migrations in an approved read-only or non-production environment before operational use.
- Company policy, signed contracts, system configuration and current law take precedence over a general field reference.
- Safety, regulated product release, customs, sanctions, tax, accounting conclusions, bank changes, legal drafting and irreversible postings require the responsible qualified specialist and documented approval.

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[START-01] How to follow any instruction in this book

You do not need to read this book from front to back. Start with one real example: an order, item, supplier, pallet, invoice, report or project that is not behaving as expected. Then open [FIND-01]. Begin with a display, preview, report or SELECT query. Change or post only after the result is proved and the local owner approves it.

1. Write the exact problem in one sentence.
2. Write one known record number and the time the problem was observed.
3. Find the matching symptom in [FIND-01].
4. Open the first route ID and read ‘Do this first.’
5. Use the filled example in a test or read-only area.
6. Replace only text marked {{LIKE_THIS}}. Never invent a code, table, column, date, unit or currency.
7. Test one known record. Compare row count, quantity, value, status and date with the source.
8. Save the evidence, owner, decision and next check.

Stop and contact the responsible company specialist for personal safety, regulated product release, customs or sanctions, bank/payment details, contract signature, data deletion, inventory write-off or an irreversible production posting.

The nine labels used in major technical recipes

Read in this order	Question it answers
1 Business question	What am I trying to decide?
2 Where to run	Which screen, editor, formula bar or worksheet?
3 Exact sample input	What does a correct input look like?
4 Filled runnable example	What can I copy without guessing?
5 Change only these	Which table, column, code, date or threshold belongs to my company?
6 Run steps	What do I click or execute?
7 Expected result	What should appear if it worked?
8 Why it works	What logic produced the answer?
9 Error and control check	What failed, and how do I prove the number?

Symbol	Meaning	Example
{{TYPE_YOUR_VALUE}}	Replace the whole marked text; do not type the braces	{{PLANT}} → 1710
'text'	A literal text value; keep the single quotes in SQL	'Open'
125	A number; normally no quotes	ordered_qty = 125
NULL	No database value. In Excel use a blank cell unless instructed otherwise	delivery_date IS NULL
[CH17]	A handbook route ID; keep the brackets when searching	Find manufacturing planning
Orders[ordered_qty]	Power BI or Excel syntax; these brackets belong to the formula	Do not replace the brackets
SELECT / FROM / WHERE	SQL words; keep them. Replace only the named table, column or value	WHERE site_id = 'CHI-DC'

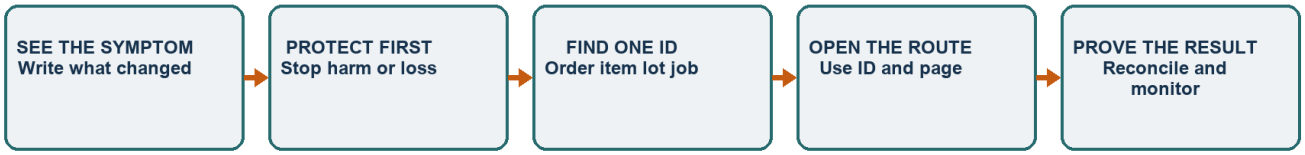
Level label	What it means
BEGINNER	Display, search, preview or calculate with the filled sample.
DAILY	Normal repeatable work after local training and access.
ADVANCED	Needs stronger data, model or process knowledge and review.
CHANGES DATA	Can post, update, delete, merge, release or commit. Test, approve and preserve recovery first.

Never paste an example into a production system until you have checked environment, permission, company/site, date, unit, currency, population and rollback. A read-only query can still expose sensitive data or consume resources.

[FIND-01] Find a problem and the right page in under one minute

You do not need to know the cause. Start with what you can see. Print: find the closest symptom, do the first safe action, and turn to the page shown. Word or PDF: press Ctrl+F on Windows or Command+F on Mac and type the exact ID, such as WH-01. If no symptom matches, start at ROAD-01.

Find the right answer without guessing



Read the arrows from left to right. The output is a verified record, a safe next action and the exact handbook route to follow.

Time	Do this	Filled example
0–10 sec	Stop if a person, product, law, payment, confidential data or system record may be harmed.	Put suspect lot L2401 on hold; do not ship.
10–20 sec	Find one exact order, item, PO, shipment, invoice, lot, job or report ID.	Order 1002; item B200.
20–30 sec	Write expected versus actual in the same unit.	Expected 20 each by 7 Oct; received 15 on 8 Oct.
30–40 sec	Write site, source/screen, time, unit and currency where relevant.	CHI-DC; ERP delivery; 10:00 CT; each.
40–50 sec	Choose the closest symptom below. A possible cause is not yet proof.	Order late or not delivered.
50–60 sec	Do only the first safe action, then open the first route.	Trace event times; open [CH14].

Fill this before changing anything	Your case
Thing or ID	{{ORDER / ITEM / LOT / INVOICE / JOB ID}}
Expected	{{WHAT SHOULD HAVE HAPPENED}}
Actual	{{WHAT ACTUALLY HAPPENED}}
Site, date, time and source	{{LOCATION / TIME / SCREEN OR REPORT}}
Risk, deadline, owner and next update	{{IMPACT / WHEN / WHO / NEXT CHECK}}

Customer, planning and inventory

What I see	Possible causes to test	First safe action	Open these routes	First proof
Customer order is late	Bad promise, shortage, release, pick or carrier delay	Trace one order through entered, released, picked, shipped and delivered times.	[CH14] p. 29 also [CH15] p. 30 also [PB-01] p. 99	Owner and last event are proved.
Customer got too little or the wrong item	Allocation, UOM, pick, pack or label error	Stop the affected dispatch; compare ordered, picked, packed and shipped item and quantity.	[CH14] p. 29 also [WH-01] p. 32 also [CH19] p. 35	Item and quantities reconcile.
Forecast error or bias is rising	History gap, event, lifecycle, override or wrong grain	Freeze the forecast version; compare baseline, forecast and actual by horizon and segment.	[CH04] p. 17 also [CH21] p. 37 also [PB-01] p. 99	Error driver and affected segment are named.

What I see	Possible causes to test	First safe action	Open these routes	First proof
S&OP plan or departments disagree	Different cutoff, assumptions, units, capacity or authority	Put demand, supply, inventory and money on one cutoff; list only decisions and gaps.	[CH05] p. 18 also [CH21] p. 37 also [CH01] p. 15	One approved plan and owner exist.
Stockout although the system shows stock	Blocked, reserved, wrong bin/UOM, expiry or stale event	Check one item-site, status, bin, lot, expiry, reservation and unit.	[CH06] p. 19 also [CH13] p. 28 also [WH-01] p. 32	Usable and unavailable stock are separated.
Too much, old or expiring stock	Forecast bias, MOQ, lead time, duplicate supply or blocked stock	List item-site stock, age, status, open supply, future demand and cost.	[CH06] p. 19 also [FIN-01] p. 53 also [WH-01] p. 32	Time-phased use, move or stop action is clear.

Suppliers, buying and contracts

What I see	Possible causes to test	First safe action	Open these routes	First proof
Supplier is late or not responding	Capacity, material, quality, cash or pickup failure	Ask for finished quantity, WIP, missing material, daily output and pickup proof.	[CH09] p. 21 also [NEG-01] p. 23 also [CH10] p. 22	Dated recovery, evidence and fallback exist.
Supplier defect repeats	Process change, tooling, material, inspection or escape	Quarantine affected lots; compare the first bad lot with the last good lot.	[CH19] p. 35 also [CH09] p. 21 also [CH10] p. 22	Population is contained and cause tested.
PO, receipt and invoice do not match	Quantity, UOM, price, tax, return or duplicate	Trace one line across PO, goods receipt, return, invoice, tax and freight.	[CH07] p. 20 also [SAP-MM] p. 41 also [QB-BUY] p. 58	One line reconciles in quantity and value.
Duplicate invoice or payment	Duplicate vendor/reference, copy, interface or manual entry	Hold the suspected payment; match supplier, invoice, amount, date and bank reference.	[QB-BUY] p. 58 also [SAP-FICO] p. 47 also [CH07] p. 20	Every payment maps to one approved liability.
Supplier choice feels risky or unfair	Unnormalized bids, weak evidence, hidden risk or bad weights	Normalize scope, unit, currency, Incoterm and service before scoring.	[CH08] p. 21 also [NEG-01] p. 23 also [CH10] p. 22	Score and award can be reproduced.
Contract, price or SLA dispute	Wrong version, vague metric, exclusions, notice or precedence	Find the signed version, definition, evidence, notice and remedy.	[CH10] p. 22 also [NEG-01] p. 23 also [CH16] p. 31	One case calculates from signed wording.
Ariba requisition, approval or PO is stuck	Missing approver, budget, supplier, catalog, accounting or integration	Open the document history; identify the exact status, failed step, owner and ERP document ID.	[SAP-ARIBA] p. 49 also [SAP-MM] p. 41 also [CH07] p. 20	One owner and one failed handoff are proved.
SAP supplier invoice is blocked or GR/IR will not clear	Receipt, quantity, price, tax, tolerance, reversal or wrong PO history	Trace one PO item through receipt, reversals, invoice and accounting documents at one cutoff.	[SAP-MM] p. 41 also [SAP-FICO] p. 47 also [CH07] p. 20	Quantity and value differences are explained.

Warehouse receiving, storage and shipping

What I see	Possible causes to test	First safe action	Open these routes	First proof
Receipt is short, damaged or unidentified	Supplier pack, ASN, count, UOM, damage or label	Separate suspect goods; compare PO/ASN, physical count, condition, lot and photos.	[CH12] p. 28 also [CH07] p. 20 also [CH19] p. 35	Accepted, rejected and missing quantities are signed.
Putaway fails or stock waits at dock	No bin, status hold, capacity, task, label or master data	Check stock status, destination rule, open task, bin capacity and label scan.	[SAP-EWM] p. 44 also [CH12] p. 28 also [WH-01] p. 32	Stock has a controlled status and destination.

What I see	Possible causes to test	First safe action	Open these routes	First proof
Physical stock differs from system	Missed move, timing, UOM, wrong bin, count or interface	Freeze the smallest area, count blind and trace the last correct movement.	[CH13] p. 28 also [SAP-EWM] p. 44 also [WH-01] p. 32	Count, status, bin and last event are proved.
Pallet or material cannot be found	Wrong scan, open task, move, lot/HU or location label	Check item, lot, handling unit, last confirmed bin and open warehouse task.	[SAP-EWM] p. 44 also [WH-01] p. 32 also [CH13] p. 28	Last scan and movement owner are known.
Warehouse is full or congested	Peak mismatch, blocked stock, poor slotting, staging or labor	Measure usable positions, cube, queues, travel and blocked stock by zone/hour.	[CH11] p. 27 also [WH-01] p. 32 also [CH06] p. 19	Largest space or flow constraint is quantified.
Pick, pack or dispatch error repeats	Slot, label, UOM, scan bypass, replenishment or training	Stop the affected shipment; compare order, pick, pack, scan and dispatch records.	[CH14] p. 29 also [WH-01] p. 32 also [CH19] p. 35	Failure point and affected shipments are known.
SAP EWM delivery, warehouse task or queue is stuck	Status, stock type, determination, resource, queue, HU or ERP integration	Use the warehouse monitor to trace delivery → request → task → confirmation and copy the first failed status/message.	[SAP-EWM] p. 44 also [WH-01] p. 32 also [SAP-X] p. 51	The first failed EWM object and owner are known.

Transport, global trade and 3PL

What I see	Possible causes to test	First safe action	Open these routes	First proof
Carrier missed pickup or delivery is late	Booking, dock, driver, capacity, route or handoff failure	Find the last physical milestone, revised ETA and accountable carrier contact.	[CH15] p. 30 also [CH14] p. 29 also [PB-01] p. 99	Location, ETA and recovery action are verified.
Shipment is damaged or lost	Pack, handling, seal, handoff, theft or carrier exception	Protect evidence; identify shipment, packages, custody events, photos and claim deadline.	[CH15] p. 30 also [CH19] p. 35 also [CH10] p. 22	Population, custody and claim owner are known.
Freight cost increased	Rate, fuel, mode, distance, cube, weight or accessorial	Bridge each cost driver from approved baseline to the current invoice.	[CH15] p. 30 also [CH21] p. 37 also [FIN-01] p. 53	Increase is separated by driver.
Customs or border hold	Document, classification, value, origin, party or permit	Get the exact hold reason from broker/authority; stop unsupported data changes.	[CH20] p. 36 also [CH10] p. 22 also [PB-01] p. 99	Reason, owner, document and release path are recorded.
Trade screening or classification concern	Wrong party, code, origin, end use or current rule	Pause release; send exact product, party and destination to the trade specialist.	[CH20] p. 36 also [CH03] p. 16 also [SRC-01] p. 107	Approved decision and source are saved.
3PL stock, service or invoice mismatch	Scope, interface, transaction timing, rate basis or exclusion	Reconcile one case between 3PL events, enterprise records and rate schedule.	[CH16] p. 31 also [CH13] p. 28 also [FIN-01] p. 53	Case, activity and charge share one ID/cutoff.

Manufacturing and quality

What I see	Possible causes to test	First safe action	Open these routes	First proof
MRP proposal is missing, too high or late	Demand, stock, UOM, BOM, lead time, lot size or calendar	Trace one item through every dated MRP input in the same unit.	[CH17] p. 34 also [SAP-X] p. 51 also [CH03] p. 16	Every input and parameter is proved.
Production is late	Material, setup, labor, maintenance, quality or priority	Trace one order from release to current operation; name the blocking resource.	[CH18] p. 34 also [CH17] p. 34 also [PB-01] p. 99	Constraint, lost time and recovery are known.

What I see	Possible causes to test	First safe action	Open these routes	First proof
Capacity is short	Bad standard, mix, yield, downtime, labor or calendar	Compare required hours with demonstrated available hours by resource and period.	[CH18] p. 34 also [OPT-01] p. 67 also [CH05] p. 18	Feasible capacity gap is quantified.
OEE, yield or scrap worsened	Availability, speed, quality, material, tool or measurement	Rebuild the measure from event time and counts; find the first change point.	[CH18] p. 34 also [CH19] p. 35 also [SAP-01] p. 38	Loss category and source records reconcile.
Cannot trace a lot or recall population	Missing lot, rework, split, merge, label or shipment link	Stop release and widen containment until genealogy reconciles.	[CH19] p. 35 also [SAP-01] p. 38 also [ROAD-01] p. 14	All affected and possible units are controlled.
BOM, routing, UOM or master data is wrong	Wrong version, validity date, conversion, plant or owner	Display the approved record and one failed transaction; do not overwrite history.	[CH03] p. 16 also [CH17] p. 34 also [SAP-01] p. 38	Correct version, date and owner are proved.

Finance, cash and business decisions

What I see	Possible causes to test	First safe action	Open these routes	First proof
Inventory value or COGS looks wrong	Quantity, valuation, date, duplicate, negative stock or posting	Trace one item through receipt, valuation, shipment and accounting entry.	[FIN-01] p. 53 also [SAP-FICO] p. 47 also [QB-BUY] p. 58	Quantity and value reconcile at one cutoff.
Cash is tied up	Inventory, receivables, payables, terms or process delay	Calculate DIO, DSO and DPO with one approved period and valuation.	[CH21] p. 37 also [FIN-01] p. 53 also [CH06] p. 19	Largest cash driver and owner are named.
Margin or price variance fell	Price, volume, mix, purchase cost, freight or currency	Bridge approved baseline to actual one driver at a time.	[CH21] p. 37 also [FIN-01] p. 53 also [CH15] p. 30	Drivers sum to the total change.
QuickBooks or bank will not reconcile	Date, duplicate, missing transaction, opening balance or direct entry	Use the same statement date; match one unmatched amount to source evidence.	[QB-CLOSE] p. 63 also [QB-01] p. 55 also [PB-01] p. 99	Statement = books plus explained timing.
Negotiation is stuck	No BATNA, authority, evidence, trade list or decision maker	Write must-have, target, fallback, walk-away, BATNA and give/get list.	[NEG-01] p. 23 also [CH10] p. 22 also [CH08] p. 21	Every concession has a condition and expiry.
No one will decide	Unclear choice, owner, authority, evidence or deadline	Write the decision, options, impact, approver and consequence of delay.	[CH01] p. 15 also [NEG-01] p. 23 also [CH29] p. 95	One approver can choose by a stated time.
QuickBooks inventory is negative or valuation changed	Sale before receipt, wrong date, missing bill, duplicate sale, costing setting or adjustment	Run Inventory Valuation Detail for one item; order every transaction by date and identify the first negative point.	[QB-BUY] p. 58 also [QB-CLOSE] p. 63 also [CH13] p. 28	Quantity and cost layers or moving average are explained.
QuickBooks customer payment is missing, duplicated or unapplied	Wrong customer, undeposited funds, duplicate bank match, credit or date	Trace invoice → receive payment → deposit → bank match for one customer and amount.	[QB-SELL] p. 61 also [QB-CLOSE] p. 63 also [FIN-01] p. 53	AR, deposit and bank agree.

Excel, data platforms and analytics

What I see	Possible causes to test	First safe action	Open these routes	First proof
Excel formula shows an error	Name, header, data type, blocked spill or version	Copy the exact error; test the formula on the filled sample first.	[XL-01] p. 69 also [DATA-01] p. 12 also [PB-01] p. 99	Sample open quantity equals 16.

What I see	Possible causes to test	First safe action	Open these routes	First proof
SQL has an error, no rows or duplicates	Wrong object/value/type, filter, grain or join	Preview ten rows, find one known ID and count before/after each join.	[SQL-01] p. 73 also [DATA-01] p. 12 also [PB-01] p. 99	Known ID returns once at intended grain.
Dashboard is wrong, blank or slow	Metric, grain, relationship, refresh, filter or model cost	Trace one known record; reconcile control totals, then inspect model/refresh.	[CH25] p. 82 also [PBI-01] p. 79 also [DATA-01] p. 12	One record and totals reconcile.
Refresh, file or interface failed	Credential, schema, path, late file, duplicate or service outage	Preserve the error and failed batch; compare last good input and row counts.	[CH25] p. 82 also [MIG-01] p. 91 also [SQL-01] p. 73	Failed layer, batch and replay plan are known.
Permission denied or object not found	Wrong role, environment, catalog/schema, case or missing grant	Confirm environment and current role; copy the exact qualified object name.	[SQL-01] p. 73 also [DBX-01] p. 83 also [SNF-01] p. 84	Approved role can preview one known ID.
Automation ran wrong or stopped	Bad scope, trigger, identity, retry, dependency or guardrail	Disable further runs if authorized; preserve run ID, input, output and error.	[CH25] p. 82 also [CH29] p. 95 also [ROAD-01] p. 14	Population and safe replay/rollback are proved.

SAP, migration, go-live and resilience

What I see	Possible causes to test	First safe action	Open these routes	First proof
SAP app, posting or status looks wrong	Organization, role, master data, document flow or period	Use Display first; copy exact message, organization, document ID and date.	[SAP-01] p. 38 also [SAP-X] p. 51 also [MIG-01] p. 91	Correct document flow and status are visible.
SAP account assignment, posting period or FI document fails	Company code, period, G/L, cost object, tax, currency or account determination	Use Check/Simulate; copy the exact message and verify company code, posting date, G/L and required cost object.	[SAP-FICO] p. 47 also [SAP-MM] p. 41 also [SAP-X] p. 51	The missing or invalid accounting field is proved.
Migration row is rejected	Mapping, type, required field, dependency or code	Keep the rejection; trace source value, rule, target and owner for one row.	[MIG-01] p. 91 also [CH28] p. 90 also [CH03] p. 16	Reason and approved correction are explicit.
Migration count or value does not match	Cutoff, duplicate, reject, transformation or rounding	Stop release; compare source, target and rejects at the same cutoff.	[MIG-01] p. 91 also [CH28] p. 90 also [FIN-01] p. 53	Source = target + approved differences.
Cutover or critical process fails at go-live	Sequence, role, interface, mapping, balance or capacity	Freeze affected postings; preserve case IDs and test the rollback trigger.	[CH28] p. 90 also [ROAD-01] p. 14 also [CH27] p. 89	Last correct event and recovery owner are known.
Users lack access or adoption is poor	Role design, training, workflow, support or local blocker	Observe one user completing one real case; record exact stop point and role.	[CH27] p. 89 also [MIG-01] p. 91 also [CH29] p. 95	User, task, blocker and owner are named.
Supplier, site, cyber or system outage	Single dependency, failed control, attack, capacity or recovery gap	Protect people/data; compare time to survive with time to recover and invoke the plan.	[CH26] p. 88 also [ROAD-01] p. 14 also [CH29] p. 95	Tested recovery protects the next deadline.

Cause not proved? Return to [ROAD-01], verify one real case, and separate evidence from assumption before changing a process or system.

[PLAIN-01] Plain words for a first-time reader

These words appear throughout the handbook. Read this page once. Return whenever a sentence feels unfamiliar.

Word	Plain meaning
Fact	Something you can show in a record, count or direct observation.

Word	Plain meaning
Assumption	Something believed but not proved yet.
Owner	The person accountable for the next result.
Evidence	A record, document number, screenshot, count or test result that proves a statement.
Population	All records included in a result.
Cutoff	The latest date and time included.
Source of truth	The approved record used when sources disagree.
Exception	A record outside the normal rule that needs attention.
Contain	Protect today without claiming the cause is fixed.
Correct	Remove the verified cause.
Prevent	Change the process so the problem does not return.
Reconcile	Explain source = target plus valid differences.
Environment	A separate system area such as development, test or production.
Post	Save a business event that changes stock, status or accounting.
Reverse	Create an approved linked correction for a posting.
Rollback	Return a release or cutover to a proven state.
Nonposting	A record such as a purchase order that does not yet change the accounts.
Subledger	Detailed records, such as item inventory or supplier balances, that must equal the related G/L account.
Item line	A transaction line using a product/service record; tracked inventory quantity can change.
Category line	A transaction line posted directly to a G/L account; it normally does not change item quantity.
Cost object	The department, order or project that receives and explains a cost.
Company file / realm	The QuickBooks company or Ariba tenant in which the record exists.

Twenty abbreviations to know first

Short form	Full form	Why you see it
3PL	Third-Party Logistics Provider	External warehouse or transport operator
AP	Accounts Payable	Money owed to suppliers
AR	Accounts Receivable	Money customers owe
ASN	Advance Shipping Notice	Supplier notice before arrival
ATP	Available to Promise	Quantity/date that can be committed
BOM	Bill of Material	Components required to make a product
COGS	Cost of Goods Sold	Cost attached to sold products
DAX	Data Analysis Expressions	Power BI calculation language
ERP	Enterprise Resource Planning	Integrated transaction and accounting system
EWM	Extended Warehouse Management	SAP advanced warehouse system
GR	Goods Receipt	Inventory or service is received
KPI	Key Performance Indicator	Measure tied to a decision
MRP	Material Requirements Planning	Net demand into planned supply
OEE	Overall Equipment Effectiveness	Availability × performance × quality
OTIF	On Time In Full	Order delivered by promise and complete

Short form	Full form	Why you see it
P2P	Procure to Pay	Need through supplier payment
PO	Purchase Order	Authorized supplier commitment
SAP	Systems, Applications and Products	Enterprise software company/platform
SKU	Stock-Keeping Unit	Distinct stocked item
UOM	Unit of Measure	EA, KG, CS and other quantity unit

SEE ALSO [ABR-01] full A–Z abbreviation list, p.104 • [DATA-01] data words and names, p.12

[MAP-01] Printed page map and clickable routes

Print: turn to the page shown and confirm the same bracketed ID appears in the destination heading. Word or PDF: click the destination name. If page numbers change after editing in Word, select the document and update fields; the stable ID still works.

ID	Destination	Page
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SNF-01	Snowflake guided execution	84
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PB-01	Problem → cause → action playbook	99
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The A–Z search list is at [INDEX-01]. The complete abbreviation list is at [ABR-01].

[DATA-01] Sample company, data and database words

Northstar Everyday Products is a fictional company used throughout the book. It buys materials, makes and stores products, sells to customers, ships from CHI-DC and uses USD. Reusing one small dataset makes the formulas and queries comparable across Excel, SQL, Power BI, Databricks and Snowflake.

Orders table: one row means	One order line for one item at one site
Key in this toy sample	order_id is unique only because the sample has one line per order. Real data normally needs order_id + line_id.
As-of date used in checks	2026-10-19
Delivered OTIF rule	delivery_date ≤ promised_date AND shipped_qty ≥ ordered_qty
Open quantity	MAX(ordered_qty − shipped_qty, 0)

order_id	order_date	customer_id	item_id	ordered_qty	site_id	status
1001	2026-10-01	C001	A100	10	CHI-DC	Delivered
1002	2026-10-02	C002	B200	20	CHI-DC	Delivered
1003	2026-10-03	C001	A100	5	CHI-DC	Open
1004	2026-10-04	C003	C300	8	CHI-DC	Delivered
1005	2026-10-05	C004	B200	12	CHI-DC	Delivered
1006	2026-10-06	C002	C300	6	CHI-DC	Open

order_id	ordered_qty	shipped_qty	promised_date	delivery_date
1001	10	10	2026-10-05	2026-10-05
1002	20	15	2026-10-07	2026-10-08
1003	5	0	2026-10-08	
1004	8	8	2026-10-09	2026-10-09
1005	12	12	2026-10-10	2026-10-12
1006	6	0	2026-10-11	

Blank delivery_date cells mean no delivery is recorded. In SQL this is NULL; before using the sample in Excel, leave the cell blank rather than typing the word NULL.

item_id	item_description	category	unit_cost	lead_time_days	pack_qty
A100	Water Bottle 1 L	Drinkware	$8.00	7	5
B200	Filter Pack	Consumables	$3.50	14	10
C300	Storage Bin	Home Storage	$12.00	10	2

Inventory sample

item_id	site_id	on_hand_qty	blocked_qty	snapshot_date
A100	CHI-DC	25	0	2026-10-19
B200	CHI-DC	4	2	2026-10-19
C300	CHI-DC	20	0	2026-10-19

The database words that remove most confusion

Word	Child-simple meaning	Sample
Database	A managed collection of related data	Northstar operations data
Schema	A named folder inside a database	analytics
Table	Rows about one kind of thing	orders
Row	One occurrence	Order 1002
Column / field	One attribute stored for every row	promised_date
Value	The content in one row and column	2026-10-07
Key	A value used to identify or connect rows	order_id or item_id
Grain	Exactly what one row represents	one order line
Filter	Keep only rows that meet a condition	status = 'Open'
Join	Connect rows through matching keys	orders.item_id = items.item_id
Aggregate	Summarize many rows	SUM ordered_qty by item
Window	Calculate across related rows while retaining each row	rank orders by customer
CTE	A named temporary query step	WITH open_orders AS (...)
Parameter	A value supplied at run time	site_id = CHI-DC
Data type	Rule for how a value behaves	date, integer, decimal, text
Name in this book	**Replace with your name**	**How to find it**
orders	Your order-line table	Data catalog, report source or system owner
item_id	SKU/material/product column	Open one known item and inspect field/metadata
site_id	Plant/DC/location column	Compare a known site in source and report
'Open'	Exact status value	List distinct statuses; spelling and case may matter
promised_date	Customer commitment date	Confirm business definition; do not assume requested date
delivery_date	Actual delivery or proof-of-delivery date	Confirm timezone and event source
ordered_qty	Committed line quantity	Confirm base/order UOM
shipped_qty	Actually shipped quantity	Confirm cancellations and returns

How to find your real table, field and code names

Do not guess. Find one known record, open its source or catalog, copy the exact object and field names, preview the value, confirm its data type and owner, then test only that record.

Tool	Where to look	Proof before replacing an example
Excel	Table Design for the table name; header row for column names	A known order is visible and the formula uses the exact headers
Power BI	Fields/Model view and Power Query Source step	The source row and model relationship keys agree
Databricks	Catalog Explorer → catalog → schema → table → Columns/Sample Data	A read-only preview returns the known ID

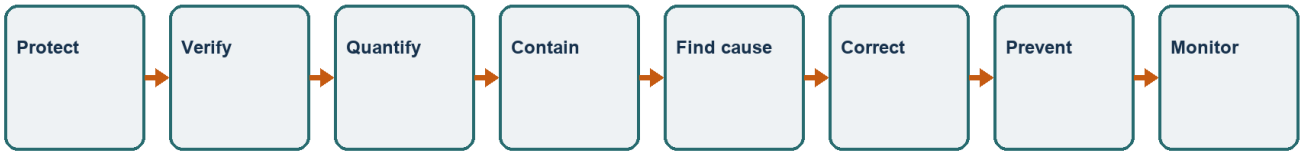
Tool	Where to look	Proof before replacing an example
Snowflake	Database Explorer or Workspace object browser → Columns/Data Preview	The current role can preview the known ID
SAP	Value help and field help/technical information where authorized	Display shows the right organization and document
QuickBooks	Report columns and the transaction form	The report and source transaction agree at one date

Case rule: SQL keywords are usually case-insensitive. String values may be case-sensitive depending on platform/collation. Unquoted identifiers are normalized differently by platforms; quoted identifiers preserve case and often become case-sensitive. Confirm the local dialect before changing names.

SEE ALSO [XL-01] Excel, p.69 • [SQL-01] SQL, p.73 • [PBI-01] Power BI, p.79 • [DBX-01] Databricks, p.83 • [SNF-01] Snowflake, p.84

[ROAD-01] Universal road map for any serious problem

Use this road map for any problem



Read left to right. Protect first, prove the facts, use a reversible containment, correct only the verified cause, and close only after monitoring shows no recurrence.

Step	Do	Evidence before moving on
1 Protect	Stop unsafe, illegal, uncontrolled or irreversible activity. Preserve product, people, data and records.	Affected area isolated; authority contacted
2 Verify	Observe one real failed case. Record time, system, item, site, status and source evidence.	Fact and assumption are separated
3 Quantify	Count affected orders, units, value, customers, time and worst credible exposure.	Population and cutoff are stated
4 Contain	Choose the smallest reversible action that protects the outcome now.	Owner, deadline, side effect and next update
5 Find cause	Trace the process backward. Ask what created the defect and what allowed escape.	Cause explains the evidence and can be tested
6 Correct	Remove the cause; update data, process, design, capacity, supplier or system.	Retest passed and records reconciled
7 Prevent	Change standard work, control, training, contract, master data or automation.	Control owner and trigger are live
8 Monitor	Watch enough future opportunities to prove no recurrence.	Period complete; closure approved

Ask this	Good answer example	Weak answer
What failed?	Order 1002 arrived one day late and five units short.	Delivery problem.
How many?	14 lines, 6 customers, $38,400 revenue; cutoff 10:00 CT.	A lot.
Where did it start?	Supplier ASN showed 20; dock count showed 15.	The system is wrong.

Step	Do	Evidence before moving on
What protects today?	Allocate five units from DAL; owner Lee; confirm by 13:00.	We are working on it.
How will we know?	All 14 lines reconciled; customer promise updated; no repeat for four receipts.	Issue closed.

Editable incident message: Issue {{SPECIFIC FAILURE}}. Impact {{ORDERS / UNITS / VALUE / DEADLINE}}. Confirmed facts {{EVIDENCE / CUTOFF}}. Containment {{ACTION / OWNER / TIME}}. Decision needed {{CHOICE}} by {{TIME}}. Next update {{TIME}}. Closure proof {{RECONCILIATION / RECURRENCE PERIOD}}.

SEE ALSO [PB-01] detailed playbook, p.99 • [CH01] rapid response, p.15 • [CH30] escalation, p.95

[CH01] 01 Start with the business need

Read first	Plain guidance
In one sentence	Turn a vague complaint into one owned decision.
Do this first	Write the failed outcome, affected orders or money, deadline, owner and next update.
What changes by company	Severity limits, approval levels and escalation contacts.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.

Field panel	Execution guidance
Use when	Starting a supply chain, taking over a role, or deciding which issue to fix first.
Five things to establish	1. What do customers need: product, quantity, location and promised date? 2. Which items stop production or create safety or compliance exposure? 3. Where is cash tied up: raw material, work in progress or finished goods? 4. Who owns each handoff and who can approve an exception? 5. Which system and timestamp are the source of truth?
Decision path	Safety, quality or trade restriction? Hold and escalate to the responsible specialist. Otherwise: delivery at risk? Identify the shortage and recovery date. Otherwise: rank cost, cash and recurring errors by business impact.
Daily operating rhythm	Start of shift: review shortages, overdue orders, blocked stock and capacity. Assign one owner and next update time to each issue. End of shift: record what changed, what remains at risk and the next action.
Problem to action	Everything is urgent -> no agreed priority -> rank by customer impact and deadline -> publish an escalation rule. Teams disagree on numbers -> different definitions -> reconcile one sample transaction -> maintain a metric dictionary.
Keep this record	Issue | affected order or item | business impact | owner | action | due time | evidence of closure. Never close an issue merely because an email was sent.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Protect people, product and transaction integrity first. Establish one verified fact base, one accountable incident lead and one decision deadline. Rank work by customer, compliance, margin and recovery-time exposure. Separate containment from correction and prevention.
Decision rules	Severity 1: safety, regulatory, cyber, traceability or enterprise outage. Activate the formal incident process now. Severity 2: material customer or financial exposure within the decision horizon. Open a cross-functional recovery cell. Severity 3: contained local issue. Assign and manage through daily control.
Control	Every issue needs a case ID, affected population, first-known time, containment, owner, next decision, next update and closure evidence. Reopen it if the outcome recurs inside the monitoring window.

[CH02] 02 Map the process and handoffs

Read first	Plain guidance
In one sentence	Show how work really moves from request to result.
Do this first	Follow one normal case and one failed case; record every wait, re-entry and approval.
What changes by company	Process names, owners, systems and accepted evidence.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Core flow	Customer need -> demand plan -> supply plan -> source or make -> receive -> store -> deliver -> invoice -> return or recover.
Minimum dependency map	Sales gives Planning demand and promotions. Planning gives Procurement quantities and dates. Engineering gives Purchasing and Manufacturing released specifications and BOMs. Quality releases material. Warehouse gives Planning usable stock. Logistics gives Customer Service delivery evidence. Finance gives credit and payment status.
Build a workable process	1. Walk one real order from request to payment. 2. Record input, activity, output, owner and system at each step. 3. Mark waiting, re-entry, rework and approval delays. 4. Define what must be true before the next team accepts the work. 5. Test one normal order and one exception.
Decision path	Missing input? Return with the exact missing field. Input valid but work stalled? Find the queue owner and capacity limit. Output rejected? Compare it with the agreed acceptance rule.
Measure	End-to-end lead time = completed timestamp - request timestamp. Example: 12 days elapsed, 2 days active work -> 10 days waiting. Investigate the largest queue before speeding up a two-minute task.
Problem to action	Repeated handoff disputes -> unclear acceptance -> agree required fields and rejection reasons -> review first-pass acceptance weekly. Duplicate approvals -> unclear authority -> assign one accountable approver -> maintain an approval matrix.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Map one real transaction and one failed transaction from customer request through cash. At every handoff define the input, acceptance rule, output, owner, system, queue and escalation. Measure waiting and rework before automating.
Decision rules	If the work is correct but waits, fix queue policy or capacity. If it returns, fix the acceptance rule or upstream quality. If it is re-keyed, integrate or designate one system of record. If nobody can approve the exception, repair decision rights.
Control	Use process mining only after agreeing case ID, activity and timestamp logic. Validate the discovered path with operators before treating it as the process.

[CH03] 03 Master data and item setup

Read first	Plain guidance
In one sentence	Keep reusable names and rules correct so transactions work.
Do this first	Check item, partner, site, unit, status and effective date before fixing the transaction.
What changes by company	Field names, code lists, ownership and local validation rules.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.

Read first	Plain guidance
Field panel	**Execution guidance**
Required fields	Item ID, description, base unit, purchase and sales units, conversion factors, site, supplier, lead time, order multiple, minimum order, lot or serial rule, shelf life, storage conditions, valuation and lifecycle status.
Setup checklist	1. Search for duplicates before creating an item. 2. Confirm specifications with Engineering and Quality. 3. Verify units: 1 case = 12 each must hold across PO, receipt and invoice. 4. Confirm supplier and site-specific planning settings. 5. Test purchase, receipt, issue, shipment and return. 6. Name a data owner and approve changes with an effective date.
Decision path	Unexpected quantity or value? Check unit conversion first, then currency, duplicate transactions and effective dates. Do not repair a planning problem by silently changing historical transactions.
Excel check	If A contains item ID: =COUNTIF($A$2:$A$1000,A2)>1 flags duplicates. If B is cases and C is each per case: =B2*C2 gives each. 100 cases x 12 each = 1,200 each; receiving 100 each understates stock by 1,100.
Problem to action	MRP recommends huge orders -> wrong unit or minimum -> correct approved settings -> validate extremes before upload. Two IDs for the same part -> uncontrolled creation -> select a surviving ID and map transactions -> require duplicate search.
Control	Keep change request, old value, new value, reason, approver and effective date. Never merge regulated or traceable lots merely because descriptions match.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Govern item, partner, location, BOM, routing, calendar, units, currency and status as products with owners and service levels. Critical fields require maker-checker control, effective dating and impact analysis.
Decision rules	Do not correct a bad result by editing historical transactions. Quarantine the bad master, quantify affected orders and replan after approval. For duplicate items, select the survivor, map open transactions, protect traceability and retire the duplicate under change control.
Control	Monitor completeness, validity, uniqueness, referential integrity, timeliness and reconciliation. Report defects by source and business impact, not only by field count.

[CH04] 04 Demand planning and forecast checks

Read first	Plain guidance
In one sentence	Estimate future demand and explain the error.
Do this first	Compare a simple baseline with the latest forecast at item, site and horizon.
What changes by company	Demand grain, event calendar, history window and error target.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Do this	Collect demand by item, location and time bucket. Separate shipments from true demand: stockouts can hide lost sales. Tag promotions, one-off orders, returns and new products. Compare a simple baseline with planner overrides.
Forecast formulas	WAPE = sum of absolute forecast errors / sum of actual demand. Bias = sum(forecast - actual) / sum(actual). Positive means overforecast. Example: actual 100, 120; forecast 110, 100. WAPE = 30/220 = 13.64%; bias = -10/220 = -4.55%. If total actual is zero, report N/A.

Read first	Plain guidance
Decision path	Stable demand? Try a moving average. Seasonal demand? Compare the same season and enough complete cycles. Intermittent demand? Review occurrence and order size separately. New item? Use an analogue, launch assumptions and frequent review.
Diagnostic checklist	Compare the forecast version available when the decision was made, not the latest revision. Check error by segment and horizon. Review stockouts, promotions and outliers before changing the model. Do not average item-level percentage errors to obtain a company total.
Problem to action	Consistent underforecast -> missing growth or promotion -> add evidence-based adjustment -> measure override benefit. Forecast swings -> tiny sample or one-off orders -> separate exceptional demand -> preserve reason codes.
Excel	With actual B2:B13 and forecast C2:C13, use D2=ABS(C2-B2), copy down; WAPE =IF(SUM(B2:B13)=0,"N/A",SUM(D2:D13)/SUM(B2:B13)).

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Segment by value, variability, intermittency and lifecycle. Compare a simple baseline with every planning layer. Store the forecast vintage used for the decision. Measure error at the decision grain and horizon; measure FVA for each override stage.
Decision rules	Data defect → repair and backfill. Lifecycle event → use launch or phase-out logic. Intermittent demand → model occurrence and size. Persistent bias → correct drivers and constraints. Negative FVA → remove or redesign the override.
Control	Never use one average accuracy target for every SKU. Pair WAPE with bias, service and inventory to prevent a low-volume segment from disappearing in the total.

[CH05] 05 Sales and operations planning

Read first	Plain guidance
In one sentence	Make one demand, supply and money plan.
Do this first	Bring only decisions: gap, choices, impact, owner and required date.
What changes by company	Time fences, planning calendar and executive authorities.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Purpose	Agree one feasible demand, supply and financial plan. Use a monthly decision cycle with weekly exception reviews; shorten the cycle when the business changes quickly.
Meeting sequence	1. Demand review: baseline, promotions, launches and risks. 2. Supply review: material, labor, equipment and supplier limits. 3. Reconciliation: quantify gaps and alternatives. 4. Executive decision: choose service, cost and cash tradeoffs. 5. Publish the approved plan, assumptions and owners.
Decision path	Demand exceeds supply? Check if timing can change. If not, compare overtime, alternate supply, allocation and customer date changes. Supply exceeds demand? Reduce releases, rebalance stock or agree a commercial action before building more.
Worked decision	Demand 12,000 units; feasible supply 10,000. Gap = 2,000. Overtime adds 1,000 at $3 extra each; a validated alternate adds 1,000 at $5 extra each. Recovery premium = $8,000. Compare with contribution at risk and customer obligations, not revenue alone.
Problem to action	Meeting becomes a data debate -> inconsistent cutoffs -> freeze a common data snapshot -> publish definitions. Same issue every month -> no accountable action -> record owner and due date -> begin next review with overdue decisions.

Read first	Plain guidance
One page decision log	Period | demand | feasible supply | gap | alternatives | incremental cost | service effect | decision | owner | review date. Finance confirms cost assumptions; Operations confirms feasibility.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Run demand, supply and financial reconciliation on one snapshot. Bring decisions, not presentations, to the executive meeting. Each scenario must state service, margin, working capital, risk, capacity and reversibility.
Decision rules	Inside the frozen horizon use allocation, approved substitution, overtime, resequencing or expedite. Outside the horizon use sourcing, capacity, network, commercial and policy changes. Do not approve a plan with unnamed constraints.
Control	Freeze assumptions and decisions by version. Track overdue decisions and compare approved plan with actual execution at the next cycle.

[CH06] 06 Inventory and replenishment

Read first	Plain guidance
In one sentence	Put the right stock in the right place at the right time.
Do this first	Separate usable, blocked, pipeline, safety, excess and expired stock.
What changes by company	Service target, lead time, review cycle, pack size and shelf life.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Choose the policy	Critical or fast-moving item? Review continuously. Predictable low-value item? Consider a periodic review or two-bin system. Uncertain or expiring item? Limit exposure and review demand and shelf life before replenishing.
Core formulas	Inventory position = usable on hand + confirmed on order - backorders. Do not subtract the same demand twice through both allocations and backorders. Reorder point = average demand during lead time + safety stock. For fixed lead time and independent daily demand: safety stock = z x daily demand standard deviation x sqrt(lead-time days).
Worked example	Daily demand 100; daily standard deviation 20; fixed lead time 9 days; z = 1.645 for a 95% normal cycle-service target. Safety stock = 98.7 -> 99 units. Reorder point = 999 units. This does not mean a 95% fill rate; validate the distribution and lead-time assumptions.
Order quantity	EOQ = sqrt(2 x annual demand x order cost / annual holding cost per unit). D=12,000, S=$50, H=$3 -> EOQ=632.46 units. Adjust for packs, MOQ, shelf life and capacity; EOQ assumes steady demand and no shortages or quantity discounts.
Problem to action	Stockouts despite high stock -> wrong mix, blocked stock or late supply -> inspect item-site availability -> maintain segmented policies. Excess stock -> obsolete demand or oversized lots -> stop unnecessary orders -> review lifecycle and minimums.
Check before releasing	Use the same time units throughout. Check usable stock, open orders, demand dates, pack rounding and expiry. Recalculate when demand or supplier lead time changes.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Set policy by item-location, not item alone. Separate cycle stock, safety stock, pipeline, prebuild, strategic buffer, blocked and excess stock. Simulate service and cash before changing targets.

Panel	What disciplined operators do
Decision rules	Stable demand and fixed lead time → statistical reorder point. Variable lead time → include lead-time variance. Intermittent or expensive demand → service-cost simulation or critical-spares logic. Shelf-life constrained → cap the order by sell-through before expiry.
Control	Recalculate only when inputs change materially or on the governed review cycle. Track manual overrides, reason, expiry date and realized service/cash effect.

[CH07] 07 Procurement from request to payment

Read first	Plain guidance
In one sentence	Buy an approved need, receive it, match the invoice and pay correctly.
Do this first	Confirm the need is not already covered; then follow PR → PO → receipt → invoice → payment.
What changes by company	Approval, supplier, tax, tolerance, account and payment rules.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.

Field panel	Execution guidance
Process	Requisition -> budget and authority approval -> supplier selection -> PO -> supplier confirmation -> receipt and inspection -> invoice match -> payment -> performance review.
Before issuing the PO	Include item and specification revision, quantity and unit, price and currency, delivery date and location, acceptance rules, applicable contract, shipping term, taxes and authorized contact. Get written confirmation of quantity, date and price.
Decision path	Invoice mismatch? Quantity: compare receipt and returns. Price: compare contract and approved changes. Tax or freight: check agreed treatment. Hold the disputed portion under the applicable process; do not invent a receipt to force payment.
Three way match example	PO: 100 units x $20. Received: 90 accepted. Invoice: 100 units. The 10-unit difference is $200 before tax. Investigate partial delivery, receipt timing or invoice error; apply approved tolerance rules.
Problem to action	PO placed after delivery -> bypassed controls -> regularize with documented approval -> track emergency buying separately. Supplier says order was never received -> missing acknowledgement -> obtain confirmation -> monitor unacknowledged POs. Late payment -> unmatched invoice or missing approval -> clear the actual block -> review aging weekly.
Control	Separate supplier creation, purchasing, receipt and payment where possible. Independently verify bank changes through a known contact. Retain requisition, approvals, PO, receipt, invoice and exception evidence.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Design procure-to-pay around demand authority, supplier commitment, receipt evidence, invoice entitlement and payment control. Use catalogs and contracts for the stable path; keep an auditable emergency path.
Decision rules	No approved need → stop. Need approved but supplier not qualified → expedite qualification or approved exception. Invoice mismatch → isolate quantity, price, tax, freight and duplicate causes. Bank change → independent callback using a trusted number.
Control	Monitor spend without PO, retrospective POs, unacknowledged POs, receipt aging, blocked invoices, duplicate invoices and emergency-buy recurrence.

[CH08] 08 Sourcing and supplier selection

Read first	Plain guidance
In one sentence	Choose a supplier on total value and risk.
Do this first	Normalize scope, unit, currency, Incoterm, lead time and quality before scoring bids.
What changes by company	Weights, mandatory criteria, baseline and award authority.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Run the event	Define specification and forecast volume. Issue the same bid pack to suppliers. Check capability and mandatory requirements first. Normalize commercial bids. Score using criteria agreed before bids arrive. Negotiate, validate and document the award.
Compare total cost	Total cost = purchase + freight + duty + handling + quality loss + inventory cost + tooling and transition cost. Avoid counting freight or duty twice when included in the quoted price.
Worked example	Supplier A: $10.00/unit + $0.80 freight + $0.20 expected quality cost = $11.00. Supplier B: $10.40 + $0.30 + $0.10 = $10.80. At 50,000 units, B is $10,000 lower before other differences.
Decision path	Mandatory quality, compliance or capacity requirement failed? Do not offset it with a low price. Both qualified? Compare total cost and delivery risk. High switching exposure? Pilot and qualify the alternate before moving volume.
Negotiation checklist	Know your target, walk-away position and alternative. Trade volume visibility for lead time, flexibility or price. Confirm index adjustment rules, MOQ, capacity, warranty and exit support. Record every concession and its condition.
Problem to action	Cheapest bid becomes expensive -> omitted charges -> rebuild a landed-cost comparison -> use a standard bid template. Single source becomes a bottleneck -> no qualified backup -> qualify an alternate -> monitor concentration by critical item.
Evidence	Retain bid pack, supplier responses, normalized comparison, qualification results, conflict declarations and award approval.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Build the category fact base before the sourcing event: demand, specifications, switching cost, supplier economics, should-cost, risk, market and exit options. Normalize every bid to the same scope, unit, currency, term and Incoterm.
Decision rules	Mandatory compliance or capability failure cannot be offset by price. If the winner depends on optimistic assumptions, run a sensitivity test. If transition risk is material, qualify and ramp through controlled milestones.
Control	Finance validates the baseline and savings method. Operations validates capacity and implementation. Procurement documents bid integrity, conflicts, award logic and concessions.

[CH09] 09 Supplier performance and recovery

Read first	Plain guidance
In one sentence	Recover a late or defective supplier with evidence.
Do this first	Verify what is physically complete, what is missing, daily output and pickup status.
What changes by company	Commitment date, evidence source, remedy and escalation path.

Read first	Plain guidance
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Score what matters	Track delivery against the agreed original commitment, accepted quality, responsiveness and corrective-action closure. Separate supplier-caused failures from buyer changes, but retain both views.
Useful measures	On-time delivery = due lines received on or before the agreed date / eligible due lines. Count overdue open lines as failures at the cutoff. Defective parts per million = defective units / inspected units x 1,000,000. Example: 8 defects in 20,000 inspected = 400 PPM. Sampling does not prove the uninspected population has exactly the same rate.
Decision path	One late order? Obtain item-level recovery dates. Repeated lateness? Compare promised capacity with actual output and constraints. Critical shortage? Escalate to leadership, validate alternate supply and allocate available stock.
Recovery checklist	Confirm actual work in progress, usable finished stock, material constraint, daily output and shipment capacity. Ask for evidence, not a percentage-complete estimate. Agree milestones, owners and update cadence. Verify the first recovered delivery.
Problem to action	Supplier keeps moving dates -> promises exceed capacity -> require a finite recovery plan -> review capacity before order acceptance. Defect recurs -> containment mistaken for root-cause correction -> verify process change -> check subsequent lots.
Corrective action record	Problem and scope | immediate containment | evidence of cause | permanent action | owner | due date | effectiveness check. Close only after evidence shows recurrence has been controlled.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Measure supplier-site-item performance against frozen obligations. Use evidence from production, WIP, material and logistics, not broad percent-complete claims. Link chronic failure to corrective action, commercial remedies and sourcing strategy.
Decision rules	Late and departed → manage ETA and border risk. Late and not departed → verify finished goods and pickup. Not produced → trace material, capacity, quality or payment constraint. Repeated miss → executive recovery plan, SCAR and alternate capacity.
Control	Scorecards show supplier-caused and total business failure separately. Close corrective action only after an effectiveness period with no recurrence.

[CH10] 10 Contracts and negotiation desk guide

Read first	Plain guidance
In one sentence	Convert business expectations into measurable obligations.
Do this first	For every risk define actor, action, timing, evidence, remedy and exit.
What changes by company	Legal entity, jurisdiction, caps, insurance and approval limits.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Build the contract pack	Agreement + scope/specification + price schedule + service levels + acceptance procedure + change process + governance and exit plan. State document precedence so a PO cannot accidentally contradict the agreement.

Read first	Plain guidance
Clause drafting examples	Acceptance: "Buyer shall inspect against Schedule A within {{INSPECTION_DAYS}} business days and notify Supplier of identified nonconformities. Latent defects remain subject to the agreed warranty." Change control: "Changes to scope, price or delivery dates require a written change order signed by authorized representatives before implementation, except as expressly provided for emergencies." Service reporting: "Supplier shall provide the monthly report defined in Schedule B by the {{REPORT_DUE_BUSINESS_DAY}} business day, including exclusions and supporting transaction records."
Negotiate and fill the gaps	Specify parties, term, scope, pricing, tax, payment, delivery, title and risk, quality, warranty, remedies, insurance, liability caps and exclusions, indemnities, IP, confidentiality, data security, audit, subcontracting, force majeure, termination, transition, governing law and dispute process.
Decision path	Ambiguous obligation? Define who does what, by when, to what standard and with what evidence. Unpriced change? Assess cost and timing before approval. Serious breach? Follow notice, cure and escalation provisions; preserve evidence.
Problem to action	SLA dispute -> unclear denominator or exclusions -> agree an example calculation -> attach a metric dictionary. Supplier lock-in -> weak exit terms -> agree data, stock and tooling handover -> test transition readiness.
Use carefully	These are drafting starters, not execution-ready legal terms. Have qualified counsel adapt remedies, liability, mandatory law and jurisdiction-specific requirements before signature.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Translate every business risk into an obligation, evidence, remedy and exit route. Align master agreement, SOW, price schedule, SLA and PO precedence. Keep negotiation targets, fallbacks and approval limits visible.
Decision rules	Ambiguous responsibility → define actor, action, timing, standard and evidence. Unbounded exposure → quantify and escalate. Operational dependency with weak exit → add transition assistance, data/tooling return and step-in rights.
Control	Maintain an obligation register with owner, frequency, evidence and remedy. Clause samples in this handbook are drafting shells for qualified legal review.

[NEG-01] Doing business, stakeholder work and negotiation

From opportunity to repeat business



Read left to right. Each commercial promise must move into approved wording, operational evidence, invoicing and a review that improves the next cycle.

Before the meeting	Write this down
Outcome	What result is needed, by when, and how it will be proved?

Before the meeting	Write this down
BATNA	Best realistic action if no agreement is reached.
Range	Must-have, target, fallback and walk-away for price, service, risk and timing.
Authority	What you can approve; what needs legal, finance, executive or technical approval.
Counterpart	Their goals, constraints, decision maker, alternatives and likely objections.
Facts	Volume, forecast, cost model, market data, quality, service, risk and switching cost.
Trade list	Low-cost things you can give and high-value things you need in return.

Meeting step	Useful words	Avoid
Open	‘Our goal today is to agree scope, economics, risks and next actions.’	Starting with a price concession
Clarify	‘Help me understand what makes that date difficult.’	Arguing before facts are clear
Diagnose	‘Which constraint is capacity, material, approval or cash?’	Accepting ‘system issue’ as a root cause
Propose	‘If you commit tested capacity and weekly evidence, we can discuss a longer term.’	Giving without receiving
Test	‘What would prevent your team from signing this wording?’	Assuming silence is agreement
Close	‘We agreed A, B and C; owners are X and Y; wording is due Friday.’	Leaving conditions unwritten

Conversation	Opening script	Decision to leave with
Supplier recovery	‘Show finished quantity, WIP, missing material, daily rate and confirmed pickup.’	Milestones, evidence, fallback and escalation
Customer shortage	‘Here is confirmed supply and the fair allocation rule; choose split, substitute or revised date.’	Written promise and communication owner
Finance	‘Here is the operational driver, accounting treatment, cash effect and reconciliation.’	Approved baseline and benefit method
Executive	‘Decision required, options, service/margin/cash/risk impact, recommendation and deadline.’	Named decision and owner
Site team	‘Show me one real transaction and where the work waits or returns.’	Process correction and local owner

Commercial offer and contract handoff

Qualify need → Define scope → Build cost → Set price → Negotiate → Approve → Contract → Onboard → Deliver → Invoice → Collect → Review

Item	Minimum definition
Scope	Who does what, where, volume, exclusions, input and output
Price	Unit, currency, taxes, freight, index, minimums and validity
Service	Formula, population, clock, source, target and evidence
Risk	Quality, continuity, data, trade, safety, insurance and liability
Change	Requester, impact, authorization, effective date and version
Exit	Notice, inventory, data, tooling, transition work and final reconciliation

Veteran rule: never trade an undefined promise. Convert every statement into actor + action + timing + standard + evidence + consequence. Contract wording in this book is a business drafting aid and must follow the company legal-review process.

Filled negotiation example

Question	Filled answer
Situation	Supplier requests an 8% price increase.
Must have	Uninterrupted supply and evidence for every cost driver.
Target	2% from the next quarter.
Fallback	Up to 4% for six months if capacity is reserved and a cost-reduction plan is signed.
Escalate / walk away	Retroactive or unsupported increase above authority.
Give	24-month nonbinding volume visibility.
Get	Verified capacity, 98% OTIF, monthly evidence and downward index adjustment.
Say	If you provide the capacity evidence and the 98% service commitment, I can take the 4% temporary option for approval.
Record	Every concession, condition, owner, expiry and final wording.

Contract architecture and negotiation cards

Build one controlled agreement pack: master terms, statement of work, specification, pricing, service levels, security/data schedule, quality agreement, change process, governance and exit plan. The clause starters below require company legal review before signature.

Commercial and service

Clause	Business risk	Plain clause starter	Check before signing
Scope and precedence	Conflicting documents or hidden work	Supplier shall perform the services and deliverables in Schedule A. If documents conflict, the agreed order of precedence applies.	Target: Complete measurable scope and buyer-controlled acceptance. Fallback: Allow later document only for the specific order and only when signed. Escalate: Unclear specification or authority.
Pricing and indexation	Uncontrolled escalation	Prices remain firm except the documented index formula using baseline month, published index, frequency, cap and downward movement.	Target: Open-book support and symmetric adjustment. Fallback: Use weighted verified inputs with cap and review trigger. Escalate: Retroactive or undefined increase.
Forecast and commitment	Forecast treated as purchase obligation	Forecasts are planning information only except for the stated firm window and authorized releases.	Target: Narrow firm window and flexibility bands. Fallback: Capacity fee tied to reserved and proven capacity. Escalate: Unlimited take-or-pay.
Service level and credit	Unmeasurable service or ineffective remedy	Supplier shall meet the metric definitions in the SLA schedule. Credits apply to validated misses and do not replace correction rights.	Target: Transaction-level evidence and chronic-failure right. Fallback: List objective exclusions with evidence and review. Escalate: Supplier controls denominator or exclusions.
Acceptance and warranty	Payment implies acceptance or latent defects excluded	Inspection within the stated period addresses visible nonconformity; latent defects remain subject to warranty.	Target: Clear criteria and correction clock. Fallback: Deemed acceptance only for visible criteria. Escalate: No remedy for latent defect.
Title risk and inventory	Unclear loss or obsolete stock ownership	Title and risk transfer at the defined event; consigned or safety stock is separately identified, insured and reconciled.	Target: Risk follows physical control and cause. Fallback: Early title with segregation, audit and insurance. Escalate: Buyer owns unusable stock without control.

Quality, continuity and change

Clause	Business risk	Plain clause starter	Check before signing
Quality recall and CAPA	Unsafe product or cost dispute	Supplier shall contain, investigate and correct nonconformity and support traceability and recall obligations.	Target: Creation and escape cause, cost allocation by cause. Fallback: Reasonable verified direct cost tied to responsibility. Escalate: No rapid containment or records.
Capacity and continuity	Supplier cannot recover	Supplier shall maintain and test the approved continuity plan and notify constraints that threaten commitments.	Target: Reserved capacity evidence and alternate plan. Fallback: Summary evidence and audit under confidentiality. Escalate: No recovery commitment for critical supply.
Change control	Silent scope price or process change	No change is effective without an authorized change order stating scope, cost, schedule, risk and validation.	Target: Buyer approval before implementation. Fallback: Temporary emergency path with notice and retrospective approval. Escalate: Supplier can change critical process unilaterally.
Data cyber and audit	Data loss unauthorized access or unverifiable charges	Supplier shall apply scheduled security controls, notify incidents and provide agreed audit evidence.	Target: Least privilege, defined notice, remediation and exit return. Fallback: Risk-based audit frequency and independent reports. Escalate: No incident duty or data return.

Legal, compliance and exit

Clause	Business risk	Plain clause starter	Check before signing
Intellectual property and tooling	Loss of designs or production tooling	Ownership and permitted use of background IP, deliverable IP, data and tooling are stated by category.	Target: Buyer-funded tooling identified and returnable. Fallback: Supplier retains background IP with required license. Escalate: Critical tooling cannot be accessed on failure.
Compliance and trade	Illegal labor trade environmental or regulatory exposure	Supplier shall comply with applicable requirements, maintain evidence and notify material breach or investigation.	Target: Flow-down to relevant subtiers. Fallback: Tie duties to performance locations and applicable regimes. Escalate: No screening or notice for controlled supply.
Liability insurance and indemnity	Exposure exceeds remedy or insurance	Liability allocation, exclusions, cap, insurance and indemnities are stated and aligned with insurable risk.	Target: Carve-outs for agreed high-severity risks. Fallback: Higher super-cap for named risks. Escalate: Cap below plausible direct loss without mitigation.
Force majeure and shortage allocation	Event becomes an excuse without mitigation	Relief requires causation, prompt notice, mitigation and periodic evidence; allocation follows agreed priorities.	Target: No relief for avoidable internal failure. Fallback: Objective causation and mitigation regardless of label. Escalate: No notice, mitigation or end point.
Termination and transition	Lock-in or interrupted supply	On termination supplier shall provide defined transition, data, inventory, tooling and knowledge assistance.	Target: Step-in for critical failure and orderly exit. Fallback: Time-boxed assistance at agreed rates unless breach-caused. Escalate: No access to critical assets/data.
Dispute and governance	Operational issue becomes legal standoff	Named operational, executive and legal escalation applies before formal proceedings where urgent relief is not required.	Target: Fast fact-based escalation. Fallback: Shorter route for service, safety and continuity. Escalate: No urgent-relief route.

Negotiation execution

Before	During	After
Rank must-have, target, fallback and walk-away. Quantify total cost, transition, service and failure exposure. Set authority.	Trade conditionally. Record concession, value, dependency and expiry. Pause if scope or risk exceeds authority.	Confirm complete wording, schedules and redlines. Recalculate the business case. Load obligations and owners.

Indexation example: 40% of price moves with published index I. Base price $100, base index 200, current index 220. Adjustment = $100 × 40% × (220 ÷ 200 − 1) = $4. New price $104 before any cap, timing or audit rule.

Service credit example: 94% actual against 98% target, schedule states 0.5% of monthly fee per percentage point missed, capped at 5%. Miss = 4 points; credit = 2% of eligible monthly fee. Define rounding, exclusions and chronic failure separately.

SEE ALSO [CH08] sourcing, p.21 • [CH09] supplier recovery, p.21 • [CH10] contracts, p.22

[CH11] 11 Warehouse layout and capacity

Read first	Plain guidance
In one sentence	Arrange space and flow for safe peak work.
Do this first	Measure item cube, order lines, handling units, travel, queues and exception space.
What changes by company	Building, equipment, safety code, product and peak profile.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.

Field panel	Execution guidance
Design around flow	Receiving -> inspection or quarantine -> storage -> replenishment -> pick -> pack -> dispatch. Separate incompatible goods, pedestrians and equipment routes through the site safety process.
Capacity checklist	Profile pallets, cases, eaches, dimensions, weights, temperature and hazardous-material needs. Check peak inventory and daily flows. Include quarantine, returns, staging, equipment and aisle space; do not allocate the entire building to storage.
Useful calculations	Storage occupancy = occupied usable locations / usable locations. Example: 850/1,000 = 85%. Investigate congestion before adopting a universal occupancy target. Required pick hours = lines / demonstrated lines per labor-hour. 3,000 lines / 60 = 50 labor-hours. At 6 productive hours per worker, round 8.33 up to 9 workers; verify skill and equipment constraints.
Decision path	Space full? Check blocked and obsolete stock first. Good stock in wrong slots? Re-slot by movement and cube. Peak inbound exceeds docks? Stagger appointments or add approved temporary staging.
Problem to action	Long travel -> fast movers far from dispatch -> slot by picks and compatibility -> review movement monthly. Floor stock everywhere -> no location discipline -> assign temporary controlled locations -> enforce putaway scans.
Before go live	Validate rack and equipment limits with qualified personnel. Walk receiving, storage, emergency access and dispatch routes. Test an oversized item, a quarantined lot and a peak-volume wave.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Design for peak flow, safe movement and exception space. Size receiving, quarantine, reserve, forward pick, packing, staging and returns separately. Validate with real SKU cube, handling unit and order-line profiles.
Decision rules	Congestion with low storage occupancy → flow or slotting problem. High occupancy with obsolete/blocked stock → disposition problem. Peak dock queue → appointment, labor, yard or staging constraint. Unsafe flow → stop and redesign with qualified safety personnel.
Control	Treat maximum usable capacity as the engineered operating limit, not total floor or rack volume. Revalidate after product mix or packaging changes.

[CH12] 12 Receiving and putaway

Read first	Plain guidance
In one sentence	Make received stock counted, inspected, recorded and findable.
Do this first	Match PO/ASN, physical item, quantity, condition, lot and destination.
What changes by company	Inspection, tolerance, stock status and putaway rules.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Process	Book appointment -> verify vehicle and shipment -> unload safely -> count -> inspect -> record receipt -> label -> quarantine or release -> put away -> confirm location.
Receiving checklist	Match PO and advance shipment notice. Verify item, quantity, unit, lot, serial, expiry, damage and required documents. Record discrepancies and photographs under company policy. Keep physical status aligned with system status.
Decision path	Wrong item or suspected damage? Segregate and notify Purchasing/Quality. Quantity difference? Recount and record actual receipt. Missing documentation? Use a controlled hold. Correct and released? Put away to the approved location.
Measure	Dock-to-stock time = available-for-use timestamp - arrival timestamp. Track inspection holds separately from physical processing. Example: arrival 08:00, available 14:30 -> 6.5 hours; 4 hours may be inspection waiting.
Problem to action	System says received, stock cannot be found -> premature receipt or missing putaway -> trace receipt and location scans -> require location confirmation. Expired product accepted -> missing shelf-life check -> hold lot and seek disposition -> validate remaining shelf life on receipt.
Close the receipt	Capture PO line, received and rejected quantities, units, lot/serial, expiry, location, timestamp and operator. Reconcile partial receipts and supplier discrepancies before closing the PO.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Use appointment, ASN, unload, count, inspect, receive, release and putaway timestamps. Keep physical status and system status synchronized. A receipt is complete only when the stock is findable and usable or deliberately quarantined.
Decision rules	Damage/wrong item → segregate. Quantity variance → recount and record actual. Missing compliance evidence → controlled hold. Valid and released → directed putaway with location confirmation.
Control	Separate carrier wait, unload, inspection and putaway time. Improve the largest delay instead of hiding it in one dock-to-stock average.

[CH13] 13 Inventory accuracy and cycle counts

Read first	Plain guidance
In one sentence	Make physical stock equal system stock.
Do this first	Freeze the count area, count blind, recount material differences and trace the last good event.
What changes by company	Tolerance, count frequency, approval and valuation policy.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**

Read first	Plain guidance
Count correctly	Choose risk-based count frequency. Freeze affected movements or use a precise transaction cutoff. Count blind without showing expected quantity. Recount material differences independently. Investigate before approved adjustment.
Useful measures	Record accuracy = records within agreed tolerance / records counted. Absolute quantity variance = sum of absolute(counted - system) quantities, only when units are comparable. Do not net positive and negative errors. Example: 96 of 100 records within tolerance -> 96% accuracy.
Decision path	Variance found? Check pending receipts, picks, transfers and unit conversion. Then check wrong location, wrong lot, damage and unrecorded consumption. Real loss remains? Follow the investigation and approval process.
Problem to action	Repeated negative stock -> issues posted before receipts or poor backflush -> repair sequence and consumption logic -> monitor negative balances daily. Count always matches after recount -> counters see system balance -> use blind counts -> independently sample recounts. Stock exists in wrong bin -> transfer not scanned -> correct location with evidence -> require source and destination confirmation.
Count sheet	Count ID | cutoff | item | location | lot | unit | system quantity | first count | recount | variance | cause | approver | adjustment ID.
Control	Never adjust stock solely to make the report green. Track recurring causes and verify the fix on the next cycle. Finance approves valuation impacts according to authority limits.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Count blind at the smallest traceable location and status. Protect the transaction cutoff. Recount significant differences independently, then trace the last known correct event through receipt, movement, consumption, pick and adjustment.
Decision rules	Same variance repeats → process defect. Random variance across many locations → discipline or system/interface issue. Quantity correct but value wrong → unit, currency or cost-layer problem. Regulated lot mismatch → quarantine and specialist escalation.
Control	Adjust only with approval and root-cause code. Measure count accuracy, absolute variance, adjustment value and repeat error by cause.

[CH14] 14 Picking packing and dispatch

Read first	Plain guidance
In one sentence	Turn demand into a correct shipment.
Do this first	Release only when stock, labor, equipment, documents and carrier capacity exist.
What changes by company	Allocation, wave, substitution, scan and cutoff rules.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Process	Release eligible order -> replenish pick face -> pick with item/location verification -> confirm quantity and lot -> pack -> label -> stage by route -> load check -> dispatch confirmation.
Before release	Check credit and quality blocks, stock availability, customer labeling rules, shelf life, route cutoff and carrier capacity. Urgent orders still need traceability and shipment evidence.

Read first	Plain guidance
Decision path	Short pick? Verify nearby and reserve locations and pending moves. Wrong or damaged item? Hold and replace. Order ready but misses cutoff? Confirm next transport option and customer commitment before paying expedite cost.
Useful measures	Pick accuracy = correct checked lines / checked lines. Order OTIF = orders delivered both on time and in full / eligible due orders. Example: 92 qualifying orders / 100 due = 92%. A split order is not fully delivered until all required quantities arrive under the agreed rule.
Problem to action	Wrong item shipped -> look-alike packaging or scan bypass -> verify item barcode -> separate look-alikes and review overrides. Missing cartons -> weak load reconciliation -> compare staged and loaded carton IDs -> require final load scan. Damage -> unsuitable packaging -> validate pack method -> review claims by item and lane.
Keep evidence	Order and line IDs, picked quantity, lot/serial, carton IDs, carrier, tracking, dispatch time and delivery proof. Keep dispatch performance separate from actual delivery performance.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Release work only when inventory, labor, equipment, carrier and document capacity exist. Allocate scarce stock through approved customer/service rules. Use scan confirmation and exception reason codes at each irreversible step.
Decision rules	Short pick → recheck location/status, alternate stock, substitute, split or revise promise. Wrong item discovered before ship → stop and repack. After ship → contain customer impact and trace affected batch.
Control	Measure perfect order at transaction level. Do not multiply separate average percentages. Audit wave release, replenishment timing, pick path and packing verification.

[CH15] 15 Transportation and freight cost

Read first	Plain guidance
In one sentence	Move freight at the promised service and controlled total cost.
Do this first	Verify location, remaining path, delivery risk and customer exposure before expediting.
What changes by company	Lane, mode, rate, fuel, accessorial and service commitments.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Select the move	Confirm delivery deadline, dimensions, weight, temperature, fragility and trade requirements. Compare feasible modes and total landed cost. Book, track milestones, capture proof of delivery and audit the invoice.
Cost checks	Chargeable weight is typically the greater of actual and dimensional weight under the carrier tariff. Example only: 60 x 40 x 40 cm / divisor 5,000 = 19.2 kg; actual 12 kg -> 19.2 kg before tariff rounding. Confirm the carrier divisor and minimum charge.
Decision path	Late shipment? Identify the last verified milestone. Not collected -> carrier dispatch. In transit -> carrier recovery. Held at border -> trade team and broker. Delivered but disputed -> proof of delivery and customer receiving check.

Read first	Plain guidance
Problem to action	Unexpected freight charges -> unpriced accessorials -> reconcile fuel, waiting and handling -> agree a complete tariff. Frequent premium freight -> late planning or release -> code the initiating cause -> fix upstream timing. Low truck utilization -> fragmented dispatch -> consolidate compatible loads -> align order cutoff and route schedule.
Shipment checklist	Origin/destination | pickup/delivery window | cartons/pallets | gross weight | dimensions | commodity and handling needs | carrier | agreed tariff | tracking | exception owner.
Compare options	Premium freight is justified by avoidable business loss, not urgency alone. Example: $800 expedite protects $3,000 contribution only if the shipment actually resolves the shortage; confirm the receiving and production window.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Select mode and carrier on total delivered economics: service probability, freight, inventory in transit, handling, damage, duty timing and carbon. Tender with clean weight, cube, equipment, windows and accessorial rules.
Decision rules	Miss imminent → confirm physical location and remaining path, then compare expedite cost with margin/service exposure. Cost spike → separate rate, mode, distance, fuel, accessorial and utilization effects. Repeated rejection → repair forecast, capacity commitment or lane design.
Control	Audit invoice against contracted rate, shipment facts and duplicate status. Track tender acceptance, pickup, delivery, dwell, claims, empty distance and cost intensity.

[CH16] 16 Managing a 3PL

Read first	Plain guidance
In one sentence	Operate a logistics partner as part of the company process.
Do this first	Reconcile provider transactions and inventory to enterprise records.
What changes by company	Scope, SLA, exclusions, credits, audit and exit duties.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Define the service	Document inbound, storage, inventory control, outbound, returns, value-added work, systems, reporting and customer-specific requirements. Agree what is included in each rate and what triggers an extra charge.
Before signing	Validate capacity, operating hours, labor plan, inventory responsibility, insurance, subcontracting, data ownership, audit access, business continuity and exit assistance. Reconcile opening stock jointly. Use the Chapter 10 legal review checklist.
SLA drafting starter	"Inventory accuracy shall be measured monthly as records within {{TOLERANCE}} divided by records counted under the agreed blind-count method. Both parties shall review discrepancies and exclusions with supporting transaction evidence." Define count selection so poor records cannot be avoided.
Worked invoice	Storage: 1,000 pallet-days x $0.50 = $500. Handling: 200 inbound pallets x $3 = $600. Picking: 800 lines x $0.40 = $320. Total = $1,420 before approved extras and taxes. Confirm whether storage is daily, average monthly or peak-based.

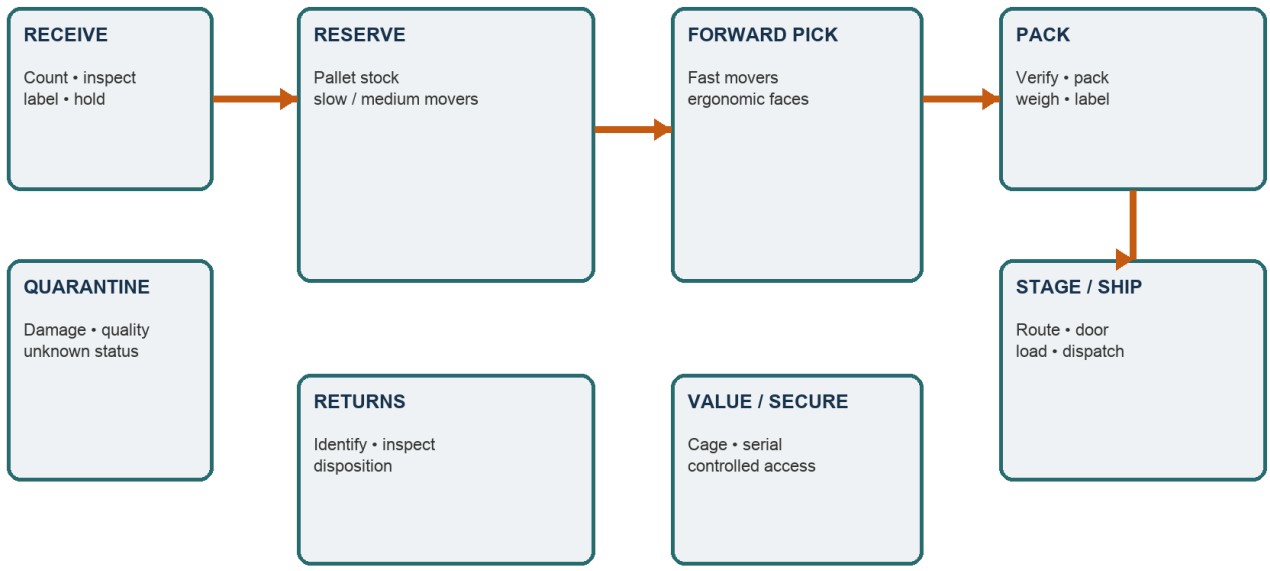
Read first	Plain guidance
Problem to action	Invoice exceeds plan -> ambiguous rate basis -> rebuild charges from activity -> require a rate-to-activity mapping. Stock mismatch -> interface timing or unrecorded move -> reconcile by item/location/lot -> review daily exceptions. Service deteriorates at peak -> unsupported forecast -> agree peak capacity and booking rules -> rehearse surge plans.
Governance	Daily exceptions; weekly service and inventory review; monthly invoice and improvement review. Maintain an exit inventory reconciliation, data export and transition schedule.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Govern a 3PL as an extension of the operating model. Contract data ownership, inventory integrity, cutoffs, controls, reporting, credits, chronic failure, change, audit and exit. Reconcile provider transactions to enterprise books.
Decision rules	Isolated miss → correct and verify. Systemic provider-controlled failure → recovery plan and remedy. Customer/data/system-controlled failure → repair owner and SLA exclusions. Persistent structural gap → redesign scope or rebid.
Control	Keep operational review, commercial review and executive governance on distinct cadences. Never let a credit replace permanent correction.

[WH-01] Warehouse placement, allocation and easy tracking

Warehouse flow and placement map



Read the arrows from receiving to reserve, forward pick, pack and ship. Returns, quarantine and secure stock stay on controlled side paths. Put each item where velocity, cube, weight, safety, temperature, expiry and security can be controlled.

Placement question	Rule
How often is it picked?	Fast movers near the golden zone and dispatch; slow movers farther away.
How large/heavy?	Protect rack/floor capacity; heavy items low; bulky items in suitable equipment paths.
Can it mix?	Separate incompatible, hazardous, allergen, odor, temperature or ownership classes.

Placement question	Rule
Does it expire?	Use FEFO, visible lot/expiry, quarantine and remaining-shelf-life controls.
Is it valuable?	Use secure access, serial control, camera/audit coverage and smaller exposure.
How is it picked?	Provide ergonomic pick face; reserve replenishes the forward location.
Is demand variable?	Keep flexible overflow and avoid permanently dedicating too much prime space.

ABC value/velocity	X stable	Y variable	Z intermittent
A high	Prime pick; tight count; small frequent replenishment	Prime/flexible; monitor event demand	Secure critical-spares policy; avoid blind forecast
B medium	Standard forward pick	Flexible slot; periodic review	Reserve/shared slot; review service need
C low	Compact shared slot	Reserve; replenish on trigger	Deep reserve or buy-to-order when suitable

Bin code	Meaning
A-03-02-B	Aisle A • bay 03 • level 02 • position B
Scan identity	Product + bin + lot/batch + serial/HU when required + stock status
Event chain	Receive → label → inspect → putaway scan → move → replenish → pick → pack → ship
Control	Each scan records what, from, to, quantity, unit, person/device and time

Allocation decision

Step	Question	Action
1	Is stock physically present and usable?	Exclude blocked, quality, expired, damaged and already committed stock.
2	Is there a hard reservation?	Honor order/project/lot reservation under policy.
3	Which rule applies?	FEFO/FIFO, customer class, margin, medical/safety criticality or contractual priority.
4	Can demand be split or substituted?	Use approved alternate, transfer, partial or revised promise.
5	What remains?	Backorder with reason, owner, next supply and customer update.

Slotting score worked example

Example score = 40% velocity rank + 20% cube fit + 15% weight/ergonomic fit + 15% affinity + 10% replenishment fit. Item A100 scores 90, 80, 100, 70, 85. Weighted score = 36 + 16 + 15 + 10.5 + 8.5 = 86. Higher scores get better eligible pick locations; mandatory safety and compatibility rules always override the score.

Problem	Look first	Action now	Prevention
Stock exists but cannot pick	Status, reservation, lot, bin block, open task	Release valid stock or create controlled alternate	Daily stuck-stock exceptions
Repeated short pick	Last moves, UOM, replenishment, wrong-slot pattern	Recount and replenish	Scan sequence and root-cause count
Congested dock	Arrivals by 15-minute interval, doors, labor, stage space	Sequence appointments and overflow	Peak capacity and appointment rule
Cannot find pallet	HU/label, last confirmed bin, open move, interface	Search controlled path; block duplicate	Mandatory source/destination scan
Expiry loss	Receipt shelf life, FEFO, demand and lot visibility	Reallocate, return or dispose under approval	Minimum remaining shelf life and alerts

SEE ALSO [CH11] layout, p.27 • [CH12] receiving, p.28 • [CH13] accuracy, p.28 • [CH14] fulfillment, p.29

[CH17] 17 Manufacturing and material planning

Read first	Plain guidance
In one sentence	Convert demand into feasible material supply.
Do this first	Check demand, usable stock, open supply, UoM, BOM, dates, lead time and lot size in order.
What changes by company	Planning type, time fence, source, calendars and parameters.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Planning flow	Released demand -> master schedule -> BOM explosion -> net requirements -> planned orders -> capacity check -> authorized release -> material issue -> production -> quality release -> completion.
MRP essentials	Use valid BOM revision, routing, yield, lead times, lot sizes, stock status and open orders. Dependent demand comes from the production plan; do not independently forecast it and count both requirements.
Worked calculation	Plan 500 good assemblies; component usage 2 each; expected component loss 5%. Gross component need = 500 x 2 / 0.95 = 1,052.63 -> 1,053. Usable stock 300 and timely confirmed receipts 200 -> net need 553. Round to order multiple only after checking dates.
Decision path	MRP shortage? Check date and usable status, then BOM/unit correctness and supply confirmation. Materials available but output late? Check the constrained work center and production sequence. Do not release more WIP into a blocked bottleneck.
Problem to action	Too many planned orders -> bad parameters or duplicate demand -> validate input and netting -> review exceptions before release. Material usage high -> scrap, BOM or backflush error -> reconcile actual consumption -> maintain engineering changes. Schedule changes daily -> unrealistic promises -> establish a controlled freeze window -> approve exceptions visibly.
Release checklist	Material, tooling, labor, equipment, quality instructions, drawings and revision are ready. Confirm the next operation can accept the output.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Stabilize demand, BOM, yield, routing, calendars, inventory status and lead times before trusting MRP. Use pegging to connect every exception to demand and time fence. Release only feasible orders.
Decision rules	Implausible MRP → validate demand, usable inventory, UOM, BOM effective date, yield/scrap, calendar, lead time, MOQ and open supply in that order. Inside freeze fence → controlled exception. Outside fence → parameter or plan correction.
Control	Monitor nervousness, expedite/defer/cancel recommendations, reschedule acceptance, shortages and planner overrides. Expire temporary parameter changes.

[CH18] 18 Capacity scheduling and OEE

Read first	Plain guidance
In one sentence	Protect the constraint and schedule work that can actually run.
Do this first	Separate theoretical, demonstrated and committed capacity; include losses.
What changes by company	Shift calendar, setup, yield, maintenance and labor assumptions.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**

Read first	Plain guidance
Start with the constraint	Identify the resource that limits total output. Protect it from shortages, avoidable changeovers and quality losses. Schedule the constraint first, then align upstream and downstream work.
Formulas and example	Takt time = available production time / customer demand. 420 minutes / 210 units = 2 minutes/unit. OEE = availability x performance x quality. Availability 0.90, performance 0.95, quality 0.98 -> OEE 83.79%. Define planned time and ideal cycle consistently. Do not multiply by OEE again if demonstrated capacity already includes those losses.
Finite capacity check	Required hours = setup hours + quantity x standard run hours per unit. 2 hours + 600 x 0.01 = 8 hours. Compare with actual available machine and qualified labor hours for the same period.
Decision path	Load exceeds capacity? Can setup or sequence improve? If insufficient, compare overtime, subcontracting or date changes. Low output with spare nominal capacity? Separate downtime, speed loss and scrap before adding equipment.
Problem to action	High local efficiency, late orders -> making wrong mix -> schedule to due dates and constraint -> measure flow and delivery. WIP grows -> releases exceed throughput -> cap release rate -> monitor queue age. Frequent changeovers -> fragmented batches -> sequence compatible products -> balance setup savings against inventory.
Daily board	Plan versus actual good output | downtime reason | scrap | constraint queue | material readiness | recovery action and owner.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Schedule the constraint first, then subordinate upstream and downstream work. Separate theoretical, demonstrated and committed capacity. Include setup, maintenance, quality, absenteeism and changeover losses.
Decision rules	Utilization high with growing queue → capacity constraint. Utilization low with low output → starvation, blocking, downtime or standard error. OEE loss → isolate availability, performance and quality before acting.
Control	A local efficiency increase can worsen total flow. Judge changes by constraint throughput, service, WIP and total cost.

[CH19] 19 Quality traceability and returns

Read first	Plain guidance
In one sentence	Contain defects, trace the population and prevent recurrence.
Do this first	Stop release, identify lots/serials, preserve evidence and obtain disposition authority.
What changes by company	Severity, traceability depth, hold and release authorities.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Contain first	Suspected nonconforming material -> identify affected lots -> physically segregate and block in the system -> trace where material went -> notify Quality and affected process owners. Only authorized personnel decide disposition.
Traceability checklist	Supplier lot -> receipt -> storage and transfers -> production consumption -> finished lot -> customer shipment. Test backward and forward tracing. Record rework and substitutions so genealogy remains complete.
Return process	Authorize return -> identify item and reason -> receive separately -> inspect -> decide restock, repair, return to vendor, recycle or scrap -> post credit and inventory change -> verify corrective action.

Read first	Plain guidance
Useful measure	First-pass yield = units passing without rework / units entering the process. 950 pass first time out of 1,000 -> 95%. Reporting 990 eventually accepted as 99% first-pass yield hides rework.
Problem to action	Recurring defect -> symptom treated, cause unverified -> test the suspected cause -> verify later lots. Returned goods mixed with good stock -> no hold status -> segregate and inspect -> enforce return disposition. Recall scope unclear -> missing lot genealogy -> reconstruct affected transactions -> require lot capture at every transformation.
Escalation	Use the applicable product-safety, recall and regulatory procedure. Do not substitute this general reference for industry-specific release, recall or reporting requirements.
Closure evidence	Disposition approval, quantities reconciled, customers or suppliers notified where required, corrective action verified and financial postings complete.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Contain affected product, preserve genealogy and decide disposition through authorized roles. Root cause must explain creation and escape. CAPA changes the process and proves effectiveness across enough future opportunities.
Decision rules	Safety/regulatory concern → stop ship and formal escalation. Traceable scope known → isolate exact lots/serials. Scope uncertain → widen containment. Rework/use-as-is/scrap/return only through approved MRB authority.
Control	Test mock recall time, population completeness, contact accuracy and reconciliation of produced, shipped, returned and destroyed quantities.

[CH20] 20 Global trade and landed cost

Read first	Plain guidance
In one sentence	Move goods across borders with correct classification, value, origin and documents.
Do this first	Obtain the exact broker or authority request before changing anything.
What changes by company	Country, product, party, regime, Incoterm and current rules.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Shipment release checklist	Confirm product classification, origin, customs value, parties, destination, end use, licenses, restricted-party checks and documents. A screening-list search alone is not a complete export authorization. Requirements vary by jurisdiction; validate with the trade specialist.
Incoterms quick check	Write the rule, precise named place or port, and "Incoterms 2020". They allocate specified delivery obligations, costs and risk; title and payment require separate contract terms. Under CPT, seller-paid carriage does not mean seller bears risk to destination.
Decision path	Documentation mismatch or possible restriction? Hold release and resolve with Trade Compliance. Broker asks for clarification? Supply verified product and transaction facts. Do not change a description or value merely to obtain clearance.
Illustrative landed cost	Goods $10,000 + freight $800 + insurance $100 + duty $500 + brokerage $150 = $11,550. At 1,000 usable units, $11.55/unit. Duty base, rates, taxes and tax recoverability depend on the transaction; this is arithmetic, not a customs valuation rule.

Read first	Plain guidance
Problem to action	Border hold -> inconsistent invoice or missing license -> reconcile documents and authorization -> pre-check before dispatch. Surprise duty -> wrong classification/origin assumption -> obtain a validated determination -> maintain classification records.
Keep the file	Commercial invoice, packing list, transport document, classification/origin evidence, permits where required, screening record, customs entry and payment evidence. Apply the jurisdiction-specific retention period.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Determine classification, value, origin, licenses, sanctions, documents and Incoterm responsibilities before shipment. The commercial term allocates costs and risks; it does not replace customs, title or payment terms.
Decision rules	Hold → identify exact customs/broker request. Missing document → correct controlled record. Classification/value/origin dispute → qualified trade review. Sanctions or license concern → stop and escalate; do not route around the control.
Control	Retain decision rationale, evidence, broker instructions and post-entry corrections. Screen relevant parties using current official rules and company procedure.

[CH21] 21 Cost cash and performance measures

Read first	Plain guidance
In one sentence	Connect service and operations to profit, cash and risk.
Do this first	Define metric formula, grain, population, source, cutoff, owner and action.
What changes by company	Accounting policy, currency, period, allocation and threshold.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Know the basis	Use agreed cost definitions, currency, period and cutoffs. Operational improvement, cash release and profit are different outcomes; avoid counting the same benefit twice.
Core formulas	Inventory turns = annual COGS / average inventory at cost. DIO = average inventory / COGS x days in period. DSO = average receivables / credit sales x days. DPO = average payables / credit purchases x days. Cash conversion cycle = DIO + DSO - DPO.
Worked example	Annual COGS $12m; average inventory $2m -> 6 turns and DIO 60.83 days using 365 days. If DSO=40 and DPO=35, cash cycle=65.83 days. Reducing inventory by $200,000 releases cash once; at a 20% annual carrying rate, modeled annual carrying benefit is $40,000.
Decision path	Cost up? Separate volume, mix, price, exchange rate and productivity. Inventory up? Split quantity change from valuation change. Savings claimed? Compare like-for-like baseline and validate with Finance.
Problem to action	Good service, poor margin -> unmeasured cost to serve -> include handling, returns and freight by customer -> review service tiers. Payment-term extension hurts supply -> cash target ignores supplier health -> assess risk -> negotiate sustainably.
Metric card	Name | purpose | numerator | denominator | grain | exclusions | source | cutoff | owner | target | escalation. Never compare percentages built from different populations.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Connect operational drivers to service, margin, cash and risk. Use cost-to-serve by customer/product/channel and reconcile it to finance. Pair each lagging outcome with a leading indicator and accountable decision.
Decision rules	Improving KPI but worsening outcome → wrong grain, gaming or missing guardrail. Savings without realized P&L/cash effect → baseline or implementation gap. High margin with high disruption exposure → add expected-loss view.
Control	Every KPI has formula, owner, grain, inclusions, exclusions, source, refresh, threshold, action, escalation and retirement rule.

[SAP-01] SAP and ERP from first click to integrated control

ERP means Enterprise Resource Planning: one controlled system connects master data, operational documents and accounting. SAP app names, fields, roles and transaction codes vary by edition, configuration and authorization. Search by business purpose, use value help instead of guessing, and record the saved document number.

Beginner run card

Question	Answer before you run
Goal	Display and understand one real document before creating or changing anything.
Need	One known document, product, supplier or plant plus the approved system and role.
Where	Sign in to SAP Fiori and search by business purpose, such as Display Purchase Orders.
Sample	Display purchase order 4500001234 for plant 1710.
Replace	Only {{DOCUMENT_NUMBER}}, {{PLANT}} and other values copied from value help or an approved record.
Run	Open a Display app, enter the value, choose Go, then open status or Document Flow.
Expected	The correct organization, items, quantities, dates, status and linked documents appear.
Why	Display follows the document chain without posting stock or accounting changes.
If wrong	Copy the exact message; check role, organization, leading zeroes, date and document type.

SAP action word	Plain meaning
Display	Read only: show a saved record without changing it.
Create / Change	Prepare a new record or edit an allowed field; may still require Save/Post.
Check / Simulate	Validate entries without committing the business posting.
Save / Post	Change stock, status or accounting and create a document number.
Reverse	Create an approved linked correction; do not hide the original posting.

Kind	Plain meaning	Examples
Organization	Where or who owns the activity	Company code, plant, storage location, purchasing organization
Master data	Reusable identity or rule	Product, supplier, customer, BOM, work center
Transaction	A business event	Purchase order, goods receipt, sales order
Document	Saved evidence of the event	Material document, accounting document, invoice

Question		Answer before you run
Configuration	Rules controlling the system	Tolerance, movement type, document type
Placeholder	**Example only**	**Find your real value**
{{COMPANY_CODE}}	US01	Finance or system owner
{{PLANT}}	1710	Product master, PO or site list
{{STORAGE_LOCATION}}	RM01	Product master or warehouse map
{{WAREHOUSE}}	W171	EWM warehouse list
{{PRODUCT}}	A100	Product/material search
{{SUPPLIER}}	1000001	Business Partner search
{{PURCH_ORG}}	P100	PO or organization map
{{QUANTITY}} / {{UOM}}	500 EA	Demand, count or document
{{DATE}}	2026-10-31	Required/planned/posting date
{{DOCUMENT_NUMBER}}	4500001234	Save confirmation/document flow

Golden rule	Reason
Use search/value help; do not guess an ID.	Codes can be valid only for a plant, company or period.
Keep leading zeroes and check uppercase/lowercase.	External IDs and integrations can treat them differently.
Check base, order and delivery UOM.	Ten cases can mean 120 each.
Check posting date, document date and timezone.	They drive accounting period and service metrics.
Save a personal variant after a correct search.	It preserves filters and visible columns.
Reverse wrong postings through approved process.	An unrelated journal entry hides the operational error.

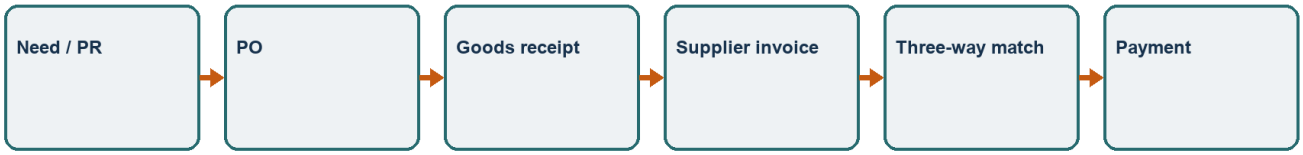
Organization and master data

Organization

Company code → Plant → Storage location → EWM warehouse → Storage bin

Master object	It answers	Minimum checks
Business Partner	Who buys, sells, pays or receives?	Legal identity, role, tax, currency, payment term
Product/material	What is bought, made, stored or sold?	Base UOM, plant, MRP, purchasing, sales, valuation
Source/info record	Who may supply it and on what terms?	Supplier, product, price, lead time, order UOM, validity
BOM	What components make it?	Header, component, quantity, UOM, validity
Routing/recipe	How is it made?	Operations, sequence, work center, standard time
Work center/resource	Where and with what capacity?	Calendar, capacity, activity, cost center
Batch/serial	Which exact lot or unit?	Identity, manufacture/expiry and status

SAP procure-to-pay document flow



Read left to right. Each step creates evidence for the next step; payment is safe only when the approved commitment, receipt and supplier invoice reconcile.

Procure to pay: exact business flow

1. Confirm the need is valid and not already covered by usable stock or open supply.
2. Create or approve the purchase requisition; select an approved source or sourcing process.
3. Create the PO. Verify product/service, quantity, UOM, plant, location, date, price, currency, tax, Incoterm and payment terms.
4. Send the approved PO; capture confirmation or ASN when used.
5. At receipt compare physical item, quantity, condition, lot/serial and documents. Post the real receipt date and location.
6. Match the supplier invoice to PO and receipt. Resolve the exact price, quantity, tax, freight or duplicate difference.
7. Pay through the authorized payment process; review old open PO and GR/IR items.

Event	Debit	Credit	Plain meaning
Purchase order	Usually none	Usually none	Commitment; exact setup can add controlling commitments
Goods receipt for stock	Inventory	GR/IR	Stock and a temporary receipt liability appear
Supplier invoice	GR/IR and applicable tax/variance	Accounts Payable	Supplier liability replaces clearing
Payment	Accounts Payable	Bank	Supplier liability and cash decrease

MRP in plain language

MRP compares dated demand with usable stock and confirmed supply. A remaining shortage becomes a proposal: commonly a purchase requisition for external supply or a planned order for internal production.

Worked example

```
Demand due Friday            1,000 EA
Safety stock                  +100 EA
Usable stock                  -250 EA
Confirmed purchase order      -300 EA
Net shortage                   550 EA
If lot size is multiples of 100, proposal may be 600 EA.
```

If MRP looks wrong	Check in this order
No proposal	Demand → product/plant extension → MRP type → procurement type → usable stock/open receipts → firming/time fence → lot size/dates → source/BOM → run scope/error log
Too much	Duplicate demand → wrong UOM → MOQ/rounding → safety stock → lot size → duplicate receipt not recognized
Too late	Requirement date → planned delivery → goods receipt/production time → calendar → source capacity → release dates

Make, store and sell

Manufacturing

Demand → MRP → Planned order → Production order → Release → GI → Confirm → GR → Settle

EWM inbound

PO → Inbound delivery → Arrival → GR → Warehouse task → Putaway → Confirm

EWM outbound

Sales order → Outbound delivery → Wave/task → Pick → Pack → Stage/load → PGI

Order to cash

Sales order → ATP/credit → Delivery → Pick/pack → PGI → Billing → AR → Payment

Common business purpose	Fiori/app search words	Classic SAP GUI example when enabled
Display product	Display Product Master	MM03
Maintain partner	Maintain Business Partner	BP
Create purchase order	Manage/Create Purchase Orders	ME21N
Stock/requirements	Manage Material Coverage	MD04
Post goods movement	Post Goods Movement	MIGO

Common business purpose	Fiori/app search words	Classic SAP GUI example when enabled
Supplier invoice	Create Supplier Invoice	MIRO
Production order	Manage/Create Production Orders	CO01
Sales order	Manage/Create Sales Orders	VA01
Outbound delivery	Manage Outbound Deliveries	VL02N
Billing	Create Billing Documents	VF01

If an app or code is missing, check role authorization, SAP edition, activated business scope and catalog assignment before concluding that the process is unavailable.

SEE ALSO [MIG-01] SAP migration, p.91 • [FIN-01] accounting, p.53 • [CH07] procurement, p.20 • [CH17] MRP, p.34

[SAP-MM] SAP MM — materials, purchasing and inventory field guide

SAP Materials Management (MM) controls what is bought, from whom, for which organization, at what quantity and value, and how the receipt reaches inventory or an expense. In SAP S/4HANA, many screens are Fiori apps; classic SAP GUI transaction codes remain common in private-cloud, on-premise and ECC estates. Use the business purpose first. Treat every code below as a locator, not a promise that your role or edition includes it.

Beginner run card

Question	Answer before you run
Goal	Trace one purchase-order line from commitment through receipt, invoice and payment evidence.
Need	A known PO number, company code, plant, supplier, product/service, quantity, UOM and cutoff date.
Where	Fiori search: Display Purchase Order; classic example when enabled: ME23N.
Sample	PO 4500001234, item 10, supplier 1000001, product A100, plant 1710, 100 EA at USD 8.
Replace	Only IDs selected from value help or copied from an approved document.
Run	Open item 10 → Delivery/Schedule → Invoice → Account Assignment → Purchase Order History.
Expected	Ordered, received, invoiced and returned quantities plus linked material/accounting documents reconcile.
Why	PO History is the evidence chain for that item; header totals alone can hide line differences.
If wrong	Check item number, UOM, reversal documents, final-delivery/final-invoice flags, service entry and cutoff.

Organization: the codes that decide where a transaction belongs

Object	Child-simple meaning	Controls or receives
Client	Top system data boundary	Shared master/configuration scope; never assume data crosses clients
Company code	Legal accounting company	G/L, AP, AR, tax, currency, posting period and financial statements
Plant	Operating site	Procurement, planning, production, inventory and delivery relevance
Storage location	Inventory subarea within a plant	Inventory Management quantity and stock status
Purchasing organization	Commercial buying unit	Supplier terms, contracts and purchasing responsibility
Purchasing group	Buyer or buyer team	Work ownership and reporting; usually not a legal organization

Object	Child-simple meaning	Controls or receives
Valuation area	Level at which material value is managed	Often plant in S/4HANA designs; verify configuration
Controlling area	Management-accounting boundary	Cost centers, internal orders, profit centers and allocations

Fast check: the same product number can behave differently by plant because MRP, purchasing, storage, accounting and warehouse views are organization-specific.

Product/material master: field-by-field effect

Field	Filled example	What changes if it is wrong
Product/material type	Trading good	Allowed views, number range, quantity/value update and field rules
Base UOM	EA	All alternate units convert to this; changing after transactions is high risk
Order UOM + conversion	1 CS = 12 EA	PO quantity and price conversion; wrong conversion multiplies stock/value
Plant / storage location	1710 / RM01	Where requirements, receipts and available stock appear
Purchasing group	N01	Buyer ownership and worklist routing
Planned delivery time	14 calendar days	MRP proposal and expected receipt dates
GR processing time	2 workdays	Time from physical arrival until stock is available
MRP type	PD	Whether and how the product is planned
Lot-sizing / rounding	EX / multiple 12	Proposal quantity; may create apparent excess
Procurement type	F external	Purchase requisition versus planned order route
Safety stock	50 EA	Shortage calculation and inventory target
Batch / shelf-life control	Batch required; 365 days	Traceability, expiry check and stock selection
Valuation class	Approved trading-goods class	Which inventory/consumption accounts automatic determination selects
Price control	S standard or V moving average	How receipt/invoice differences affect inventory or variance
Standard or moving price	USD 8.00 / EA	Inventory value and COGS basis under the configured method

Supplier Business Partner: one identity, several organizational views

Level	Fields to prove	Typical failure when missing/wrong
General	Legal name, address, tax IDs, language, bank change control	Duplicate identity, bad remittance or compliance risk
Company code	Reconciliation account, payment terms/methods, withholding tax, dunning	Invoice cannot post or pays on wrong terms
Purchasing organization	Order currency, Incoterms, partner functions, tolerances, schema group	PO defaults or pricing/communication fail
Plant/source	Source list, quota, info record, lead time and order UOM	MRP finds no source or selects the wrong one

Purchasing object	What it means	When it wins
Purchase requisition (PR)	Internal request; not yet the supplier commitment	Demand is approved for sourcing or ordering
Request for quotation	Collect comparable supplier offers	Competition or new price/terms are required
Purchasing info record	Supplier-product purchasing relationship	Defaults price, lead time, order unit and history
Source list	Allowed/fixed sources for product-plant and validity	MRP/source determination must know eligible suppliers

Level	Fields to prove	Typical failure when missing/wrong
Quota arrangement	Splits requirements across approved sources	Business wants a governed source share
Contract	Value or quantity agreement over a validity period	PO releases consume negotiated terms
Scheduling agreement	Long-term agreement with schedule lines	Repeated deliveries need dated call-offs
Purchase order (PO)	Authorized external commitment	Supplier can fulfill the exact approved terms

Create or review a PO without guessing

Area	Enter or verify	Filled example / test
Header	Supplier, company code, purchasing org/group, currency, document date	1000001; US01; P100; N01; USD; 2026-10-21
Item identity	Product/service, short text, item category	A100; Water Bottle 1 L; standard item
Quantity	Order quantity, order UOM and conversion to base UOM	100 EA; conversion 1 EA = 1 EA
Delivery	Plant, storage location, delivery date, tolerance, reminder	1710; RM01; 2026-11-04; approved tolerance
Commercial	Net price, price unit, currency, tax, Incoterms, freight condition	USD 8 per 1 EA; tax code per policy
Account assignment	Category, G/L and cost object when consumption	Blank for valuated stock; K + cost center for expense
Invoice	GR-based invoice, invoice receipt, final invoice and tolerance	Match invoice to actual item receipts
Evidence	Approval/release, messages/output, confirmation and attachments	Approved workflow plus supplier confirmation

Do not enter an expense account on a valuated-stock PO merely to force posting. The material's valuation class and configured automatic account determination normally drive the inventory account.

Goods movements: common standard meanings

Common movement type	Business meaning	Check before posting / reversal
101 / 102	PO or order receipt / reversal	PO/order, quantity, UOM, plant, location, batch, posting date
122 / 123	Return to supplier / reversal	Reference receipt, reason, physical shipment and supplier agreement
201 / 202	Issue to cost center / reversal	Cost center, reason, available stock and approval
261 / 262	Issue to production order / reversal	Order, component, batch, backflush and actual quantity
311	Storage-location transfer in one plant	Source/destination, stock type and physical move
551 / 552	Scrap / reversal	Disposition, approval, reason and financial impact
601 / 602	Goods issue for outbound delivery / reversal	Delivery status, picked quantity, batch and billing impact

These are common SAP standard defaults. Your system can restrict, replace or configure their behavior. Preview/check first and use the movement generated by the approved business process.

Worked P2P posting — 100 units at standard price

Document and accounting chain

```
ASSUMPTIONS
Product A100; 100 EA; standard price USD 8.00; PO price USD 8.00; no tax.

PO saved                        No FI posting in this simple case
101 goods receipt USD 800       Dr Inventory 800 | Cr GR/IR 800
Supplier invoice USD 800        Dr GR/IR 800     | Cr Accounts Payable 800
Payment USD 800                 Dr Accounts Payable 800 | Cr Bank 800
```

```
IF invoice is USD 820 and standard-price rules apply:
Dr GR/IR 800 + Dr price difference 20 | Cr Accounts Payable 820
Actual accounts and capitalization depend on valuation, stock coverage and configuration.
```

Mismatch	Prove in this order	Normal correction path
Received less than PO	Physical count → delivery note → GR quantity/UOM → open PO	Post actual receipt; keep remainder open or formally close/cancel
Wrong receipt	Material document, stock location/batch and physical stock	Approved reversal with reference, then correct receipt
Invoice quantity blocked	PO history, receipts/reversals, returns and invoice quantity	Correct source document or obtain approved tolerance/release
Invoice price blocked	PO conditions, price unit, currency, tax, freight and invoice	Correct PO/invoice or approve legitimate variance
GR/IR old balance	Open receipt/invoice pairs by PO item and cutoff	Post missing document, reverse wrong one, close residual under policy
No account determined	Valuation class, transaction key, valuation area and account modifier	Master/configuration owner corrects rule; do not guess a G/L

MM app and transaction quick finder

Business purpose	Fiori search words	Classic example when enabled
Display/change product	Display/Manage Product Master	MM03 / MM02
Create/display requisition	Manage Purchase Requisitions	ME51N / ME53N
Create/display PO	Manage/Create/Display Purchase Orders	ME21N / ME23N
PO worklist	Monitor Purchase Order Items	ME2N
Post goods movement	Post Goods Movement or Post Goods Receipt	MIGO
Stock by product/site	Stock — Multiple Materials / Stock Overview	MMBE / MB52
Create supplier invoice	Create Supplier Invoice	MIRO
Review blocked invoice	Release Blocked Invoices	MRBR
Stock/requirements	Manage Material Coverage	MD04

SEE ALSO [CH07] procure to pay, p.20 • [CH17] MRP, p.34 • [SAP-FICO] accounting, p.47 • [SAP-ARIBA] source to contract, p.49

[SAP-EWM] SAP EWM — receiving, storage, picking and shipping field guide

Extended Warehouse Management (EWM) controls warehouse work at a finer level than Inventory Management: doors, staging areas, handling units, storage bins, stock types, warehouse tasks, resources, queues, waves and labor-relevant execution. EWM may be embedded in S/4HANA or decentralized. Never assume ERP stock and EWM stock are the same object merely because totals match.

Beginner run card

Question	Answer before you run
Goal	Find why an inbound or outbound delivery is not moving.
Need	Delivery or handling-unit ID, warehouse number, product, quantity, status and observed time.
Where	Warehouse Monitor; classic example when enabled: /SCWM/MON.
Sample	Inbound delivery 180000123, warehouse W171, product A100, 100 EA.
Replace	Only IDs copied from ERP/EWM document flow, scanner or label.
Run	Search delivery → open status/warehouse request → tasks/orders → stock/HU → application log.

Question	Answer before you run
Expected	The first incomplete object, exact status/message, queue/resource and next owner are visible.
Why	Later statuses cannot complete until the first required object or confirmation succeeds.
If wrong	Check warehouse number, document category, time range, leading zeroes and ERP/EWM ID mapping.

Warehouse structure and execution objects

EWM object	Plain meaning	Example / control
Warehouse number	Top warehouse execution unit	W171; links structure, resources and processes
Storage type	Area with common storage behavior	High rack, staging, work center, bulk, pick face
Storage section	Subdivision used by placement/search	Fast mover, heavy, hazardous or temperature zone
Storage bin	Exact physical address	A-03-02-B; barcode must match physical sign
Activity area	Bins grouped for an activity	Picking or counting route
Work center	Place for packing, deconsolidation or value-add	PACK-01 with valid layout/process
Door / staging area	Where vehicle and goods meet	Door D05; staging group OUT-1
Warehouse product	Product settings for this warehouse	Putaway, removal, replenishment and slotting data
Handling unit (HU)	Identified pallet/carton/tote plus contents	HU 8000123456; label and status controlled
Warehouse request	Demand for warehouse work	Created from an inbound/outbound delivery
Warehouse task (WT)	One source-to-destination action	Move 12 EA from A-01-01 to PICK-01
Warehouse order (WO)	Group of tasks assigned for execution	Tasks grouped by rules for a worker/resource
Queue / resource	Work stream and person/equipment	FORKLIFT-IN / FL-07

Inbound: exact evidence chain

Step	Operator or system action	Evidence / next status
1 Plan	Create/receive inbound delivery; verify supplier, product, quantity, date and warehouse	Inbound document accepted with ERP/EWM IDs
2 Arrive	Record vehicle/transport arrival and door when used	Arrival status and timestamp
3 Unload	Unload and identify every HU/package; record exception	Physical count, label, damage and discrepancy
4 Receive	Post goods receipt only for actual accepted quantity/status	GR document and stock appears in receiving/interim area
5 Follow-up	Deconsolidate, inspect, label, batch/serial capture as designed	Stock type and follow-on task are valid
6 Put away	Create/confirm WT from source to determined bin	Destination bin/HU and confirmed quantity
7 Close	Reconcile delivery, GR, open tasks, HUs and final bin	No unexplained open quantity or orphan HU

Outbound: exact evidence chain

Step	Action	Proof before the next step
1 Delivery	ERP creates outbound delivery; EWM receives warehouse request	Product, quantity, batch rules, route and dates agree
2 Plan	Release wave or create WTs using process/stock-removal rules	Eligible stock and source bins are determined

Step	Action	Proof before the next step
3 Pick	Resource confirms product, HU/bin, quantity and exception	Picked quantity and difference reason are recorded
4 Pack	Pack into outbound HU; weight, labels and documents	HU contents equal picked/delivery quantity
5 Stage/load	Move to correct staging area/door and load vehicle	Door, route, HU and loading status agree
6 PGI	Post goods issue after approved physical dispatch	Stock decreases; ERP material/FI document and delivery status update
7 Close	Reconcile delivery, HUs, tasks, PGI and carrier handoff	No open task, quantity or interface message

Special process	Trigger	Control
Replenishment	Pick-face falls below minimum or planned need	Source stock, destination capacity, open WTs and duplicate trigger
Physical inventory	Scheduled/cycle/event count	Freeze/count rules, blind count, recount, difference approval and posting
Scrapping	Approved unfit stock	Stock type, disposition, reason, quantity, value and ERP posting
Quality inspection	Product/source/process rule	Inspection stock is not allocatable until usage decision/release
Cross-docking	Inbound demand can flow directly to outbound	Eligibility, quantity, timing and fallback destination
Value-added service	Kitting, labeling or packaging step	Work instruction, components, confirmation and traceability

EWM failure tree — locate the first broken object

Symptom	Look first	Likely route
No EWM delivery	ERP output/qRFC/integration log, warehouse determination, mapping	Correct integration or master mapping; do not recreate blindly
No warehouse task	Warehouse request status, process type, stock/bin determination, hold	Fix determination/master/status, then create/reprocess once
Task cannot confirm	Source stock/HU, actual bin, resource/queue, lock and exception code	Resolve physical/system fact; confirm actual move only
No destination bin	Putaway control, storage type search, capacity, section/bin type, product data	Correct eligible master/capacity or approved exception bin
Cannot pick	Stock type, batch/serial, entitlement/owner, HU, allocation and open task	Release/choose valid stock or correct upstream rule
HU not found	Last confirmed WT, HU status/nesting, label, open task and physical scan	Control search; prevent duplicate HU/label
PGI fails	Open pick/pack/load tasks, quantity, delivery block, ERP queue and posting period	Complete/reverse correct object and reprocess interface
ERP and EWM differ	Cutoff, queues, open documents, posting/reversal pair, stock type/UOM	Freeze affected flow; reconcile object-by-object before correction

EWM correction rule: correct the first wrong business object. Recreating a delivery, HU or task without closing the original often creates duplicates and makes stock reconciliation harder.

Purpose	Common place
Monitor deliveries, tasks, stock, resources and queues	Warehouse Monitor; /SCWM/MON when enabled
Execute scanner work	RF menu; /SCWM/RFUI when enabled
Inspect inbound delivery	Manage/Process Inbound Deliveries or /SCWM/PRDI when enabled
Inspect outbound delivery	Manage/Process Outbound Deliveries or /SCWM/PRDO when enabled
Check integration failure	Application log, qRFC monitor and interface monitoring under support ownership

SEE ALSO [WH-01] warehouse placement, p.32 • [CH12] receiving, p.28 • [CH13] accuracy, p.28 • [CH14] fulfillment, p.29

[SAP-FICO] SAP FI/CO — financial integration and control field guide

Financial Accounting (FI) produces legal books: general ledger, accounts payable, accounts receivable, assets and bank. Controlling (CO) explains internal cost and profit: cost centers, internal orders, production orders, projects, profit centers and profitability. Supply-chain postings usually create FI and CO evidence automatically; fix the operational source instead of hiding it with an unrelated journal entry.

Area	Question answered	Common master/object
FI General Ledger	Which account, company, period and currency changed?	G/L account, company code, ledger, document
Accounts Payable	What do we owe each supplier and when?	Supplier BP, reconciliation account, payment term
Accounts Receivable	What do customers owe and what is overdue?	Customer BP, invoice, payment, credit
Asset Accounting	What long-lived asset was acquired/depreciated?	Asset class, asset number, useful life
Cost Center Accounting	Which function consumed the cost?	Cost center, activity type, responsible manager
Internal Order / WBS	Which temporary initiative or project consumed it?	Internal order or project/WBS element
Product Cost Controlling	What should/does a product or order cost?	Cost estimate, production/process order, variance
Profit Center / Margin	Which responsibility unit/product/customer earned profit?	Profit center and profitability characteristics

The six fields to inspect on every failed posting

Field	Filled example	Why it matters
Company code	US01	Legal books, currency, tax, periods and reconciliation accounts
Posting date	2026-10-21	Fiscal period and exchange-rate date under policy
Document type	Supplier invoice / material document	Number range, control and reversal behavior
G/L account	Inventory, GR/IR, expense or revenue	Financial classification; often determined automatically
Amount/currency	800.00 USD	Document, company-code and group currencies may all exist
Cost object	Cost center CC110 or production order	Required for many P&L accounts and CO reporting

If the cost is for...	Use only when policy/configuration says	Avoid
A department's recurring work	Cost center	Using a supplier or employee as the cost object
A temporary campaign/repair/event	Internal order	Permanent accumulation with no settlement/closure
A project deliverable	WBS element	Posting outside the approved project structure
Making a product	Production/process order or product cost collector	Manual COGS entry disconnected from material flow
Stock for future sale/use	Inventory account via material valuation	Direct expense when the item should remain an asset

SCM event → FI/CO effect

Business event	Typical debit	Typical credit	Trace key
Stock PO goods receipt	Inventory	GR/IR	PO item + material document + accounting document
Consumption receipt	Expense/cost object	GR/IR	PO account assignment + service/GR evidence

Business event	Typical debit	Typical credit	Trace key
Supplier invoice	GR/IR, tax, variance/expense	Accounts Payable	Supplier invoice + PO history + FI document
Supplier payment	Accounts Payable	Bank clearing/bank	Payment document + bank statement
Issue to production	Production order/WIP	Inventory	Material document + production order
Production receipt	Finished-goods inventory	Production order/WIP	Order confirmation/GR + cost object
Scrap/adjustment	Approved loss/variance	Inventory	Count/disposition + material/FI document
Post goods issue for sale	COGS	Inventory	Outbound delivery + material/FI document
Customer billing	Accounts Receivable	Revenue and tax	Billing document + FI document
Customer receipt	Bank clearing/bank	Accounts Receivable	Payment allocation + bank statement

The words ‘typical’ matter. Account determination, ledgers, taxes, Material Ledger, actual costing, valuation and local accounting rules can add or change lines.

Automatic account determination — what the configuration team checks

Common key	Common purpose	Business data that may select the G/L
BSX	Inventory posting	Valuation area, valuation class and account grouping
WRX	GR/IR clearing	Chart of accounts and configured transaction key
GBB	Offsetting entry for inventory postings	Movement/account modifier, valuation class and transaction
PRD	Price differences	Valuation grouping/class and price-difference configuration

These keys explain diagnosis; they are not instructions to change production configuration. When ‘account determination’ fails, capture company/plant, product, movement, valuation class, transaction and exact message for the configuration owner.

GR/IR and invoice-block cleanup

1. Choose one cutoff and one PO item. List every receipt, reversal, return, invoice, credit memo and currency/UOM conversion.
2. Reconcile ordered quantity, received quantity, accepted service quantity, invoiced quantity and open quantity in the same unit.
3. Reconcile PO value, GR/IR value, invoice value, tax, freight and price/FX variance.
4. Classify the difference: timing, missing document, wrong quantity/UOM, wrong price, duplicate, return, cancellation or rounding.
5. Correct the originating document with reference when possible. Release a block only after the variance is legitimate and approved.
6. For aged residuals, use the approved GR/IR maintenance/clearing process with evidence; never clear merely to make aging disappear.

Close area	Required reconciliation	Proof
Inventory	Subledger quantity/value = G/L plus explained in-transit/timing	Stock valuation, material documents, G/L balance
GR/IR	Receipts and invoices paired; aged items owned	PO history and GR/IR aging
AP	Supplier open items, duplicate/blocked invoices and payment proposals reviewed	Supplier line items and payment log
AR	Billing complete; receipts applied; disputed/old items owned	Customer line items and aging
Production	Confirmations, failed movements, WIP, variance and settlement complete	Order status/cost and settlement log
Freight/accrual	Service received but unbilled cost estimated and reversed correctly	Carrier evidence, accrual calculation and approval

Close area	Required reconciliation	Proof
Period	Subledgers closed in sequence; late postings governed	Close checklist, locks and sign-off

FI/CO app and transaction quick finder

Purpose	Fiori search words	Classic example when enabled
Display accounting document	Display Journal Entries / Manage Journal Entries	FB03
Supplier open/cleared items	Manage Supplier Line Items	FBL1N
Customer open/cleared items	Manage Customer Line Items	FBL5N
G/L line items	Display Line Items in General Ledger	FAGLL03 / FAGLL03H
Cost center actuals	Cost Centers — Actuals / Plan-Actual	KSB1
Production order cost	Production Cost Analysis / Analyze Costs by Work Center	CO03 and cost reports
Blocked supplier invoices	Release Blocked Invoices	MRBR
Payment run	Manage Automatic Payments	F110

SEE ALSO [FIN-01] finance basics, p.53 • [SAP-MM] purchasing postings, p.41 • [CH21] cost and cash, p.37 • [MIG-01] balances, p.91

[SAP-ARIBA] SAP Ariba — sourcing, contracts, buying and supplier network field guide

SAP Ariba solutions manage supplier discovery/onboarding, sourcing, contracts, guided buying, purchase collaboration, invoicing and spend visibility. SAP Business Network connects trading partners. Exact product names, forms, realm configuration and integration vary; follow the document status and integration ID rather than assuming the screen name.

Current SAP integration name: SAP Integration Suite, managed gateway for spend management and SAP Business Network. Older project documents may call it Cloud Integration Gateway or CIG. Confirm which integration project, test/production connection, document type and message ID handled the transaction before reprocessing it.

Capability	What the business does	Main output
Supplier Lifecycle and Performance	Request, qualify, segment, evaluate and monitor suppliers	Approved supplier identity, qualification and risk/performance evidence
Sourcing	Run RFI/RFP/auction and compare bids	Award decision with bids, scoring, approvals and savings basis
Contracts	Author, negotiate, approve and manage obligations	Signed contract workspace, terms, dates, owners and linked spend
Buying / Guided Buying	Request catalog, noncatalog, service or limit items	Approved requisition and purchase order route
Invoicing	Receive, validate, reconcile and approve supplier invoices	Reconciled invoice or exception with owner
SAP Business Network	Exchange PO, confirmation, ship notice, service entry and invoice	Shared transaction status and supplier response
Spend analysis	Classify and analyze spend	Category, supplier, contract and compliance insight

Sourcing event: from business need to defensible award

Step	Do	Evidence
1 Intake	Define scope, volume, sites, service, term, incumbent baseline, constraints and decision owner	Approved sourcing brief
2 Supplier set	Prequalify capability, compliance, financial/operational risk and conflicts	Invite list and exclusion reasons
3 Event design	Build questions, line items, currency/UOM, price terms, lots, scoring and timetable	Published event version

Step	Do	Evidence
4 Bid period	Issue event, answer all material questions consistently and control amendments	Message/Q&A and amendment log
5 Normalize	Convert currency, UOM, Incoterm, freight, duties, rebates, payment terms and risk	Comparable total-cost table
6 Evaluate	Apply mandatory gates and approved weighted scoring; test scenario/award split	Reproducible score and scenario
7 Negotiate	Use BATNA, give/get log and approval limits	Conditional concessions and final offer
8 Award	Approve decision, notify suppliers and preserve audit trail	Award, approvals and debrief
9 Contract/implement	Create contract, operational terms/catalog and transition plan	Signed version and enabled buying channel

Line-item field	Filled example	Why it is required
Item / specification	A100 Water Bottle 1 L, approved drawing rev C	Prevents suppliers pricing different scope
Quantity / UOM	120,000 EA annual estimate	Normalizes bid and capacity
Ship-to / Incoterm	Chicago DC; DDP named place	Defines logistics cost/risk assumption
Price components	Unit, tooling, freight, duty, surcharge, rebate	Separates recurring and one-time cost
Currency / validity	USD; 2027-01-01 to 2027-12-31	Makes timing and conversion explicit
Lead/service/quality	14 days; 98% OTIF; ≤500 PPM	Makes non-price performance measurable
Capacity / contingency	12,000 EA/month; alternate line tested	Proves supply feasibility and resilience

Guided buying to payment

Ariba + SAP

Need → Catalog/noncatalog request → Accounting + approval → PO → Supplier confirm/ASN → Receipt/service entry → Invoice reconciliation → SAP FI payment

Document / handoff	Minimum fields	Control
Requisition	Requester, entity/site, supplier, item, quantity/UOM, date, accounting, contract	Budget/policy, preferred channel and approver
Purchase order	Buyer/supplier IDs, line, price/currency, ship-to, tax, Incoterm, terms	ERP PO ID returned and supplier delivery confirmed
Order confirmation	Accepted/rejected quantity, price and committed date	Difference routed before shipment
Ship notice	PO/line, package/HU, quantity, lot/serial, carrier and ETA	Receiving can identify and plan arrival
Receipt/service entry	Accepted quantity or service evidence, date, receiver and exception	Only actual accepted value becomes match evidence
Invoice	Supplier invoice/reference, PO/line, quantity, tax, freight, currency and remittance	Duplicate check and PO/receipt/contract validation
Reconciliation exception	Variance type, amount, owner, comment and resolution	Approval follows authority; correction remains linked

Contract workspace versus operational buying object

Object	Purpose	Do not confuse it with
Contract workspace	Govern documents, authors, approvals, negotiation, signature, dates and obligations	An ERP outline agreement that PO creation consumes
Contract terms / pricing content	Structured items, rates, limits and validity	The signed attachment if structured fields disagree
Catalog	Controlled shopping content and price	A supplier's unrestricted marketing catalog
SAP MM contract/scheduling agreement	Operational source for releases/call-offs	Ariba workspace unless integration explicitly links them
Purchase order	One authorized commitment under terms	The master agreement itself

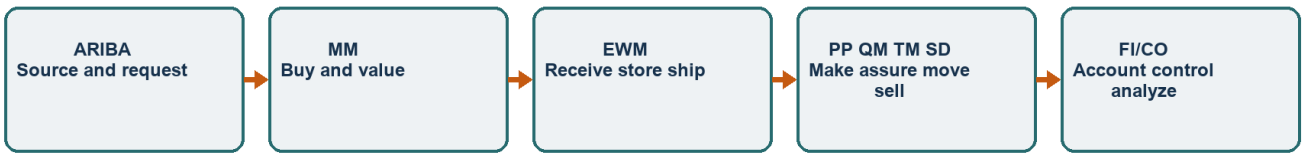
Integration map and first-failure checks

Ariba / Network	SAP S/4HANA target	Key IDs to retain
Supplier request/registration	Business Partner and supplier roles	Request ID, supplier/network ID, BP/supplier ID
Catalog/requisition	PR or PO with account assignment	Requisition ID, ERP PR/PO number and line
Sourcing award/contract	Info record, source list, contract or scheduling agreement	Event, award, workspace and ERP outline-agreement IDs
PO / change order	MM purchase order/version	Network order ID, ERP PO, line and version
Ship notice	Inbound delivery/ASN	PO line, ASN, delivery and package/HU IDs
Receipt/service sheet	Goods receipt or service entry	Receipt/SES and material/accounting document IDs
Invoice	Supplier invoice and FI document	Supplier invoice, reconciliation and ERP invoice/FI IDs
Payment status	Cleared supplier item/payment	Invoice, payment and remittance references
Symptom	**First check**	**Do next**
Requisition approval stuck	Approval history, active user/delegate, threshold, accounting and budget	Assign correct owner or correct source field; preserve audit history
PO not at supplier	ERP PO/output status, version, network routing and supplier relationship	Repair/reprocess one message; prevent duplicate PO
Supplier cannot invoice	PO status, invoicing rule, receipt/service requirement, supplier account	Complete prerequisite or correct rule; do not bypass three-way match
Invoice rejected/held	Duplicate key, tax, PO line, quantity, price, receipt and tolerance	Correct supplier/ERP source or route approved exception
Contract price not used	Validity, commodity/item, supplier, purchasing org/site and catalog/source linkage	Correct operational linkage/version, then reprice under control
Supplier duplicated	Legal/tax ID, network account, BP mapping and registration request	Stop creation; merge/remediate through master-data governance
Status differs across systems	Last successful integration message and IDs on both sides	Reprocess from failed handoff once; reconcile before manual recreation

SEE ALSO [NEG-01] negotiation, p.23 • [CH08] sourcing, p.21 • [CH10] contracts, p.22 • [SAP-MM] operational buying, p.41

[SAP-X] SAP cross-module integration, implementation and practice lab

How the SAP supply-chain and finance pieces connect



Follow the business object from left to right. A complete process keeps the same product, partner, organization, quantity, unit, value, date and reference IDs through every handoff.

Adjacent area	It owns	Critical integration
PP / MRP	Requirements, planned orders, production orders and capacity	MM sources components; EWM stages/receives; CO collects order cost

Adjacent area	It owns	Critical integration
SD	Sales order, availability, delivery, billing and customer flow	EWM executes delivery; FI posts AR/revenue; MM/PP provide supply
QM	Inspection lot, results, defects and usage decision	MM/EWM stock status and supplier/production quality
TM	Freight planning, tender, execution, settlement and charges	SD/MM deliveries, EWM loading and FI freight settlement/accrual
IBP / planning	Demand, supply, inventory and response plans	Approved plan feeds S/4 execution; actuals return for replan
MDG / BP	Governed master creation/change and duplicate prevention	Replicates trusted partner, product and finance master data

One complete purchase-to-sale scenario

Step	Document / module	Quantity/value and proof
1 Source	Ariba sourcing event and contract	A100 awarded to supplier 1000001 at USD 8/EA; approved award
2 Request	Ariba requisition → SAP MM PR/PO	100 EA for plant 1710; PO 4500001234; workflow approved
3 Confirm	Business Network confirmation/ASN	100 EA due 2026-11-04; ASN/package IDs returned
4 Receive	EWM inbound + MM 101 GR	100 EA accepted and put away; material/FI documents saved
5 Invoice	Ariba invoice → SAP invoice/FI	USD 800 matched; AP liability created
6 Pay	SAP FI payment	AP cleared against bank/remittance reference
7 Sell	SAP SD order/delivery	20 EA at USD 15; ATP and delivery created
8 Ship	EWM pick/pack/load + PGI	20 EA leaves; inventory credited and COGS debited
9 Bill/collect	SD billing + FI AR/payment	AR USD 300, revenue USD 300, then customer receipt clears AR
10 Reconcile	MM/EWM/FI/CO analytics	QOH 80 EA; operational quantity/value and ledgers tie at cutoff

Cross-module diagnostic key

If the wrong thing is...	Start in	Trace next
Supplier, product, UOM or organization	BP/product master or MDG	Replicated views, validity and downstream document snapshot
Need, source, price or PO	Ariba/MM	Approval, source/contract, PO version and supplier confirmation
Physical location or warehouse status	EWM	Delivery, HU, stock type, WT/WO, queue and ERP interface
MRP date or quantity	PP/MRP	Demand, stock, receipts, master parameters, calendar and source
Quality release	QM	Inspection lot, results, defects, usage decision and stock posting
Shipment or freight	SD/TM/EWM	Delivery, freight order, loading, PGI, carrier event and settlement
Account, value or period	FI/CO	Originating logistics document, account determination and cost object

Greenfield implementation — evidence gates

Gate	Required decisions	Exit evidence
1 Scope/design	Entities, plants, warehouses, processes, editions, integrations, countries and outcomes	Signed scope, process inventory and design principles

Gate	Required decisions	Exit evidence
2 Fit-to-standard	Standard process, true gaps, controls, roles, data and reporting	Approved process flows, backlog and no-customize decisions
3 Configure/build	Organizations, document/movement types, tolerances, determination, workflows and interfaces	Configuration record, unit tests and transport/release trace
4 Data	Owners, field mapping, cleansing, deduplication, dependencies and retention	Mock load reconciles count, quantity and value
5 Test	Positive, negative, security, volume, cutover and recovery scenarios	Unit + SIT + UAT evidence; defects accepted or closed
6 Deploy	Sequence, freeze, extraction, load, interfaces, roles, opening balance and rollback	Go/no-go checklist and signed reconciliation
7 Stabilize	Command center, severity/ownership, monitoring, backlog and handover	KPIs stable; BAU accepts support and controls

Test case must say	Filled example
Precondition	A100 is extended to plant 1710 and warehouse W171; supplier is approved
Input	PO 100 EA at USD 8, delivery 2026-11-04, storage location RM01
Steps	Create/approve PO → ASN → partial GR 60 → final GR 40 → invoice 100 → pay
Expected documents	PO, two GR/material documents, one invoice/FI document and payment clearing
Expected balances	QOH +100; GR/IR zero after invoice; AP zero after payment
Negative test	Invoice 110 EA or wrong price exceeds tolerance and is blocked
Evidence	Screens/export, IDs, timestamps, tester, result and defect link

Practice without SAP access

1. Draw five columns labeled Ariba, MM, EWM, FI/CO and Evidence. Use the ten-step scenario above.
2. Create paper IDs: event EVT-01, PO 4500001234, inbound 180000123, HU 8000123456, GR 5000007777 and FI 1900000888.
3. Change the receipt to 95 EA. Mark exactly which documents, quantities and balances must remain open.
4. Change the invoice to 100 EA at USD 8.20. Calculate the USD 20 difference and decide whether it is legitimate, blocked or corrected under the stated tolerance.
5. Simulate a failed EWM task: write the last correct object, first failed status, physical evidence, containment, owner and safe replay step.
6. Finish by reconciling ordered, received, put away, invoiced, paid, shipped and remaining quantities. If one cannot be explained, the exercise is not complete.

This paper lab teaches document logic, field dependencies and accounting without a live subscription. A company system is still required to learn its specific configuration, roles, screens and approvals.

SEE ALSO [MIG-01] migration/cutover, p.91 • [CH27] greenfield, p.89 • [CH28] testing and go-live, p.90 • [QB-01] QuickBooks crosswalk, p.55

[FIN-01] Finance, accounting and working-capital control

Debit and credit are the two sides of one accounting entry; neither means good or bad. Assets and expenses usually increase with a debit. Liabilities, revenue and equity usually increase with a credit.

QuickBooks word	Plain meaning
Purchase order	Buying intention; normally no accounting entry yet.
Bill	The supplier has charged you; record what is owed and pay later.
Expense	Record a purchase paid now; use only when this matches the real event.
Bill payment	Pay the recorded bill and clear Accounts Payable.
Report	A read-only view used to inspect and reconcile before changing transactions.

Beginner run card

Question	Answer before you run
Goal	Safely inspect one inventory balance before changing any transaction.
Need	Company, report date, accounting basis and one known item.
Where	QuickBooks Online → Reports → search ‘Inventory Valuation Summary’.
Sample	Report date 2026-10-19; accrual basis; item A100.
Replace	Only {{REPORT_DATE}}, {{BASIS}} and {{ITEM}} with approved values.
Run	Choose Customize, set date/basis, run report, then open the item detail.
Expected	Quantity and asset value agree with the item transactions at the same date.
Why	The report summarizes dated item transactions under the selected basis.
If wrong	Check opening balance, inactive items, direct G/L entries, future dates, duplicates and negative stock.

Statement	Child-simple meaning	Question
Balance Sheet	A photograph on one date	What do we own, owe and have left?
Profit and Loss	A movie for a period	Did revenue exceed cost and expenses?
Cash Flow	A water-flow report	Why did cash rise or fall?

Core equations

```
Assets = Liabilities + Equity
Gross Profit = Revenue − Cost of Goods Sold
Net Profit = Revenue − COGS − Operating Expenses − Other Costs
Change in Cash = Operating + Investing + Financing Cash Flow
```

QuickBooks event	Debit	Credit	Control
Buy inventory on a bill	Inventory Asset	Accounts Payable	Use item, quantity, cost and supplier
Sell inventory	AR or Cash	Sales Revenue	Customer, item, quantity, price and tax
Recognize sold inventory	COGS	Inventory Asset	Driven by tracked item cost
Pay supplier	Accounts Payable	Bank	Match bill and bank reconciliation
Shortage adjustment	Shrinkage/COGS under policy	Inventory Asset	Count evidence and approval

QuickBooks purchase and inventory flow

Purchase order → Supplier ships → Receipt/bill records stock → Inventory and AP update → Pay bill → Reconcile bank

QuickBooks inventory costing is method-sensitive and current accounts may offer First In, First Out (FIFO) or Moving Average Cost (MAC); Desktop editions and older company files can differ. Example: buy 10 at $8 and 10 at $12, total $200. Sell 6. FIFO COGS is $48 and remaining value $152; MAC COGS is $60 and remaining value $140. Verify the company file's chosen method before explaining margin or changing dates.

Inventory does not match Balance Sheet	Check
Report basis/date	Use the same date and accrual basis.
Direct G/L postings	Find entries to Inventory Asset without an inventory item.
Inactive items	Include inactive items with quantity or value.
Future/duplicate transactions	Search future adjustments, duplicate bills and sales.
Negative stock	Later receipts can change cost calculations; inspect transaction order.

Inventory does not match Balance Sheet	Check
Opening balance	Confirm opening quantity, cost and date.

Working-capital worked example

Example

```
Annual revenue          $120,000
Annual COGS              $72,000
Average inventory        $18,000 → turns 4.0; DIO 91 days
Average AR               $15,000 → DSO 46 days
Annual purchases         $60,000
Average AP               $12,000 → DPO 73 days
Cash conversion cycle = 91 + 46 − 73 = 64 days
```

QuickBooks idea	SAP S/4HANA idea	Do not assume a one-to-one match
Vendor	Business Partner with supplier role	SAP has company-code and purchasing-organization data
Customer	Business Partner with customer role	SAP adds sales-area and partner functions
Product/Service	Product/material	SAP adds plant, MRP, valuation, sales and warehouse views
Bill	Supplier invoice	SAP often matches PO, GR and invoice
Invoice/Sales receipt	Sales order/delivery/billing chain	Operational and accounting documents are separated
Location/Class	Plant, profit center, cost center, segment	Design the mapping from reporting purpose

Month-end SCM finance check	Evidence
Inventory	Goods movements complete; count adjustments approved; subledger = G/L
Procurement	GR/IR aging, invoice blocks, open commitments and accruals reviewed
Production	Confirmations, GR, failed movements, WIP, variance and settlement complete
Sales	Deliveries, PGI, billing and AR exceptions reviewed
Freight	Carrier accrual, invoice audit and claims reflected
Reserves	Excess, obsolete, expiry and quality exposure reviewed
Close	Reports saved, reconciliations signed and period closed by policy

SEE ALSO [CH21] cost and cash, p.37 • [SAP-01] SAP postings, p.38 • [MIG-01] financial migration, p.91

[QB-01] QuickBooks Online and Desktop — zero-to-expert field guide

You can learn this logic without buying software. The product is a user interface around customers, suppliers, items, transactions, accounts and reports. Menus and features vary by country, plan, company-file age and rollout. Use the in-product search when a menu moves, and prove the accounting result with reports rather than trusting the form alone.

QuickBooks operational and accounting evidence chain



Need	QuickBooks Online	QuickBooks Desktop / Enterprise
Access	Browser; continuous interface and feature rollout	Installed company file; release/edition and hosting matter

Need	QuickBooks Online	QuickBooks Desktop / Enterprise
Inventory	Available only in eligible plans/add-ons; current companies may use FIFO or MAC	Core editions commonly use average cost; Enterprise Advanced Inventory can add FIFO and advanced controls
Purchase orders	Eligible plans; PO itself is nonposting	Available by edition/configuration; PO itself is nonposting
Sales orders	Availability can depend on current edition/rollout	Common in Premier/Enterprise; not every Desktop edition
Assemblies/manufacturing	Current availability depends on plan/rollout; verify BOM/order functions	Inventory assemblies/BOM are strongest in manufacturing editions
Sites/bins/lots	Location/class tracking is not automatically warehouse-bin stock	Enterprise Advanced Inventory can support sites/bins/lot/serial features
Landed cost	Use current supported feature/app or controlled item-cost method	Enterprise Advanced Inventory can provide native landed-cost allocation
Audit/close	Audit log and close-books controls	Audit trail, closing date and file/user controls

Feature labels change. Before following a recipe, record: Online or Desktop, country, plan/edition, company name, fiscal year, inventory costing method and whether you are in a copy/test file.

2026 receiving-method checkpoint

Mode	What records physical receipt	Implementation warning
QBO bill-receiving method	Bill/Expense/Check with Item details updates inventory	PO remains nonposting; do not also create a receipt for the same quantity
QBO Item receipts when offered	Item Receipt records received inventory; linked Bill completes payable behavior	Current guidance says changing to this method can be permanent; test partials, returns, price differences and backdating
QBD Receive without bill	Item Receipt, followed later by Enter Bills against Inventory	Link bill to receipt so inventory is not counted twice
QBD Enterprise EIR	Receipt debits Inventory and credits Inventory Offset; bill clears offset to AP	Enhanced Inventory Receiving cannot simply be switched off; back up and test first

Before first live inventory: write the selected costing method and receiving method in the accounting policy, save the configuration evidence, and run a PO → partial receipt → bill → return → payment test.

Setup order: build the controls before transactions

Step	Decide and enter	Proof before continuing
1 Company	Legal name, tax identity, address, fiscal year, home currency and accounting basis	Settings agree with approved legal/accounting profile
2 Users	Named users, least-privilege roles, approvers and accountant access	No shared admin; payment/vendor changes segregated
3 Close policy	Month-end timetable, lock date and who may override	Prior period cannot change silently
4 Chart of accounts	Only accounts needed for reporting, tax and control	Account type/detail type and naming reviewed
5 Tracking dimensions	Class, location, project/customer and product category purpose	Each dimension answers one stable question
6 Lists	Payment terms, tax, customers, vendors, products/services and opening dates	Duplicates removed; owners and IDs defined
7 Forms/workflows	Estimate/order, PO, bill, invoice, payment, credit and approval path	One normal and one exception case tested
8 Bank/feeds	Accounts, opening balance, connection, rule governance and review owner	First statement reconciles to zero
9 Opening data	AR, AP, inventory, bank, fixed assets, loans, tax and equity	Trial balance and subledgers reconcile
10 Reports	Management pack, filters, basis, cutoff and distribution	Each report ties to source and decision

Chart of accounts for a product business

Account family	Example accounts	Use
Assets	Bank; Accounts Receivable; Inventory Asset; Prepaids; Fixed Assets	Resources owned or controlled
Liabilities	Accounts Payable; credit card; tax payable; accrued freight; loan	Amounts owed or obligations
Equity	Owner capital; retained earnings; opening balance clearing	Net owner interest; opening clearing must be resolved
Income	Product sales; service revenue; freight income; returns/discounts	Revenue and contra-revenue by useful category
COGS	Product COGS; freight-in if expensed by policy; production variance	Direct cost recognized with sales or approved policy
Operating expense	Warehouse labor; outbound freight; software; rent; professional fees	Period cost not carried in inventory
Other	Interest; FX; gain/loss on disposal	Nonoperating or separately presented activity
Dimension	**Use it for**	**Do not use it for**
Product/service	What was bought or sold; quantity, income and cost behavior	Department or legal entity
Category	Group related products/services	A second chart of accounts
Class	Department, product line or another management segment	Warehouse bin or customer
Location	Office/store/region or other configured transaction location	Assuming quantity by site unless inventory explicitly supports it
Customer/project	Customer, job and project profitability	Supplier or internal department
Account	Financial statement classification	Item-level SKU detail

Product and service master: every important field

Field	Filled example	What it controls
Type	Inventory	Quantity tracking and automatic Inventory Asset/COGS behavior
Name/SKU	A100 / WB-1L	Stable transaction identity; do not recycle an old code
Category	Drinkware	Product grouping and reporting
Sales description	Water Bottle 1 L	Default customer form text
Sales price/rate	USD 15.00	Default selling price; form may override under authority
Income account	Product Sales	Revenue account when sold
Purchase description	A100 bottle, approved spec	Default supplier form text
Cost	USD 8.00	Default purchase cost; not always the valuation method result
Expense/COGS account	Product COGS	Expense recognized under item behavior
Inventory asset account	Inventory Asset	Balance-sheet value for tracked item
Initial quantity/date	100 EA as of 2026-10-01	Opening layer/balance; blocks or complicates earlier transactions
Reorder point	25 EA	Low-stock warning; not a full MRP calculation
Preferred vendor	Northstar Supplier	Purchasing default/reporting where supported
Choose type	**When**	**Accounting implication**
Inventory	You own and sell stock and need quantity/value tracking	Purchase adds Inventory Asset; sale creates revenue plus COGS/inventory entry

Field	Filled example	What it controls
Non-inventory	Bought/sold item whose quantity is not tracked in QuickBooks	Posts to selected income/expense accounts; no QOH
Service	Time or service rather than stocked goods	Income/expense behavior; no inventory quantity
Bundle/group	Convenient set of items on a sales form	Usually components retain their own accounting; verify current behavior
Assembly	Manufactured item with a BOM in an eligible edition	Build consumes components and creates finished inventory

Vendor and customer master fields

Record	Minimum fields	Control
Vendor	Unique display ID/name; legal name; remit-to; order/AP contacts; terms; currency; tax setup; account reference	Bank/remit change verified through an independently known contact and second approval
Customer	Unique ID/name; legal name; bill-to; ship-to; terms; currency; tax status; credit limit; delivery/payment preference	Tax exemption evidence and credit approval remain current
Both	Stable external ID for WMS/commerce/SAP connection	Names are for people; IDs are for interfaces and must never be reused

Useful shortcut	Action
QBO Ctrl+Alt+? / Mac Cmd+Option+?	Open the current shortcut guide
Tab / Shift+Tab	Next / previous field
QBD Ctrl+I	Create invoice
QBD Ctrl+J	Customer Center
QBD Ctrl+A	Chart of Accounts
QBD F2 or Ctrl+1	Product and company-file information

Roles and controls that prevent expensive errors

Risk	Separate or review	Evidence
Fake/changed supplier	Vendor creation/bank change from bill approval/payment	Independent callback, change log and approval
Duplicate bill/payment	Bill entry from payment release	Vendor + invoice number + amount + date duplicate check
Unauthorized discount/credit	Sales entry from credit approval	Reason, authority and linked customer document
Inventory write-off	Counter from adjustment approver	Count sheet, reason, value and approval
Bank-feed error	Suggested category from reviewer acceptance	Source document and match rather than duplicate Add
Prior-period change	Bookkeeper from close-date override	Lock, password/approval and audit log review

SEE ALSO [FIN-01] accounting basics, p.53 • [QB-BUY] purchasing, p.58 • [QB-SELL] sales, p.61 • [QB-CLOSE] close, p.63

[QB-BUY] QuickBooks purchasing, bills, inventory and landed cost

Beginner run card

Question	Answer before you run
Goal	Buy 100 A100 units, recognize the supplier liability and prove inventory.
Need	Vendor, item, quantity/UOM, date, unit cost, tax/freight treatment, due date and approver.

Question	Answer before you run
Where	Online: + Create → Purchase order, then Bill/Expense or supported receipt workflow. Desktop: Vendors/Purchase Orders and Receive Items/Bills.
Sample	Vendor Northstar Supplier; A100; 100 EA at USD 8; terms Net 30.
Replace	Only company-approved vendor/item/account/tax values.
Run	Create PO → record actual receipt/bill → inspect item transaction and Inventory Valuation Detail.
Expected	QOH +100, Inventory Asset +USD 800, AP +USD 800; PO receipt/open status is correct.
Why	The PO is nonposting; the receipt/bill/expense is the quantity and accounting event.
If wrong	Check item-detail versus category-detail line, date, quantity, cost, duplicate form and inactive item.

Use	When	Typical posting
Purchase order	Commitment/order before receipt	No accounting entry by itself
Bill	Goods/services received now; pay supplier later	Dr Inventory/expense; Cr Accounts Payable
Expense/check/card	Goods/services received and paid now	Dr Inventory/expense; Cr bank/card
Bill payment	Pay an existing bill	Dr Accounts Payable; Cr bank
Vendor credit	Supplier reduces what you owe or goods are returned	Reduces AP and inventory/expense according to linked item/process
Item receipt	Separate quantity receipt before bill where supported	Edition/workflow determines receipt and later bill accounting

Partial receipt example

Purchase and receipt control

```
PO: 100 EA at USD 8 = USD 800 (no posting)
Receipt/bill 1: 60 EA at USD 8 = USD 480
Open PO: 40 EA = USD 320
Receipt/bill 2: 40 EA at USD 8 = USD 320
Final: QOH +100; Inventory Asset +800; AP +800 before payment

Control: never enter 100 on the first bill merely because the PO says 100. Record what was actually
accepted.
```

Three-way comparison	Question	Exception owner/action
PO vs physical receipt	Did we receive the ordered item, quantity, condition and UOM?	Receiving documents actual accepted/returned quantity
PO vs supplier bill	Are price, price unit, tax, freight and terms correct?	Buyer/AP resolves commercial variance
Receipt vs bill	Are we paying only for accepted goods/services?	AP holds unmatched quantity unless approved prepayment policy applies

If Item receipts are enabled: create the receipt from the PO for actual quantity, then create the Bill and link the suggested PO/receipt. Prove QOH, Inventory Asset, AP and PO status after each step. In QBD Enterprise EIR, also reconcile Inventory Offset by vendor.

Purchasing field	What to enter	Common mistake
Vendor	Legal payee that supplied the item	Selecting a duplicate or similarly named vendor
PO/bill date	Actual commitment and accounting/receipt dates	Changing date to force a period result
Item details	Inventory/noninventory/service item, quantity and rate	Using Category details for tracked inventory
Category details	G/L expense/asset with no item quantity	Posting directly to Inventory Asset and breaking item subledger
Customer/project	Only if cost is job-related or billable	Using customer as a department

Purchasing field	What to enter	Common mistake
Class/location	Approved reporting dimension	Leaving blank or inventing new near-duplicate values
Terms/due date	Contracted payment terms	Using invoice date when receipt or terms say otherwise
Bill number	Supplier's exact invoice/reference	Omitting it and weakening duplicate detection

Inventory costing — know which method your company uses

Method	Worked sale after 10@$8 + 10@$12	Operational effect
FIFO	Sell 6 → COGS USD 48; remaining 4@$8 + 10@$12 = USD 152	Transaction dates/layers matter; backdating can recalculate later COGS
Moving Average Cost	Average USD 10; sell 6 → COGS USD 60; remaining 14 = USD 140	Every receipt can change average for later sales
Desktop average cost	Often behaves as weighted average unless Enterprise FIFO is enabled	Company preference/start date and file history matter

Current QuickBooks Online inventory features can expose FIFO or Moving Average Cost choices for eligible companies. The selection may be constrained once inventory items exist. Record the actual company setting; do not infer the method from a report column named Average Cost.

Negative inventory: why profit changes later

1. Run Inventory Valuation Detail for one item over all relevant dates. Sort by transaction date and identify the first row where quantity becomes negative.
2. Look for a sale dated before receipt, PO never converted to bill/receipt, duplicate invoice/sales receipt, missing opening quantity or wrong item.
3. Correct the real document and date with approval. Later receipts can recalculate COGS under the costing method.
4. Re-run item valuation, Balance Sheet Inventory Asset and Profit and Loss COGS at the same cutoff.
5. Do not fix the G/L alone. A journal entry to Inventory Asset can make the Balance Sheet look right while the item quantity/value remains wrong.

Landed cost without breaking the item subledger

Landed-cost example

```
Merchandise: 100 EA × USD 8.00 = USD 800
Inbound freight: USD 100
Policy basis: allocate by units
Landed amount per unit = (800 + 100) / 100 = USD 9.00

If 20 units sell after correct capitalization:
COGS uses the company's costing method based on the USD 900 inventory value.
Remaining inventory value before other changes = USD 720.
```

Situation	Preferred method	Control
Initial inventory setup	Include eligible freight/duty in approved initial unit cost	Opening quantity × unit cost = approved inventory value
Native landed-cost feature	Allocate approved freight/duty/insurance through the supported workflow	Allocation basis and receiving bills reconcile
Desktop Enterprise feature	Map landed-cost items/account and allocate to inventory/assembly bill	Allocated total equals selected shipping/handling bills
No native item-level feature	Calculate landed unit cost before item receipt/bill or use an integrated inventory system	Item valuation and G/L stay equal
Late freight invoice	Use accountant-approved method that updates both item valuation and G/L	Never post only to Inventory Asset without subledger proof

Warehouse limitations and escalation point

Need	QuickBooks may handle	Use WMS/inventory app or Enterprise capability when
QOH by item	Basic tracked inventory	Multiple systems/channels need one availability service

Need	QuickBooks may handle	Use WMS/inventory app or Enterprise capability when
Reorder warning	Simple reorder point	MRP, safety stock, lead-time variability or multi-echelon planning is needed
Class/location reporting	Management segmentation	You need actual quantity by site/bin rather than report classification
Lot/serial/expiry	Edition/add-on dependent	Regulated traceability, FEFO, recall or genealogy is required
Pick/pack/ship	Sales transaction and simple fulfillment	Waves, RF scanning, bins, HUs, replenishment or labor control is required
Manufacturing	Assembly/BOM features where available	Routing, capacity, WIP, yield, backflush or shop-floor control is required

Inventory problem	Root cause to test	First safe action
QOH differs from count	Missing/duplicate transaction, wrong item/date, theft/damage or UOM	Freeze item, blind count, trace from last correct transaction
Inventory Asset differs from valuation	Direct G/L entry, inactive item, date/basis, negative stock or migration	Run both reports same date/accrual; list difference transactions
PO still open	Partial receipt, bill not linked, wrong vendor/item or close flag	Open PO detail and linked bills; correct relationship
Cost/margin changed	Costing method, backdated receipt, negative inventory, credit or landed cost	Trace one item's valuation detail chronologically
Duplicate bill	Missing supplier invoice number, copied form or bank-feed Add	Hold payment; compare vendor/reference/amount/date
Bill remains unpaid after cash left	Expense/check used instead of Bill Payment or payment unapplied	Trace bill → payment → bank match; do not record a second payment
PO does not appear on Bill	Wrong vendor, PO closed, no remaining quantity or Item details disabled	Verify vendor, open status, remaining quantity and form settings
COGS spikes after late receipt	Negative stock caused later cost recalculation	Find first negative row and correct real receipt/sale sequence under policy
Item missing from report	Inactive item or filter/date	Include inactive items and clear filters before creating a duplicate
Landed-cost clearing remains	Allocation incomplete, wrong bill/account or late freight	Reconcile every clearing entry to shipment/item allocation

SEE ALSO [CH07] procurement, p.20 • [CH13] inventory accuracy, p.28 • [SAP-MM] enterprise P2P, p.41 • [QB-CLOSE] reconciliation, p.63

[QB-SELL] QuickBooks sales, customer payments, returns and cash

Form	Use when	Posting
Estimate/quote	Customer is considering price/scope	Normally nonposting; can convert to later transaction
Sales order	Customer committed but fulfillment/invoice is later; only where available	Usually nonposting operational commitment
Invoice	Customer owes now and will pay later	Dr Accounts Receivable; Cr revenue/tax; tracked inventory also Dr COGS/Cr Inventory
Sales receipt	Customer pays at the time of sale	Dr cash/Undeposited Funds; Cr revenue/tax; plus inventory/COGS
Receive payment	Apply customer money to invoice	Dr bank/Undeposited Funds; Cr Accounts Receivable
Bank deposit	Group received payments into the real bank deposit	Dr bank; Cr Undeposited Funds
Credit memo/refund	Reduce/return customer amount under approval	Reduces AR/revenue and may restore inventory under item/process rules

Beginner run card

Question	Answer before you run
Goal	Sell 20 A100 units on account, receive payment and reconcile the deposit.
Need	Customer, item, quantity, price, tax, invoice/due date, ship evidence and payment method.
Where	+ Create → Invoice; later + Create → Receive payment; then Bank deposit if using Undeposited Funds.
Sample	C001; A100; 20 EA at USD 15; invoice USD 300; Net 30.
Replace	Approved customer/item/tax/terms/deposit account only.
Run	Create invoice → inspect AR/revenue and item valuation → receive payment → group exact bank deposit.
Expected	AR +300 then clears; revenue +300; inventory -20; COGS follows selected costing method; bank deposit matches statement.
Why	Each form represents a different real-world event; combining or duplicating them distorts AR and cash.
If wrong	Check invoice versus sales receipt, payment application, deposit account, duplicate bank-feed Add and date.

Invoice field decoder

Field	Filled example	Control
Customer/project	C001 / Retail rollout	Correct legal customer and job profitability
Invoice / due date	2026-10-21 / 2026-11-20	Revenue/tax period and credit collection date
Product/service	A100	Uses configured income, asset and COGS accounts
Quantity/rate	20 × USD 15	UOM, price authority and extended amount
Tax	Approved code/rate	Jurisdiction/product/customer rule; do not guess
Class/location	Consumer / Chicago	Management reporting under a governed design
Deposit/Undeposited Funds	Undeposited Funds	Allows several receipts to equal one bank deposit
Memo/attachment	Shipment 70000123 / proof	Trace operational fulfillment without replacing line detail

Payment and deposit worked example

Why Undeposited Funds exists

```
Invoice INV-1001                AR +300 | Revenue +300
Receive customer payment          Undeposited Funds +300 | AR -300
Two other receipts                Undeposited Funds +500
Bank deposit screen groups all     Bank +800 | Undeposited Funds -800
Bank statement shows one USD 800 deposit. Match it; do not add another income transaction.
```

Symptom	Trace	Correction principle
Invoice still unpaid	Customer → invoice → payment → applied-to field	Apply existing payment/credit; do not create a second receipt
Bank feed duplicates income	Downloaded amount → existing deposit/sales receipt	Match to existing transaction instead of Add
Undeposited Funds grows	Old receipts by date/customer → actual deposit batches	Create/correct deposit grouping; investigate missing bank event
Customer has credit and overdue invoice	Credit memo/payment and open invoice by customer	Apply under customer instruction/policy; retain audit link
Return did not restore stock	Credit/refund line type, item, quantity and date	Use correct item-based return process; prove QOH and value

Symptom	Trace	Correction principle
Sales and deposit differ	Gross sales, tax, fees, refunds, timing and payment processor settlement	Build a settlement bridge; do not force-match net to gross

Customer and sales control pack

Report	Question	Action
Open Invoices / AR Aging	Who owes what and how late?	Assign collection/dispute owner and promised date
Sales by Product/Service	Which items create revenue?	Reconcile quantity and returns to fulfillment
Sales by Customer	Which customers/mix drive revenue?	Review concentration, margin and credit exposure
Profit and Loss by Class/Location	Which segment appears profitable?	Confirm all transactions are tagged before concluding
Transaction List by Customer	Which forms make this balance?	Trace duplicates, credits and payment application
Deposit detail	What customer receipts make the bank amount?	Match exact batch to statement

SEE ALSO [CH14] fulfillment, p.29 • [CH21] margin/cash, p.37 • [QB-CLOSE] bank close, p.63 • [SAP-X] order-to-cash, p.51

[QB-CLOSE] QuickBooks bank reconciliation, month-end close and reports

Beginner run card

Question	Answer before you run
Goal	Reconcile one bank statement to exactly USD 0.00 difference.
Need	Official statement, account, beginning/ending dates and balance, and all period transactions.
Where	Online: Accounting → Reconcile; Desktop: Banking → Reconcile.
Sample	Checking; ending 2026-10-31; statement ending balance USD 25,400.
Replace	The exact account, ending date and balance printed on the statement.
Run	Confirm beginning balance → enter ending balance/date → check only statement transactions → finish at zero.
Expected	Difference USD 0.00 and a saved reconciliation report.
Why	The process proves the books contain every cleared bank event once.
If wrong	Check beginning balance change, ending input, missing/duplicate, wrong amount/date/account and uncleared timing.

Difference cause	How to test	Safe response
Wrong beginning balance	Compare prior reconciliation report and change/discrepancy report	Find who changed/deleted a reconciled transaction; restore/correct with audit evidence
Wrong ending input	Compare ending date/balance to statement	Edit reconciliation information
Missing book transaction	Tick statement rows and search exact amount/date/reference	Enter real source transaction once, then reconcile
Duplicate book transaction	Search amount/vendor/customer/date and bank-feed history	Remove/reverse duplicate with approval; keep real source
Amount/date/account wrong	Open source transaction and compare statement/support	Correct source; do not enter a plug
Outstanding check/deposit	Exists in books but not statement	Leave uncleared and investigate old items
Fees/interest	Statement has real charge/credit absent from books	Enter to approved account and date

Never force a reconciliation by posting an unexplained reconciliation discrepancy. A zero difference is meaningful only when every checked row represents the same real bank event.

Tax and multicurrency controls

Area	Maintain	Never assume
Sales tax	Registrations/effective dates, ship-to, item tax category, customer exemption evidence and overrides	Software decides where you must register or what is legally taxable
Purchase/use tax	Vendor tax, self-assessed use tax, import duty and recoverable-tax treatment	Every tax belongs in one generic expense account
Multicurrency	Home currency, party transaction currency, rate/date/source and realized/unrealized FX	Home currency or party currency can be casually changed after transactions
Close	Sales Tax Liability and FX reports tied to filed/approved schedules	A zero bank difference proves tax or currency treatment

Month-end close in dependency order

Order	Complete	Reconcile
1 Cutoff	Set period, timezone and last transaction acceptance time	Late items and next-period list
2 Sales/AR	Invoices, credits, receipts, deposits and processor settlements	AR aging to Balance Sheet AR
3 Purchasing/AP	Bills, vendor credits, payments, received-not-billed accrual	AP aging to Balance Sheet AP
4 Inventory	Receipts, sales, counts, returns, negative items and landed cost	Inventory Valuation to Inventory Asset G/L
5 Payroll/fixed assets	Payroll liabilities, capital additions, depreciation	Subsidiary schedules to G/L
6 Bank/cards	Every statement through cutoff	Each reconciliation difference = zero
7 Accruals/tax/FX	Approved entries with reversal where required	Schedules and tax reports to G/L
8 Review	Balance Sheet, P&L, cash flow, trend and unusual balances	Explanations and sign-off
9 Lock	Close books through approved date; password/warning policy	Close evidence and audit-log review

Inventory-to-G/L reconciliation

Reconciliation example

```
Inventory Valuation Summary at 2026-10-31 (accrual)     USD 42,500
Balance Sheet Inventory Asset at 2026-10-31               USD 43,200
Difference                                                   USD 700

Find: direct journals/category lines to Inventory Asset, inactive items, report date/basis, opening
conversion entries, negative inventory and future/backdated transactions.
Close only when USD 42,500 = USD 43,200 + individually approved reconciling items.
```

Core report	Set these before reading	Decision
Balance Sheet	As-of date, basis, comparative period	Liquidity, inventory, debt, AR/AP and equity
Profit and Loss	Period, basis, class/location and comparison	Revenue, gross margin, expense and profit drivers
Cash Flow	Period and classified accounts	Operating, investing and financing cash movement
Trial Balance	Same cutoff/basis as close	Every account's debit/credit balance and unusual sign
AP/AR Aging	Report date, aging method and currency	Overdue exposure, disputes, collection/payment plan
Inventory Valuation Detail	Item, all relevant dates and method	Quantity/value genealogy and first negative point
Audit log/trail	User, event and close period	Who changed what, when and from which process

Supply-chain KPIs from QuickBooks data

KPI	Formula	Limit
Inventory turns	Annual COGS ÷ average inventory	Use enough monthly points for seasonal stock
DIO	Average inventory ÷ annual COGS × 365	Financial days, not item availability/service
Gross margin %	(Sales − COGS) ÷ Sales	Missing landed cost or negative inventory distorts it
GMROI	Gross margin dollars ÷ average inventory cost	Compare like categories and periods
DSO	Average AR ÷ credit sales × days	Separate material cash sales
DPO	Average AP ÷ credit purchases × days	Purchases are better than COGS when available
Cash conversion cycle	DIO + DSO − DPO	Diagnostic; do not improve by harming suppliers/service
Supplier OTD / customer OTIF	On-time receipts/deliveries ÷ eligible lines	Needs PO promise and WMS/carrier proof; invoice date alone is insufficient

Close review: numbers that should not be accepted blindly

Signal	Question	Drill
Negative asset or liability	Is it a real credit/overpayment or wrong posting?	Transaction detail and source document
Large Uncategorized/Ask My Accountant	Why has coding not been resolved?	Owner, source evidence and correction date
Opening Balance Equity remains	Which converted/opening balance was not assigned?	Migration reconciliation by account
Inventory with zero/negative quantity but value	Date, cost layer or direct G/L problem?	Item valuation from first anomaly
Old uncleared bank items	Void, stale, lost, duplicate or still valid?	Bank evidence and payee/customer confirmation
AR/AP negative or very old	Unapplied credit/payment or dispute?	Customer/vendor transaction chain
Gross margin jumps	Price, volume, mix, cost method, backdate or classification?	Sales and item-cost bridge

SEE ALSO [FIN-01] statements, p.53 • [CH21] working capital, p.37 • [QB-BUY] inventory, p.58 • [PB-01] root cause, p.99

[QB-MIG] QuickBooks imports, migration, automation and SAP crosswalk

Import in dependency order

Sequence	Object	Control before next load
1	Chart of accounts	Unique names/numbers, correct account types and no placeholder balance
2	Customers and vendors	Legal identity, tax/contact, terms and duplicate review
3	Employees/users if in scope	Identity, role and privacy access
4	Classes and locations	Governed list and reporting purpose
5	Products and services	Type, SKU, accounts, costing method, opening date, QOH and value
6	Open transactions	Open invoices, bills, credits, checks and deposits retain references
7	Opening balances	Bank, inventory, assets, debt, tax and equity complete the trial balance
8	History only if justified	Volume, audit, editability and reporting retention are understood

Import field	Transform rule	Reject when
Name/ID	Trim spaces; preserve leading zeroes in a separate stable external-ID field	Blank, duplicate or reused identity

Sequence	Object	Control before next load
Date	Use one documented YYYY-MM-DD interpretation	Ambiguous or outside allowed period
Amount	Decimal with documented sign/currency	Text, unsupported currency or control total differs
Account/item	Map source code to one approved active target	No owner-approved mapping
Customer/vendor	Match governed identity, not display name alone	Possible duplicate or missing required fields
Class/location/project	Map only if dimension purpose is retained	Invented target or excessive hierarchy
Opening inventory	Quantity × method-consistent unit value at one cutoff	Subledger value does not equal approved G/L

Migration reconciliation pack

Control	Source	Target	Pass rule
List counts	Active/inactive customers, vendors, items, accounts	Imported + rejected	Source = target + approved rejects/merges
AR/AP	Open document count and value by party/currency	Aging and G/L	Document and G/L totals agree
Inventory	Item-location QOH/value and costing cutoff	Valuation and Inventory Asset	Count, quantity and value reconcile
Bank/cards	Statement/reconciled balances	Register and first reconciliation	First reconciliation reaches zero
Trial balance	Approved closing TB	Opening TB	Every account and total debit/credit agrees
Tax	Liability/receivable and filed-period status	Tax center/G/L	Accountant-approved transition
Reports	Baseline Balance Sheet/P&L/Aging	Same date/basis target reports	Differences explained and signed

Automation ladder

Level	Example	Guardrail
1 Memorized/template	Recurring form or saved report	Review dates, amounts and parties before posting
2 Bank rule	Suggest category/payee for a stable pattern	Prefer review; never let vague text create uncontrolled postings
3 Recurring transaction	Monthly rent bill/invoice	Owner, end date, approval and exception review
4 Import	Approved batch of products or invoices	Template version, control totals, reject log and rollback
5 Connector/app	Shop, WMS, payroll or expense integration	System of record, unique ID, retry/idempotency and daily reconciliation
6 API/workflow	Create/update objects from governed source	Least privilege, sandbox test, rate/error handling and audit log
7 Autonomous suggestion	Anomaly/category recommendation	Human approval for payment, vendor, tax, write-off and close changes

Integration rule: every external transaction needs a stable source ID stored in QuickBooks or the connector ledger. A retry must update/ignore the same business event, not create a second bill, invoice, payment or item.

SAP ↔ QuickBooks terminology crosswalk

QuickBooks	SAP S/4HANA	Critical difference
Company	Company code plus broader enterprise structure	One QuickBooks company can be much simpler than SAP legal/controlling design
Vendor	Business Partner with supplier/FI roles	SAP adds purchasing-org and company-code views

QuickBooks	SAP S/4HANA	Critical difference
Customer	Business Partner with customer/FI roles	SAP adds sales-area and partner-function views
Product/service	Product/material and service master	SAP adds plant, MRP, valuation, sales, QM and warehouse views
Class/location	Cost/profit center, segment, plant, sales area or other characteristic	Map from reporting purpose; there is no universal one-to-one match
Purchase order	MM purchase order	Both are commitments; SAP commonly separates GR and invoice more rigorously
Bill	Supplier invoice + FI document	SAP invoice may match PO/GR and create blocks
Inventory adjustment	Goods movement + FI/CO document	SAP requires movement/reason/organization and deeper stock status
Invoice/sales receipt	Sales order/delivery/PGI/billing/payment chain	SAP splits operational fulfillment and accounting documents
Project	WBS/project, internal order or customer order	Choose based on cost/revenue/control purpose
Audit log	Change documents, application logs and document flow	SAP evidence is distributed across business objects/logs

Practice without QuickBooks access

1. On paper create six accounts: Bank, AR, Inventory Asset, AP, Product Sales and COGS.
2. Write a PO for 100 A100 units at USD 8. Mark it nonposting.
3. Record the bill: debit Inventory Asset 800, credit AP 800. Draw QOH 100.
4. Record an invoice for 20 units at USD 15: debit AR 300 and credit Sales 300. Under MAC at USD 8, debit COGS 160 and credit Inventory 160. Draw QOH 80.
5. Record customer payment and deposit, supplier payment, then calculate final Bank, AR, AP, inventory quantity/value, revenue, COGS and gross profit.
6. Now create one duplicate bank-feed income entry. Use the reconciliation and report rules to find and remove the duplicate without changing the real invoice/payment chain.
7. Repeat under FIFO using purchases at two costs. Explain why COGS differs even though physical units are the same.

Practice answer key

```
Expected before cash timing:
Revenue = USD 300
COGS = USD 160 (if all 20 units carried at USD 8 under the stated method/history)
Gross profit = USD 140
Remaining QOH = 80 EA
Remaining inventory value = USD 640
AP = USD 800 before supplier payment
AR = USD 300 before customer payment
```

SEE ALSO [MIG-01] migration/cutover, p.91 • [SAP-X] enterprise integration, p.51 • [XL-01] Excel controls, p.69 • [CH25] automation, p.82

[OPT-01] Optimization from business question to usable answer

Word	Plain meaning	Example
Decision variable	What the model may choose	Units shipped from plant to customer
Objective	Number to minimize or maximize	Minimize freight cost
Constraint	Rule the answer must obey	Do not exceed plant capacity
Feasible	Obeys every mandatory constraint	All demand met within capacity
Optimal	Best feasible answer for the stated objective	Lowest cost among feasible plans
Sensitivity	How answer changes when an input changes	Cost if Dallas capacity rises by 100

Two-plant example	Customer East	Customer West	Capacity / demand
Chicago cost per unit	$4	$7	Capacity 100
Dallas cost per unit	$6	$3	Capacity 80

Word	Plain meaning	Example	
Demand	90	70	Total 160; capacity 180
One best plan	CHI→East 90; DAL→East 0	CHI→West 0; DAL→West 70	Cost = 90×4 + 70×3 = $570; 10 capacity remains at each site

Excel Solver: exact build

1. Enter costs in B3:C4, capacities in D3:D4 and demand in B5:C5.
2. Enter decision cells B8:C9 initially as zero: CHI-East, CHI-West, DAL-East, DAL-West.
3. In D8:D9 sum each plant row. In B10:C10 sum each customer column.
4. In B12 enter =SUMPRODUCT(B3:C4,B8:C9). This is total cost.
5. Data → Solver. Set Objective $B$12 to Min. By Changing $B$8:$C$9.
6. Add constraints $D$8:$D$9 <= $D$3:$D$4; $B$10:$C$10 = $B$5:$C$5; $B$8:$C$9 >= 0.
7. Choose Simplex LP; Solve. Keep solution only after checking all totals and units.

If the result looks wrong	Check
No feasible solution	Demand, capacity, mandatory lanes, minimums and equality signs
Fractional units	Add integer constraints only when business units cannot split
Unrealistically cheap	Missing lane, fixed, handling, duty, service or risk cost
One site overloaded	Capacity cell formula or wrong constraint direction
Plan is operationally unusable	Add calendar, batch, lane, service and policy constraints deliberately

Optimization does not decide what is fair, legal or safe. Those conditions must be explicit constraints or reviewed outside the model. Always compare with the current plan, explain the drivers and test several scenarios.

[CH22] 22 Excel daily shortcuts and checks

Read first	Plain guidance
In one sentence	Use spreadsheets as controlled calculators and exception lists.
Do this first	Turn raw ranges into Tables, set data types, isolate inputs and add checks.
What changes by company	Workbook path, table/column names, date format and Excel version.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Windows quick keys	Ctrl+T: create table. Ctrl+Shift+L: toggle filters. Ctrl+Arrow: jump to edge of data. Ctrl+Shift+Arrow: extend selection. Ctrl+1: format cells. Ctrl+;: insert date. F4: cycle reference type while editing a formula. Alt+=: AutoSum.
Mac quick keys	Command+T: create table. Command+1: format cells. Command+Arrow: jump to edge. Command+Shift+Arrow: extend selection. Control+;: insert date. Command+Shift+T: AutoSum. Keyboard settings and function keys can differ.
Useful formulas	Exact lookup: =XLOOKUP(A2,Items[Item],Items[LeadDays],"CHECK") Sum by item: =SUMIFS(Stock[Qty],Stock[Item],A2) Late open order: =IF(AND(D2="Open",C2<TODAY()),"LATE","") Pack rounding: =CEILING.MATH(B2,C2) Clean text: =TRIM(CLEAN(A2)) Missing/duplicate key: =IF(COUNTIF(Items[Item],A2)=1,"OK","CHECK")
Before using the result	Confirm dates are dates, numbers are numeric, units match and source ranges include new rows. XLOOKUP requires a supported Excel version; use an exact-match INDEX/MATCH fallback on older versions. A blank lookup result is not automatically zero.
Problem to action	SUMIFS gives zero -> mismatched text/type or hidden spaces -> inspect sample keys -> normalize import types. Formula changes when copied -> relative reference -> use $A$1 where fixed -> check first and last rows. Filtered total looks wrong -> SUM includes hidden rows -> use =SUBTOTAL(109,B2:B1000) for visible-row sum.

Read first	Plain guidance
Habit	Keep raw data separate, use Tables, avoid merged data cells and save a dated version before bulk edits.

[XL-01] Excel guided execution

Beginner run card

Question	Answer before you run
Goal	Calculate open quantity from the handbook sample without changing source data.
Need	Excel and the two sample Orders tables from [DATA-01]. Leave blank delivery dates blank.
Where	A new workbook → cell A1 → Table Design → Table Name.
Sample	Name the order table tblOrders; create a new column named open_qty.
Replace	Only table/header names after you confirm the exact spelling in your workbook.
Run	Type the formula in the first data row and press Enter; Excel fills the column.
Expected	Open quantities 0, 5, 5, 0, 0, 6; total 16.
Why	Each row subtracts shipped quantity from ordered quantity; MAX prevents a negative result.
If wrong	Check numeric types, table name, headers, blank dates, hidden rows and copied sample values.

1. Create a blank sheet named Orders. Copy the first Orders table from [DATA-01] into cells A1:G7, including its header row.
2. In H1:J1 type shipped_qty, promised_date and delivery_date. From the second Orders table, copy only those three columns for order IDs 1001–1006 into H2:J7. Do not copy its repeated order_id or ordered_qty columns.
3. Confirm A2:A7 reads 1001, 1002, 1003, 1004, 1005, 1006 and SUM(E2:E7) equals 61. Leave the two missing delivery dates blank; do not type NULL.
4. Select A1:J7, press Ctrl+T, confirm ‘My table has headers’, then Table Design → Table Name → type tblOrders.
5. Create a second sheet named Items. Paste the Items table from [DATA-01], press Ctrl+T and name it tblItems.

Formula part	Plain meaning
tblOrders	The whole Excel table.
[open_qty]	The complete open_qty column.
[@ordered_qty]	ordered_qty in this row only; @ means this row.
MAX(number,0)	Use the larger number, so open quantity never becomes negative.
=	Tell Excel that the cell contains a formula.

Action	Formula — type or paste exactly
Make open_qty in a new table column	=MAX([@ordered_qty]-[@shipped_qty],0)
Make otif_flag in a new table column	=--AND([@status]="Delivered",[@delivery_date]<=[@promised_date], [@shipped_qty]>=[@ordered_qty])
Sum open quantity in any output cell	=SUM(tblOrders[open_qty])
Count open orders in any output cell	=COUNTIFS(tblOrders[status],"Open")
List exceptions in an empty Microsoft 365 area	=FILTER(tblOrders[[order_id]:[status]],tblOrders[open_qty]>0,"None")
Lookup item cost in a new table column	=XLOOKUP([@item_id],tblItems[item_id],tblItems[unit_cost],"Missing")

Check	Expected result
open_qty for all six rows	0, 5, 5, 0, 0, 6; total 16
otif_flag	1 only for orders 1001 and 1004

Formula part	Plain meaning
open-order count	2
FILTER output	Three rows: orders 1002, 1003 and 1006
item cost	A100 $8.00; B200 $3.50; C300 $12.00
Change only these	**How**
tblOrders	Replace with the exact name shown in Table Design.
column names	Use the spelling in the header; Excel inserts structured names when you click the column.
status text	Use a real source value such as Open, OPEN or Released.
date fields	Confirm Excel stores dates as numbers; changing display format does not convert text.
semicolon/comma	Some regional Excel settings use semicolons between arguments.

Error	Meaning	Fix
#NAME?	Function/table/column name not recognized	Check spelling, Excel version and table name
#N/A	Lookup did not find a match	Trim spaces; check data type, case-sensitive method and key
#VALUE!	Wrong data type or array shape	Test dates/numbers; inspect each part with F9 carefully
#SPILL!	Dynamic result area is blocked	Clear cells in the spill range; tables cannot contain spilled arrays
Wrong total	Hidden filters, duplicates, text numbers or wrong population	Reconcile row count and SUM to source controls

Shortcuts worth memorizing

Shortcut	Action
Ctrl+T	Create Table
Ctrl+Shift+L	Toggle filters
Ctrl+Arrow	Jump to data edge
Ctrl+Shift+Arrow	Select to data edge
Ctrl+1	Format Cells
F2	Edit active cell
F4	Repeat action or lock A1 reference while editing
Ctrl+`	Show/hide formulas
Alt+=	AutoSum
Ctrl+;	Current date
Ctrl+Space	Select table column
Shift+Space	Select row

ADVANCED: the next section includes Power Query, Office Scripts, Power Automate and VBA. These can refresh, write or automate workbook actions. Use a copy, verify the population and keep a recovery path before running them.

Advanced Excel execution system

Use five layers: Inputs, Parameters, Calculations, Checks and Output. Convert inputs to Tables, apply data types, protect formulas, name assumptions, show refresh time and reconcile control totals. Avoid merged data cells and hidden manual constants.

Task	Formula or method — then check
Open PO value	=SUMIFS(tblPO[OpenValue], tblPO[Buyer], $B$2, tblPO[Site], $B$3, tblPO[Status], "Open") Check: reconcile to the source cutoff.

Task	Formula or method — then check
Overdue exceptions	=SORT(FILTER(tblPO[[PO]:[Buyer]], (tblPO[OpenQty]>0)*(tblPO[PromiseDate]<TODAY()), "None"), 3, 1) Check: dates must be real dates.
Latest matching price	=XLOOKUP(1, (tblPrice[SKU]=[@SKU])*(tblPrice[Supplier]=[@Supplier]), tblPrice[UnitPrice], "Missing", 0, -1) Check: sort by effective timestamp.
Visible total	=SUBTOTAL(109, tblData[Value]) Check: use only when a filtered view is intended.
Pack rounding	=CEILING.MATH(RequiredQty, PackQty) Check: confirm MOQ and shelf life.
WAPE	=LET(a, tblF[Actual], f, tblF[Forecast], d, SUM(ABS(a)), IF(d=0, "", SUM(ABS(a-f))/d)) Check: blank means undefined when total actual is zero.
Bias	=LET(a, tblF[Actual], f, tblF[Forecast], d, SUM(a), IF(d=0, "", SUM(f-a)/d)) Check: blank means undefined; state the sign convention.
Variable LT safety stock	=LET(z, $B$2, d, AVERAGE(Demand), sd, STDEV.S(Demand), L, AVERAGE(LeadTime), sL, STDEV.S(LeadTime), z*SQRT(L*sd^2+d^2*sL^2)) Check: demand and lead-time units must match.

Reusable named LAMBDA

Save as ROP in Name Manager then call =ROP(B2,C2,D2,E2,F2)

```
=LAMBDA(avgD,avgLT,z,sdD,sdLT,
  avgD*avgLT + z*SQRT(avgLT*sdD^2 + avgD^2*sdLT^2))
```

Dynamic arrays and model checks

Need	Pattern	Typical failure
Distinct list	=SORT(UNIQUE(tblOrders[Supplier]))	Blank or inconsistent names
Two-dimensional summary	=PIVOTBY(tbl[Site],tbl[Month],tbl[Qty],SUM)	Function availability varies by release
Select fields	=CHOOSECOLS(FILTER(tbl,tbl[Risk]>0.6),1,3,7)	Blocked spill range
Stack extracts	=VSTACK(Jan!A2:F999,Feb!A2:F999)	Headers or data types differ
Row check	=IF(COUNTIFS(tbl[Key],[@Key])=1,"OK","CHECK")	Key is not the intended grain
Reconciliation	=SourceTotal-CalculatedTotal	Round only for display, not logic

Power Query field method

1. Use the purpose-built connector and keep credentials outside the workbook logic.
2. Set data types immediately. Remove unused columns and filter rows early.
3. Keep staging queries separate from business outputs. Disable load for staging where appropriate.
4. Preserve query folding through the expensive steps. Inspect folding indicators or query plan.
5. Join only at declared grain. Aggregate before joining two transaction tables.
6. Create rejected-row and reconciliation outputs. Refresh must fail visibly when controls fail.
7. Parameterize environment, file path and cutoff. Document privacy and gateway requirements.

Power Query M pattern

```
let
  Source = Excel.CurrentWorkbook(){[Name="tblPO"]}[Content],
  Typed = Table.TransformColumnTypes(Source,
      {{"PO", type text}, {"PromiseDate", type date}, {"OpenQty", type number}}),
  OpenLate = Table.SelectRows(Typed,
      each [OpenQty] > 0 and [PromiseDate] < Date.From(DateTime.LocalNow())),
  Checked = Table.AddColumn(OpenLate, "DataCheck",
      each if [PO] = null then "MISSING KEY" else "OK", type text)
in
  Checked
```

Power Pivot Solver and automation

Tool	Use it for	Execution rule
Power Pivot	Large relational model and DAX inside Excel	Star schema, Date table, measures, no fact-to-fact joins
Solver	Allocation, mix, transport or capacity optimization	Define objective, decision cells, constraints, units and feasible baseline
What-if Data Table	Sensitivity of one or two assumptions	Keep formula reference and scenario units visible
Office Scripts	Repeatable cloud workbook steps	Validate table and columns; write run timestamp and exception result
Power Automate	Schedule, approval, file and notification workflow	Send exceptions only; include retry, owner and failure path
VBA	Desktop legacy automation when approved	Digitally sign, avoid hard-coded paths, add error handling and rollback

Office Script control pattern

```
function main(workbook: ExcelScript.Workbook) {
  const table = workbook.getTable("tblExceptions");
  if (!table) throw new Error("tblExceptions missing");
  const rows = table.getRowCount();
  workbook.getWorksheet("Checks").getRange("B2").setValue(rows);
  workbook.getWorksheet("Checks").getRange("B3").setValue(new Date().toISOString());
}
```

Excel shortcuts

Action	Windows shortcut	Why it matters
Create Table	Ctrl T	Expands references and supports structured formulas
Toggle filters	Ctrl Shift L	Fast exception review
Refresh all	Ctrl Alt F5	Refresh connections and models
Flash Fill	Ctrl E	Pattern extraction with manual verification
AutoSum	Alt =	Fast totals
Edit cell	F2	Audit formula in place
Toggle absolute reference	F4 while editing	Control copied formulas
Show formulas	Ctrl `	Find constants and inconsistent formulas
Go to precedent/dependent	Ctrl+[or Ctrl+]	Trace model logic
Format cells	Ctrl 1	Set types and custom formats
Select to data edge	Ctrl Shift Arrow	Fast range inspection
Paste values	Ctrl Alt V then V Enter	Freeze approved output
Full recalculation	Ctrl Alt F9	Rebuild calculation chain when required

Shortcut behavior varies by operating system, keyboard and Excel release. Keep raw inputs immutable and save a dated version before bulk changes.

SEE ALSO [DATA-01] sample data, p.12 • [OPT-01] Solver, p.67 • [PBI-01] Power BI, p.79

[CH23] 23 SQL for supply chain exceptions

Read first	Plain guidance
In one sentence	Ask a database a precise question and return auditable rows.
Do this first	State the row you expect, then select only needed columns and filter early.
What changes by company	Catalog, schema, table, column, data type and SQL dialect.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.

Read first	Plain guidance
Field panel	**Execution guidance**
Assumed schema	PostgreSQL-style examples. po_line has one row per po_id and line_id: item_id, due_date, ordered_qty, status. receipt has one row per receipt event with po_id, line_id and accepted_qty. All quantities use the PO base unit.
Code	WITH received AS (SELECT po_id, line_id, SUM(accepted_qty) AS qty FROM receipt GROUP BY po_id, line_id) SELECT p.po_id, p.line_id, p.item_id, p.due_date, p.ordered_qty - COALESCE(r.qty,0) AS open_qty FROM po_line p LEFT JOIN received r ON p.po_id=r.po_id AND p.line_id=r.line_id WHERE p.status <> 'Cancelled' AND p.due_date < CURRENT_DATE AND p.ordered_qty > COALESCE(r.qty,0);
Why this works	Aggregate receipts before joining so multiple receipt events do not multiply PO quantities. LEFT JOIN retains lines with no receipt. COALESCE treats missing receipt totals as zero. Returns, reversals and closed-short rules must be incorporated into the source definition.
Duplicate item check	SELECT item_id, COUNT(*) AS n FROM item_master GROUP BY item_id HAVING COUNT(*) > 1;
Problem to action	Totals double after join -> grain mismatch -> aggregate to common keys -> compare row counts and totals before/after. Missing records -> INNER JOIN drops unmatched keys -> inspect with LEFT JOIN -> repair master data.
Validate	Test a PO for 100 with receipts 30 and 20: open quantity must be 50 once, not two separate balances. Test no receipt, fully received and cancelled lines. Use read-only access for investigation.

[SQL-01] SQL guided execution

Beginner run card

Question	Answer before you run
Goal	List incomplete orders at CHI-DC without changing data.
Need	A read-only development/analytics editor, approved access and one known order.
Where	Your SQL editor or tool query window; select the approved development connection.
Sample	Use the filled SELECT query below with the Northstar sample.
Replace	Only catalog/schema/table, column names, quoted site value and approved date/threshold.
Run	Highlight only the SELECT statement and choose Run/Execute.
Expected	Orders 1002, 1003 and 1006 with open quantities 5, 5 and 6; total 16.
Why	WHERE keeps incomplete CHI-DC rows; subtraction finds remaining quantity; ORDER BY sorts output.
If wrong	Preview ten rows, find one known ID, inspect column names/types and list real site/status values.

Run SQL in a read-only development or analytics workspace first. A SELECT query reads data; INSERT, UPDATE, DELETE, MERGE, CREATE, ALTER and DROP change data or objects. Do not run a changing statement without authorization, a tested population and recovery plan.

Read-only ladder before any calculation

Step	Example purpose	Do not move on until
1 Preview	Return ten rows from the approved object	You recognize columns and data types
2 Values	List distinct status or site values	You copy a real value exactly
3 Known ID	Filter one order you already know	It returns once at the intended grain
4 Count	Count rows and sum a control quantity	The result ties to the source cutoff
5 Calculate	Add business logic and joins	Every step keeps expected rows and totals

Keep exactly	Replace only after discovery
SELECT, FROM, WHERE, AND, ORDER BY, commas and parentheses	catalog, schema, table, column, quoted value, date and threshold

SQL word	Plain meaning
Alias	A short temporary name for a table or output column.
CASE	If/then logic inside a query.
COUNT(*)	Number of returned rows.
DISTINCT	Remove repeated combinations from the displayed result.
COALESCE	Use the first value that is not NULL.
NULLIF	Return NULL when two values are equal; often prevents divide-by-zero.
ROW_NUMBER	Number related rows in a chosen order.
PARTITION BY	Restart a window calculation for each group.

Need	Common syntax	Possible alternative
First ten rows	LIMIT 10 at the end	SQL Server: SELECT TOP (10) …
Date literal	DATE '2026-10-19'	CAST('2026-10-19' AS date)

SQL part	Plain meaning	Example
SELECT	Columns to return	order_id, item_id
FROM	Table to read	orders
WHERE	Rows to keep	status = 'Open'
GROUP BY	Grain of summary	item_id
HAVING	Filter after summary	SUM(open_qty) > 5
ORDER BY	Display order	open_qty DESC
JOIN ... ON	How two tables connect	orders.item_id = items.item_id

Filled runnable example — open quantity by order

```
SELECT
    order_id,
    item_id,
    ordered_qty - shipped_qty AS open_qty,
    promised_date
FROM orders
WHERE site_id = 'CHI-DC'
  AND ordered_qty > shipped_qty
ORDER BY promised_date;
```

Change only these	What it means	Example replacement
orders	Your catalog.schema.table or view	prod_erp.analytics.sales_order_line
order_id/item_id	Your order and SKU/material fields	sales_order_number/material_number
ordered_qty/shipped_qty	Comparable quantities in one UOM	ordered_each/shipped_each
site_id	Site/plant/location field	plant_code
'CHI-DC'	Real site value; keep SQL single quotes	'1710'
promised_date	Approved commitment field	confirmed_delivery_date

Change only these	What it means	Example replacement
Expected output	**order_id 1002 open 5; 1003 open 5; 1006 open 6, sorted by promised date**	
Why	WHERE keeps incomplete rows at CHI-DC. Subtraction calculates remaining units. ORDER BY only changes display order.	
Control check	Expected open total for those rows is 16. Compare row count and sum with source report.	
If no rows	Check table/environment, exact site value, status/date/UOM and whether shipped quantity includes cancellations.	
If duplicates	Confirm the table grain. It may contain order schedules, deliveries or change history rather than one order line.	

OTIF teaching query — line grain

```
WITH line_status AS (
  SELECT
      order_id,
      CASE WHEN status = 'Delivered'
                AND delivery_date IS NOT NULL
                AND delivery_date <= promised_date
                AND shipped_qty >= ordered_qty
           THEN 1 ELSE 0 END AS otif_flag
  FROM orders
  WHERE promised_date <= DATE '2026-10-19'
)
SELECT
    COUNT(*) AS eligible_lines,
    SUM(otif_flag) AS otif_lines,
    1.0 * SUM(otif_flag) / NULLIF(COUNT(*), 0) AS otif_rate
FROM line_status;
```

Expected sample result: eligible_lines 6, otif_lines 2, otif_rate 0.3333. The cutoff includes every due line; a due line with a blank delivery date or incomplete quantity fails OTIF. Business policy may exclude cancelled, held or customer-caused lines; define exclusions before changing SQL. Aggregate lines to one order result before calling the output order-level OTIF.

Join and summarize

```
SELECT
    o.item_id,
    i.item_description,
    SUM(o.ordered_qty - o.shipped_qty) AS open_qty
FROM orders o
JOIN items i
  ON o.item_id = i.item_id
WHERE o.ordered_qty > o.shipped_qty
GROUP BY o.item_id, i.item_description
ORDER BY open_qty DESC, o.item_id;
```

Expected open quantity: C300 = 6, A100 = 5, B200 = 5. The second sort key makes tied quantities repeatable. Before trusting a JOIN, count rows before and after it. If the row count multiplies unexpectedly, the joining key is not unique at the assumed grain.

Error	Likely meaning	First fix
Object not found	Wrong catalog/schema/table or case/quote	Copy the qualified name from the catalog
Column not found	Wrong alias/spelling or column not in view	Inspect columns; prefix alias
Invalid type	Text/date/number mismatch	Cast deliberately and find bad values
Divide by zero	No eligible rows	NULLIF denominator; investigate population
Permission denied	Role lacks read or object use	Request least privilege; do not bypass
Query slow	Too many rows/columns, bad join, skew or no pruning	Filter early; inspect plan/profile

ADVANCED — CHANGES DATA OR OBJECTS: MERGE, CREATE, ALTER, streams and tasks can write or change persistent objects. A beginner should not run them in production. Require review, non-production test, population proof, approval and recovery.

Advanced SQL supply chain patterns

Declare the grain before writing SQL. Profile keys and nulls. Pre-aggregate each fact to common keys before joining. Reconcile row count, quantity and value after every material transformation. Examples use portable SQL where practical; date and MERGE syntax vary by engine.

Logical processing

FROM JOIN → WHERE → GROUP BY → HAVING → WINDOW → SELECT → DISTINCT → ORDER BY → LIMIT

Latest valid record per business key

```
WITH ranked AS (
  SELECT s.*,
         ROW_NUMBER() OVER (
           PARTITION BY item_id, site_id
           ORDER BY source_updated_at DESC, ingested_at DESC
         ) AS rn
  FROM stage_inventory s
)
SELECT * FROM ranked WHERE rn = 1;
```

Use QUALIFY rn = 1 in engines that support it. Keep a deterministic tie-breaker. Deduplicate before MERGE.

Order grain OTIF from line events

```
WITH order_result AS (
  SELECT order_id,
         MAX(CASE WHEN actual_delivery_date IS NULL
                       OR actual_delivery_date > promised_date
                  THEN 1 ELSE 0 END) late_flag,
         MAX(CASE WHEN COALESCE(delivered_qty,0) < ordered_qty THEN 1 ELSE 0 END) short_flag
  FROM fact_order_line
  WHERE cancelled_flag = 0
    AND promised_date <= DATE '2026-10-19'
  GROUP BY order_id
)
SELECT 100.0 * AVG(CASE WHEN late_flag=0 AND short_flag=0 THEN 1.0 ELSE 0 END) otif_pct
FROM order_result;
```

Replace 2026-10-19 with the approved reporting cutoff. The cutoff keeps only due lines; a due line with no delivery date fails on-time. Freeze requested/confirmed date, early/late tolerance, cancellations, splits and open-overdue treatment in the metric contract.

Deterministic running balance

```
SELECT item_id, site_id, event_ts, event_id, qty_delta,
       SUM(qty_delta) OVER (
         PARTITION BY item_id, site_id
         ORDER BY event_ts, event_id
         ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
       ) AS calculated_on_hand
FROM inventory_events;
```

ABC by annual consumption value

```
WITH sku AS (
  SELECT item_id, SUM(issue_qty * standard_cost) annual_value
  FROM fact_issue
  WHERE issue_date >= CURRENT_DATE - INTERVAL '365' DAY
  GROUP BY item_id
), ranked AS (
  SELECT item_id, annual_value,
         SUM(annual_value) OVER (ORDER BY annual_value DESC ROWS UNBOUNDED PRECEDING)
         / NULLIF(SUM(annual_value) OVER (),0) cumulative_share
  FROM sku
)
SELECT *, CASE WHEN cumulative_share<=0.80 THEN 'A'
               WHEN cumulative_share<=0.95 THEN 'B' ELSE 'C' END abc_class
FROM ranked;
```

Recursive BOM explosion with illustrative cycle guard

```
WITH RECURSIVE exploded AS (
  SELECT parent_item, component_item, qty_per, 1 bom_level,
         CAST(parent_item || '>' || component_item AS VARCHAR(4000)) path
  FROM bom WHERE parent_item = :finished_good
  UNION ALL
  SELECT e.parent_item, b.component_item, e.qty_per*b.qty_per, e.bom_level+1,
         e.path || '>' || b.component_item
  FROM exploded e JOIN bom b ON b.parent_item=e.component_item
  WHERE e.bom_level < 30 AND e.path NOT LIKE '%' || b.component_item || '%'
)
SELECT component_item, SUM(qty_per) total_qty_per_fg
FROM exploded GROUP BY component_item;
```

Use engine-specific arrays or graph logic for robust cycle detection. The text check is illustrative and can create false matches for similar IDs.

Advanced exception recipes

Problem	SQL pattern	Control
As of inventory	Join event or snapshot to latest timestamp not after cutoff	Cutoff timezone and late-arriving data
Gaps and islands	Flag change with LAG then cumulative SUM group	Stable ordering and inclusive dates

Problem	SQL pattern	Control
Lead-time percentile	PERCENTILE_CONT or engine equivalent by supplier item	Remove invalid dates and segment lanes
Demand intermittency	Calendar spine LEFT JOIN demand; count zero periods	Do not drop zero-demand buckets
Slowly changing dimension	Hash tracked attributes; expire old row and insert new row	One current row per business key
Anti-join	NOT EXISTS correlated subquery	Safer than NOT IN when NULL exists
Fact-to-fact	Aggregate each fact to common conformed dimensions	Check fanout and control totals
Incremental load	High-water mark plus overlap; dedupe and idempotent MERGE	Late updates and replay

Gaps and islands for status dwell

```
WITH changes AS (
  SELECT t.*,
         CASE WHEN LAG(status) OVER (PARTITION BY order_id ORDER BY event_ts,event_id)=status
              THEN 0 ELSE 1 END AS new_island
  FROM order_status_event t
), grouped AS (
  SELECT *, SUM(new_island) OVER (
      PARTITION BY order_id ORDER BY event_ts,event_id ROWS UNBOUNDED PRECEDING) grp
  FROM changes
)
SELECT order_id, status, MIN(event_ts) start_ts, MAX(event_ts) end_ts
FROM grouped GROUP BY order_id,status,grp;
```

Supplier lead-time tail risk

```
SELECT supplier_id,
       PERCENTILE_CONT(0.95) WITHIN GROUP (ORDER BY receipt_date - po_date) p95_lead_days,
       AVG(receipt_date - po_date) avg_lead_days
FROM receipt
WHERE receipt_date >= po_date
GROUP BY supplier_id;
```

Idempotent merge and data controls

Generic MERGE pattern

```
MERGE INTO curated_order t
USING (
  SELECT * FROM (
    SELECT s.*, ROW_NUMBER() OVER (
      PARTITION BY order_id ORDER BY source_updated_at DESC, ingested_at DESC) rn
    FROM stage_order s WHERE batch_id = :batch_id
  ) x WHERE rn=1
) s
ON t.order_id=s.order_id
WHEN MATCHED AND s.source_updated_at>=t.source_updated_at THEN UPDATE SET
  status=s.status, source_updated_at=s.source_updated_at
WHEN NOT MATCHED THEN INSERT (order_id,status,source_updated_at)
VALUES (s.order_id,s.status,s.source_updated_at);
```

Control	Test
Uniqueness	No more than one source row per declared key after deduplication
Completeness	Expected source rows and required fields present
Referential integrity	All critical fact keys map to a valid or governed unknown dimension
Reconciliation	Source and target counts, quantities, values and status populations within signed tolerance
Freshness	Maximum source event time and successful load time within SLA
Replay	Same batch rerun produces the same state without duplicate side effects

Query tuning and production support

1. Read the actual plan or query profile. Compare estimated and actual rows; locate scans, fanout, spill, skew and queueing.
2. Reduce rows and columns early while preserving partition or micro-partition pruning. Do not wrap the filtered raw date column unnecessarily.

3. Fix grain and join cardinality before adding compute. Use UNION ALL when deduplication is unnecessary.
4. Choose indexes, clustering or layout for measured selective predicates and joins. Refresh statistics after material change.
5. Keep transactions short, update objects in a consistent order and retry approved transient isolation or deadlock failures.
6. Parameterize values. Apply least privilege. Log batch, version, row counts, checks and error without exposing protected data.
7. Test normal, zero, null, duplicate, late, reversed, cancelled, extreme and rerun cases before release.

Symptom	Likely cause	First test	Correction
Totals multiply	Fact fanout	Rows before/after join	Aggregate or bridge at declared grain
NOT IN returns nothing	NULL in subquery	Find NULL key	Use NOT EXISTS and reject null key
Running result changes	Tied ordering	Duplicate timestamp	Add deterministic sequence
Full scan	Non-sargable filter or layout	Plan partitions/bytes scanned	Filter raw column and improve layout
MERGE fails	Duplicate source key	Count by key	Rank and keep one valid row
Deadlock	Conflicting lock order	Transaction graph/log	Consistent order, shorter transaction, retry

SEE ALSO [DATA-01] sample data, p.12 • [PBI-01] DAX, p.79 • [DBX-01] Databricks, p.83 • [SNF-01] Snowflake, p.84

[CH24] 24 Power BI and DAX quick reference

Read first	Plain guidance
In one sentence	Build one governed model and measures that answer business questions.
Do this first	Set data types and relationships before writing DAX.
What changes by company	Source, refresh, relationship keys, metric definitions and security.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Model first	Use dimensions for Item, Supplier, Site and Date with unique keys and one-to-many relationships to facts. Keep each fact at a consistent grain. Do not directly join receipt events to order lines and sum order quantities.
Assumed fact	OrderDue contains one row per eligible due order at the reporting cutoff: OrderID, DueDate, OTIF_Flag (1 or 0). Include overdue undelivered orders as 0; apply the agreed cancellation rules. Relate Calendar[Date] to OrderDue[DueDate].
Code	Due Orders = COUNTROWS(OrderDue) OTIF Orders = CALCULATE([Due Orders], OrderDue[OTIF_Flag] = 1) OTIF % = DIVIDE([OTIF Orders], [Due Orders]) Open Qty = SUM(OpenPOLine[OpenQty])
Read the result	DIVIDE returns blank on a zero denominator unless an alternate is supplied. Blank can correctly mean no eligible orders. Format OTIF as a percentage. 92 passing rows / 100 eligible rows = 92%; test this manually. OpenPOLine must already contain one row per open PO line.
Problem to action	Same value for every supplier -> missing/inactive relationship -> inspect filter path -> test a single supplier. Grand total differs from average of rows -> weighted ratio -> compute total numerator/denominator -> label the metric. Old figures -> failed refresh -> show refresh timestamp -> assign refresh ownership.

Read first	Plain guidance
Release checklist	Reconcile totals with source; test filters and date roles; verify access controls with actual roles; check missing keys and refresh failures. Inventory snapshots must not be summed across dates as if they were flows.

[PBI-01] Power BI and DAX guided execution

Beginner run card

Question	Answer before you run
Goal	Build one OTIF card from the sample and prove it against source rows.
Need	Power BI Desktop and the sample Orders data from [DATA-01].
Where	Desktop: Get data → Transform Data → Model view → Modeling → New measure.
Sample	Paste one DAX measure at a time, then add it to a Card in Report view.
Replace	Only table/column names and the approved cutoff after checking the Fields pane.
Run	Close & Apply, create the measure, add a Card, then use slicers for site/item/date.
Expected	Ordered 61, shipped 45, open 16, eligible lines 6, OTIF lines 2, OTIF 33.33%.
Why	Measures calculate from source rows under the filters selected in the report.
If wrong	Check data types, row grain, relationships, filters, blank dates and source control totals.

Power BI place	Plain meaning
Desktop	Build data, model, measures and report pages.
Service	Publish, refresh, secure and share approved reports.
Report view	Arrange cards, tables, charts and slicers.
Data view	Inspect loaded rows and columns.
Model view	See tables, keys and relationships.
Measure	A calculation evaluated under report filters.
Slicer	A visible control that filters the report.
Relationship	A governed connection between table keys.

1. Power BI Desktop → Get data → Excel/Text/CSV/SQL source. Choose the sample Orders table.
2. Select Transform Data. Set IDs to Text, quantities to Whole Number, and dates to Date. Close & Apply.
3. In Model view, confirm one fact row means one order line. Create separate Item, Customer, Site and Date dimensions when building a real model.
4. Modeling → New measure. Paste each DAX measure below one at a time.
5. Add a Card for OTIF %, a table for order exceptions, and slicers for site, item and date.
6. Reconcile total row count and quantity to the source before publishing.

Basic measures

```
Ordered Qty = SUM(Orders[ordered_qty])

Shipped Qty = SUM(Orders[shipped_qty])

Open Qty =
SUMX(Orders, MAX(Orders[ordered_qty] - Orders[shipped_qty], 0))
```

OTIF line measures for the toy sample

```
Eligible Lines =
CALCULATE(
```

```
    COUNTROWS(Orders),
    Orders[promised_date] <= DATE(2026, 10, 19)
)

OTIF Lines =
COUNTROWS(
    FILTER(
        Orders,
        Orders[promised_date] <= DATE(2026, 10, 19) &&
        Orders[status] = "Delivered" &&
        Orders[delivery_date] <= Orders[promised_date] &&
        Orders[shipped_qty] >= Orders[ordered_qty]
    )
)

OTIF % = DIVIDE([OTIF Lines], [Eligible Lines])
```

The toy data has one line per order, so line OTIF and order OTIF happen to match. If real orders have several lines, aggregate every line to one governed order result before reporting order-level OTIF.

Expected sample card	Value	If different
Ordered Qty	61	Check all six rows and numeric type
Shipped Qty	45	Check blank/zero and duplicate rows
Open Qty	16	Check calculation at row grain
Eligible Lines	6	Check date type and current filter context
OTIF Lines	2	Check promise/delivery and complete quantity rule
OTIF %	33.33%	Format as percent; inspect slicers and exclusions

DAX word	Meaning
Measure	Calculated at query time under report filters
Calculated column	Stored row-by-row at data refresh
Filter context	Filters from slicers, visuals, relationships and CALCULATE
Row context	The current row used by an iterator or calculated column
CALCULATE	Re-evaluates an expression after changing filter context
SUMX	Iterates rows, calculates an expression, then sums
DIVIDE	Safe division with controlled zero/blank behavior
VAR	Names an intermediate result for readable, faster DAX

Common problem	Root cause	Fix
Total differs from source	Wrong grain, duplicate join, filter or data type	Reconcile row count and key uniqueness
Same value on every row	Relationship/filter direction or wrong measure	Inspect model path; use dimension fields
Blank date measures	No proper Date table/relationship	Create marked Date table and active relationship
Slow visual	High-cardinality columns, complex iterator or too many visuals	Use star schema, measures and Performance Analyzer
User sees forbidden rows	RLS missing or not tested	Define roles and test as role in Service

Power BI semantic model and DAX

Model before visualizing. Dimensions filter; facts summarize. Each fact has one declared grain. Prefer one-to-many, single-direction relationships, integer surrogate keys and a conformed Date dimension. Hide technical columns and publish certified measures.

Model decision	Default	Escalate when
Fact grain	Lowest stable business event needed for decisions	Two processes need different grains; use separate facts
Relationships	One-to-many single direction from dimension	Legitimate bridge or role-playing date is required
Calculation	Measure for dynamic aggregation	Stored row attribute is required and stable

Model decision	Default	Escalate when
Security	RLS on dimensions	Object-level security or separate model is required
Refresh	Incremental refresh for large facts	Near-real-time composite design is justified
Release	Development test production pipeline	Emergency fix uses controlled hotfix path

Core quantity measures

```
Ordered Qty = SUM ( FactOrderLine[OrderedQty] )

Filled Qty =
SUMX ( FactOrderLine,
       MIN ( FactOrderLine[ShippedQty], FactOrderLine[OrderedQty] ) )

Fill Rate % = DIVIDE ( [Filled Qty], [Ordered Qty] )
```

Governed order grain OTIF

```
Delivered Orders = DISTINCTCOUNT ( FactOrder[OrderID] )

OTIF Orders =
CALCULATE (
  DISTINCTCOUNT ( FactOrder[OrderID] ),
  KEEPFILTERS ( FactOrder[IsOnTime] = TRUE () ),
  KEEPFILTERS ( FactOrder[IsInFull] = TRUE () )
)

OTIF % = DIVIDE ( [OTIF Orders], [Delivered Orders] )
```

Rolling demand inventory and forecast

```
Demand Qty L28D =
CALCULATE ( [Demand Qty],
  DATESINPERIOD ( 'Date'[Date], MAX('Date'[Date]), -28, DAY ) )

Days of Supply = DIVIDE ( [On Hand Qty], DIVIDE([Demand Qty L28D],28) )

WAPE % =
DIVIDE (
  SUMX ( FactForecast, ABS(FactForecast[ActualQty]-FactForecast[ForecastQty]) ),
  SUMX ( FactForecast, ABS(FactForecast[ActualQty]) )
)
```

Tail risk and predictive value

```
Supplier P95 Lead Time =
PERCENTILEX.INC (
  FactReceipt,
  DATEDIFF ( FactReceipt[PODate], FactReceipt[ReceiptDate], DAY ),
  0.95
)

Revenue at Risk =
SUMX ( FactOpenOrder,
  FactOpenOrder[OpenValue] * FactOpenOrder[LateProbability]
  * FactOpenOrder[ImpactFraction] )
```

DAX performance and correctness

Rule	Reason	Check
Use VAR for repeated expressions	Readable and avoids repeated evaluation	Inspect measure output under key filters
Prefer Boolean CALCULATE filters	Gives engine efficient options	Avoid FILTER over whole large fact when simple predicate works
Use DIVIDE	Safe denominator and blank behavior	Decide whether blank or zero is semantically correct
Avoid casual bidirectional or many-to-many	Ambiguous filters and poor performance	Test filter path with one member
Remove unused high-cardinality columns	Smaller model and faster scans	Model size and column cardinality
Do not sum snapshots across dates	Inventory is semi-additive	Use last date or time-weighted logic

Report design and control tower

Page	Questions	Required behavior
Executive	Service cash margin risk and top decisions	Only decision KPIs; each red state links to owner and deadline
Exception	Which orders items suppliers or lanes need action	Rank by expected avoidable loss and remaining action time
Root cause	When and where did the change start	Drill by cause, segment and transaction evidence
Operations	What must happen today	Owner, due time, current state and write-back status
Trust	Can the user rely on this	Refresh, source cutoff, reconciliation, lineage and data-quality status

Limit visuals per page, restrict large tables by default and test with Performance Analyzer. A control tower is a decision workflow, not a wall of charts.

Refresh security and release

1. Preserve Power Query folding and configure incremental refresh ranges for large transaction facts.
2. Put RLS rules on dimensions and test actual roles, including users with multiple roles and unmatched keys.
3. Promote through development, test and production. Keep parameters and connection bindings outside report logic.
4. Reconcile measures to approved source totals by representative slices. Test blanks, totals, date roles and daylight-saving boundaries.
5. Monitor refresh duration, failures, model size, capacity load, slow visuals and data freshness. Assign an owner and recovery runbook.
6. Publish metric definitions and certification status. Retire duplicate datasets and measures.

Symptom	Check in order
Same number for every supplier	Relationship active → unique dimension key → direction → measure removes filter
Grand total surprises	Ratio at total grain → SUMX context → expected weighted result
Blank dimension member	Orphan fact key → unknown-member policy → refresh order
Slow visual	Performance Analyzer → DAX query → cardinality/model → source folding → visual count
Old data	Dataset refresh → gateway/credential → source event time → upstream pipeline

SEE ALSO [DATA-01] sample data, p.12 • [SQL-01] source query, p.73 • [CH21] metric governance, p.37

[CH25] 25 Dashboards and root cause analysis

Read first	Plain guidance
In one sentence	Diagnose the driver behind a result, then automate safely.
Do this first	Reconcile totals, drill to transactions and separate signal from data failure.
What changes by company	Alert threshold, action authority, model version and guardrail.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.

Field panel	Execution guidance
Make the dashboard actionable	Show service, shortages, inventory, supplier delivery and capacity exceptions. Each exception needs an owner, due time and drill-through to the underlying transaction. Display the data cutoff and refresh time.
Diagnostic sequence	1. Confirm the metric definition and affected population. 2. Locate when the change started. 3. Split by item, site, supplier, customer and cause. 4. Inspect actual failed transactions. 5. Test the suspected cause against evidence. 6. Change the process and check recurrence.

Read first	Plain guidance
Choose the chart	Trend line: change over time. Pareto: largest recurring causes. Scatter: possible relationship between two measures. Aging bands: queues and overdue work. A correlation does not establish the cause.
Worked Pareto	100 late orders: material shortage 45, capacity 30, carrier 15, other 10. The first two causes account for 75%. Review the underlying orders: a production delay caused by missing material should not be counted twice.
Problem to action	Too many alerts -> no thresholds or ownership -> prioritize impact and timing -> review false alarms. Root cause says "human error" -> process not examined -> inspect task design, training and controls -> verify the prevention works. Metric improves but customers complain -> wrong denominator -> compare complaint orders with reported population -> revise the definition.
Action register	Problem | baseline | verified cause | containment | permanent action | owner | due date | result | recurrence check. Keep observations separate from assumptions.

[DBX-01] Databricks guided execution

Beginner run card

Question	Answer before you run
Goal	Run the sample as a read-only Databricks SQL query.
Need	Approved workspace, running SQL warehouse, catalog access and one known order.
Where	Workspace sidebar → SQL Editor → select SQL warehouse, catalog and schema.
Sample	Use northstar_ops.silver.orders and site CHI-DC.
Replace	Only {{CATALOG}}, {{SCHEMA}}, {{TABLE}} and quoted site value copied from Catalog Explorer.
Run	Highlight the SELECT query and choose Run.
Expected	C300 open 6; A100 open 5; B200 open 5.
Why	The fully qualified name chooses the object; WHERE filters; GROUP BY totals by item.
If wrong	Check compute, current catalog/schema, exact object name, role/grant, column types and real site values.

Names usually follow catalog.schema.table. In the workspace sidebar, open SQL Editor, select a running SQL warehouse and catalog/schema, paste a read-only query, then Run. In a notebook, attach compute and use a %sql cell. Workspace labels vary by account and release.

Filled example

```
SELECT
  item_id,
  SUM(ordered_qty - shipped_qty) AS open_qty
FROM northstar_ops.silver.orders
WHERE site_id = 'CHI-DC'
  AND ordered_qty > shipped_qty
GROUP BY item_id
ORDER BY open_qty DESC, item_id;
```

Change only	Example	Find it
catalog	northstar_ops	Catalog Explorer
schema/layer	silver	Catalog Explorer; governed curated layer
table	orders	Open table details and Preview
warehouse	Analyst Warehouse	SQL warehouse selector; use approved size
site value	CHI-DC	Preview DISTINCT site_id values

Change only	Example	Find it
Layer	**Purpose**	**Do**
Bronze	Faithful source landing	Append source, ingestion time, file/batch ID; keep raw evidence
Silver	Clean conformed records	Types, valid codes, deduplication, quarantine, business keys
Gold	Decision-ready product	Stable grain, metric definitions, dimensions and service expectations

ADVANCED — CHANGES DATA: the MERGE example below writes to a persistent table. Do not run it as a beginner or in production without a deduplicated test source, approval, reconciliation and recovery plan.

Advanced pattern — test in non-production; source must be one row per key

```
MERGE INTO northstar_ops.silver.orders AS t
USING northstar_ops.bronze.orders_clean AS s
ON t.order_id = s.order_id
WHEN MATCHED AND s.source_updated_at > t.source_updated_at THEN UPDATE SET *
WHEN NOT MATCHED THEN INSERT *;
```

Problem	Check
Table not found	Catalog/schema selection, three-part name, case and permission
Permission denied	USE CATALOG/SCHEMA and SELECT grants through approved role
MERGE matches multiple rows	Deduplicate source with ROW_NUMBER before MERGE
Slow/costly	Query profile, scanned data, filters, file size, skew, join and compute
Wrong rows after stream	Checkpoint, event time, late data, schema evolution and idempotency

[SNF-01] Snowflake guided execution

Beginner run card

Question	Answer before you run
Goal	Run the sample as a read-only Snowflake query.
Need	Approved role, warehouse, database/schema access and one known order.
Where	Snowsight → Projects → Workspaces → + Add New → SQL File.
Sample	Role ANALYST_ROLE; warehouse ANALYST_WH; database NORTHSTAR_OPS; schema SILVER.
Replace	Only {{ROLE}}, {{WAREHOUSE}}, {{DATABASE}}, {{SCHEMA}}, {{TABLE}} and real quoted filter values.
Run	Run the four approved USE statements first to set session context. Then select only the SELECT query and press Ctrl+Enter on Windows or Command+Return on Mac.
Expected	C300 open 6; A100 open 5; B200 open 5.
Why	Session context locates the object; the SELECT remains read only.
If wrong	Check current role/context, warehouse state, exact case/quotes, grants, columns and filter values.

Open Snowsight → Projects → Workspaces → + Add New → SQL File. Choose the approved role and warehouse, then the database and schema. Run selected SQL with Ctrl+Enter on Windows or Command+Return on macOS. Workspaces has replaced Legacy Worksheets. Snowflake commonly stores unquoted identifiers in uppercase; quoted mixed-case names must be referenced exactly.

Filled example

```
USE ROLE ANALYST_ROLE;
USE WAREHOUSE ANALYST_WH;
USE DATABASE NORTHSTAR_OPS;
USE SCHEMA SILVER;

SELECT
  item_id,
  SUM(ordered_qty - shipped_qty) AS open_qty
```

```
FROM orders
WHERE site_id = 'CHI-DC'
  AND ordered_qty > shipped_qty
GROUP BY item_id
ORDER BY open_qty DESC, item_id;
```

Name	Meaning	Example
Role	Permissions being used	ANALYST_ROLE
Warehouse	Compute that runs the query	ANALYST_WH
Database	Top-level data container	NORTHSTAR_OPS
Schema	Named group of objects	SILVER
Table/view	Rows to query	ORDERS

If	Then
Object exists but query says not found	Check role, current database/schema and quoted case; copy fully qualified name.
Warehouse suspended	Resume only if authorized; set auto-suspend and appropriate size.
Query spills or scans heavily	Open Query Profile; reduce columns/rows, fix joins and improve pruning.
Need repeated transformation	Choose dynamic table for declarative freshness, or streams/tasks for procedural control after verifying current feature limits.
Need safe test copy	Use zero-copy clone where policy permits; protect sensitive data and retention.
Need cost control	Resource monitors/budgets, auto-suspend, tags, workload isolation and cost-per-product review.

ADVANCED — CHANGES DATA OR OBJECTS: dynamic tables, pipes, streams, tasks and MERGE can create or change persistent objects. Require non-production proof, owner, cost limit, monitoring, reconciliation and recovery.

Data architecture from source to decision

Stage	Required output	Typical tool
Define	KPI, grain, owner, source, SLA, tolerance and action	Metric registry
Profile	Counts, nulls, duplicates, keys, ranges and drift	SQL and Power Query
Ingest	Immutable raw payload plus source and load metadata	Auto Loader or Snowpipe
Conform	Types, UOM, currency, timezone, keys, dedupe and quarantine	SQL Spark
Model	Facts, dimensions, snapshots and action-ready exceptions	Delta or Snowflake
Publish	Certified semantic model and reusable measures	Power BI
Automate	Schedule, retry, alert, backfill and write-back	Jobs Tasks workflow
Govern	Ownership, least privilege, masking, lineage and retention	Catalog and policies
Operate	Freshness, quality, runtime, cost and business-impact SLO	Platform observability

Carry source_system, source_record_id, source_updated_at, ingested_at, batch_id and a deterministic business key. Every pipeline needs replay, reconciliation, quarantine, owner, SLA and recovery. If both platforms exist, nominate one master for each gold dataset and KPI.

Databricks field guide

Layer	Purpose	Exit control
Bronze	Immutable payload and ingestion metadata	File or event traceable; schema drift visible
Silver	Typed standardized deduplicated reconciled data	Rejected rows quarantined; keys and units valid

Layer	Purpose	Exit control
Gold	Facts dimensions KPI aggregates and exceptions	Business owner signs grain and reconciliation
Unity Catalog	Ownership grants tags masks lineage and audit	Group-based least privilege
Lakeflow Jobs	Orchestration parameters retry alert and backfill	Idempotent task and recovery runbook

Auto Loader ingestion

```
(
  spark.readStream
       .format("cloudFiles")
       .option("cloudFiles.format", "json")
       .option("cloudFiles.schemaLocation", schema_path)
       .option("rescuedDataColumn", "_rescued_data")
       .load(source_path)
       .writeStream
       .option("checkpointLocation", checkpoint_path)
       .trigger(availableNow=True)
       .toTable("scm_bronze.order_events")
)
```

Idempotent Delta upsert

```
MERGE INTO scm_silver.inventory t
USING (
  SELECT * FROM (
    SELECT b.*, ROW_NUMBER() OVER (
      PARTITION BY item_id,site_id,source_event_id
      ORDER BY source_updated_at DESC,ingested_at DESC) rn
    FROM scm_bronze.inventory_events b WHERE batch_id=:batch_id
  ) WHERE rn=1
) s
ON t.item_id=s.item_id AND t.site_id=s.site_id
AND t.source_event_id=s.source_event_id
WHEN MATCHED AND s.source_updated_at>=t.source_updated_at THEN UPDATE SET *
WHEN NOT MATCHED THEN INSERT *;
```

Need	Practice
Performance	Select required columns, filter early, use built-in functions, inspect adaptive plan, handle skew, avoid collect on large data
Layout	Use managed Delta and current clustering guidance; avoid tiny files; run OPTIMIZE when workload evidence supports it
Incremental	Use change data feed for consumers; keep permanent audit separately because feed follows retention
Recovery	Repair reruns from task start; append-only custom code can duplicate partial output, so use MERGE or partition replacement
Governance	Grant groups and service principals; tag sensitive data; apply masks and row filters; preserve lineage
Cost	Monitor duration, input/output rows, shuffle, spill, skew, file count and cost per successful load

Snowflake field guide

Separate warehouses for ingestion, transformation, BI and data science when workloads differ. Use auto-suspend and auto-resume. Resource monitors cover user-managed warehouse credit usage; budgets cover broader serverless spend.

Continuous file ingestion

```
CREATE OR REPLACE PIPE raw.orders_pipe
  AUTO_INGEST = TRUE
AS COPY INTO raw.orders
FROM @raw.orders_stage
FILE_FORMAT = (TYPE = JSON);
```

Declarative latest-state pipeline

```
CREATE OR REPLACE DYNAMIC TABLE curated.order_latest
  TARGET_LAG = '15 minutes'
  WAREHOUSE = transform_wh
AS
SELECT * FROM raw.orders
QUALIFY ROW_NUMBER() OVER (
  PARTITION BY order_id
```

```
    ORDER BY source_updated_at DESC,ingested_at DESC)=1;
```

Stream-triggered merge

```
CREATE OR REPLACE STREAM raw.order_stream ON TABLE raw.orders;

CREATE OR REPLACE TASK transform.merge_orders
  WAREHOUSE = transform_wh
  WHEN SYSTEM$STREAM_HAS_DATA('RAW.ORDER_STREAM')
AS
MERGE INTO curated.orders t
USING (
  SELECT * FROM raw.order_stream
  QUALIFY ROW_NUMBER() OVER (
    PARTITION BY order_id ORDER BY source_updated_at DESC,ingested_at DESC)=1
) s
ON t.order_id=s.order_id
WHEN MATCHED THEN UPDATE SET status=s.status,source_updated_at=s.source_updated_at
WHEN NOT MATCHED THEN INSERT (order_id,status,source_updated_at)
VALUES (s.order_id,s.status,s.source_updated_at);

ALTER TASK transform.merge_orders RESUME;
```

Choose	When	Watch
Dynamic table	Declarative SQL pipeline with target freshness	Refresh mode, unsupported constructs and warehouse cost
Stream and task	Procedural MERGE, stored procedure, external call or custom orchestration	Stream staleness, task graph and retry
Materialized view	Repeated acceleration on one base table	Maintenance cost and eligibility
Snowpark	Governed Python Java or Scala data application	Pushdown, package controls and observability

1. Open Query Profile. Inspect rows, partitions scanned, queueing, local/remote spill and dominant operator.
2. Reduce rows and columns; fix joins and repeated calculations; confirm micro-partition pruning.
3. Resize up for a large compute-bound query; use multi-cluster for concurrency rather than one query.
4. Use clustering, search optimization, query acceleration or materialized views only after the base query is sound and benefit is measured.
5. Apply RBAC, row access, masking and tags. Separate policy ownership from data ownership.
6. Monitor idle warehouses, metering, serverless spend, failures, freshness and cost per data product.

Automation process mining and AI

Level	Capability	Release gate
Observe	Read governed data and explain status	Correct scope citations and access
Recommend	Rank actions with evidence confidence and expected value	Back-test, owner and workflow
Execute with approval	Prepare transaction; authorized person releases	Identity, segregation and post-check
Bounded autonomous	Execute reversible low-risk action inside limits	Value/quantity limits, idempotency, audit, kill switch and rollback

Keep vendor bank changes, payments, nonstandard contracts, export-control decisions, inventory write-offs and safety-critical releases under authorized human control.

Process mining event field	Minimum meaning
Case ID	One order shipment invoice return or other complete process instance
Activity	Governed business event name
Timestamp	Comparable event time with timezone rule
Actor or system	Responsible role or automation
Value and outcome	Business materiality and result
Exception code	Reason for deviation, rework or control path

Process mining loop

Discover actual variants → Confirm with operators → Remove waste → Standardize → Automate stable path → Recheck conformance

Future KPI generation

Generation	Question	Example
G1 Descriptive	What happened	OTIF was 91 percent
G2 Diagnostic	Why	Supplier delay caused 48 percent of misses
G3 Predictive	What will happen	$2.4M orders have greater than 60 percent late risk
G4 Prescriptive	What should we do	Reallocate stock and change carrier
G5 Bounded autonomous	Can the system act safely	Rebook within approved cost and service limits
G6 Adaptive	Which policy should change	Update safety stock after verified learning

1. Name the outcome and the decision that changes it.
2. Define failure mode, leading signal and action window.
3. Specify formula, grain, horizon, exclusions, units and authoritative fields.
4. Set an economic, statistical, contractual or compliance threshold by segment.
5. Back-test and reconcile. State confidence, model version and refresh time.
6. Pair the KPI with a guardrail that prevents gaming.
7. Assign owner, escalation, evidence and retirement rule.

Predictive or autonomous KPI	Formula or meaning	Guardrail
Expected net value of action	P(exception) × impact × mitigation effectiveness × actionability − action cost − execution risk	Safety and compliance override economics
Stockout risk 30 days	Simulations where demand exceeds available supply ÷ simulations	Calibration and scenario assumptions
Planning nervousness	Σ|new plan − prior plan| ÷ Σ prior plan inside frozen horizon	Service and constraint feasibility
Decision yield	Accepted actions with verified incremental benefit ÷ executed actions	Counterfactual and finance validation
Autonomous resolution	Eligible exceptions closed without human touch ÷ eligible	Override and escape rates
Constraint breach rate	Automated actions violating policy ÷ automated actions	Target zero for mandatory controls
Digital twin robustness	Scenarios meeting mandatory constraints ÷ tested scenarios	Simulation error and model expiry
Carbon adjusted cost to serve	Delivered cost plus approved carbon/compliance cost	Service and legal constraints

SEE ALSO [SQL-01] SQL logic, p.73 • [MIG-01] data migration, p.91 • [CH25] automation, p.82

[CH26] 26 Risk resilience and continuity

Read first	Plain guidance
In one sentence	Prepare for disruption before stock or time runs out.
Do this first	Compare time to recover with time to survive for each dependency.
What changes by company	Scenario, probability, alternate capacity and customer priority.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Map critical exposure	List single-source parts, concentrated sites, long lead times, key systems, utilities, transport lanes and specialist skills. Link each dependency to customer orders and recovery options.
Decision path	Disruption occurs -> protect people and follow site emergency procedures -> establish reliable facts -> assess usable stock and time to stockout -> activate approved recovery -> communicate validated commitments.

Read first	Plain guidance
Simple calculation	Time to stockout = usable uncommitted stock / average daily demand, if demand is stable and no supply arrives. 900 units / 100 per day = 9 days. If recovery takes 14 days, the planning gap is 5 days; use a daily time-phased balance when receipts or demand vary.
Continuity checklist	Named incident lead; deputies and contacts; alternate suppliers and routes; data backup and restore test; manual transaction process; stock allocation rule; customer update cadence; recovery exit criteria.
Problem to action	Backup supplier cannot start -> only commercially approved -> complete technical and capacity qualification -> run periodic test orders. Manual transactions lost during outage -> no controlled log -> number and reconcile records -> rehearse recovery posting. All alternatives share one sub-tier -> hidden concentration -> map critical sub-tiers -> diversify actual exposure.
Control	Prioritize risks using impact and plausible recovery time, not a score alone. Rehearse one realistic scenario and record decisions, failures and improvements. Verify restoration before closing the incident.

Veteran practice note

Panel	What disciplined operators do
Veteran operating method	Map dependencies to customer and cash outcomes. Quantify time to survive, time to recover, alternate capacity and expected loss. Rehearse scenarios that cross suppliers, sites, systems, people and transport.
Decision rules	TTR ≤ TTS → monitor recovery. TTR > TTS → close the gap with inventory, alternate source, substitute, capacity, route or demand action. Correlated risks require scenario analysis, not independent score addition.
Control	Backups count only when technically qualified, contractually available and periodically tested. Record manual transactions during outages and reconcile them after restoration.

[CH27] 27 Greenfield implementation gates

Read first	Plain guidance
In one sentence	Build a new operation through evidence-based gates.
Do this first	Name outcome, scope, sponsor, owners, dependencies, acceptance and no-go rules.
What changes by company	Deployment model, site, process, systems and regulatory scope.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Gate 1 Define	Approve scope, customer promise, volumes, sites, budget, owners and success measures. Output: charter and requirement register. Do not buy systems before defining operating needs.
Gate 2 Design	Approve process maps, layout, organization, controls, data model and system fit. Output: signed design and gap decisions. Test high-volume and exception flows on paper.
Gate 3 Build	Configure systems; recruit and train; qualify suppliers; install and validate infrastructure; load clean master data. Output: configured process, training records and controlled data loads.
Gate 4 Prove	Execute end-to-end tests: purchase to pay, make to stock, order to cash, returns, traceability and outage recovery. Output: test evidence, defect log and reconciled balances.
Gate 5 Launch	Approve cutover, ownership, rollback decision point and support coverage. Output: go/no-go record and controlled opening stock. No unresolved critical safety, legal or transaction-integrity defect.

Read first	Plain guidance
Gate 6 Stabilize	Review service, inventory accuracy, backlog, system defects and support demand daily. Exit only after agreed performance and ownership conditions are met.
Problem to action	Launch date fixed, readiness weak -> schedule overriding evidence -> escalate gate exceptions and recovery options -> require named risk acceptance. Scope grows -> no change control -> assess time/cost/benefit -> maintain approved baseline.
Workbook row	Gate | deliverable | accountable owner | dependency | planned date | actual date | evidence link | pass/fail | issue | decision. Each gate needs evidence, not a presentation-only status.

[CH28] 28 Cutover testing and data migration

Read first	Plain guidance
In one sentence	Move clean data and operations into production without losing control.
Do this first	Load by dependency, reconcile count/quantity/value and rehearse rollback.
What changes by company	Migration object, mapping, cutoff, tolerance and environment.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Migration sequence	Extract -> clean -> map -> validate -> trial load -> reconcile -> approve -> freeze -> final delta load -> reconcile again. Assign owners for master data, opening stock, open POs, sales orders and balances.
Test the hard cases	Partial receipt; over/under delivery; rejected lot; unit conversion; split shipment; cancelled line; credit hold; expired stock; return; invoice mismatch; interface retry; outage and recovery. Verify permissions and audit history.
Decision path	Balance mismatch? Stop dependent release; compare source-to-target counts, quantities and value at the same cutoff. Interface retry? Check transaction IDs before replaying to avoid duplicates. Critical defect? Apply agreed no-go or rollback criteria.
Worked reconciliation	Source stock: 10,000 each valued at $50,000. Target: 10,000 each at $48,000. Quantity passes; value fails by $2,000. Check cost method, currency, rounding and excluded costs with Finance before signoff.
Cutover checklist	Freeze time; transaction cutoff; backups; extraction owner; load sequence; validation queries; approval contacts; business restart; support rota; rollback deadline; recovery transaction log.
Problem to action	Opening stock changes during load -> movements not frozen or captured -> reconcile delta transactions -> enforce cutoff control. Test passes in isolation, fails end-to-end -> handoffs omitted -> test real scenarios across teams -> require joint signoff.
Test record	Scenario | inputs | expected outcome | actual outcome | evidence | tester | defect ID | retest | owner approval. Passing means matching the expected business result.

[MIG-01] Migration, implementation and cutover with no loose handoff

Migration evidence chain



Read the arrows from source scope to stable operation. Do not advance because a file loaded; advance only when counts, quantities, values, identities and business transactions are proved.

Project word	Plain meaning
Sandbox	A safe learning area that is not trusted as production.
Development	Where configuration or code is built.
Test	Where integrated business scenarios are proved with controlled data.
Production	The live system that runs real operations and accounting.
Configuration	System settings that control allowed process behavior.
Interface	A controlled connection that exchanges data between systems.
Fit-to-standard	Use the standard process unless a proved requirement justifies a change.
Cutover	The timed move from old operation to the new production operation.
Hypercare	Short, intensive support after launch.
BAU	Business-as-usual ownership after the project stabilizes.

Migration term	Plain meaning	Required proof
Object	A business thing moved together	Product, supplier, stock or open order
Source	Where the old value comes from	System/table/file and extraction cutoff
Target	Where the new value goes	Object/field and target environment
Mapping	Rule connecting source to target	Approved crosswalk with owner
Transformation	How a value is changed	EACH → EA; DAL → 1710
Default	Approved value when source lacks one	Reason, owner and affected count
Reject	Record that fails a mandatory rule	Error code, owner and disposition
Reconciliation	Proof nothing material was lost or changed	Count, quantity, value and sampled identity

Typical load order	Why
1 Organization/configuration	Later objects need valid company, plant, currency, UOM, tax and calendar rules.
2 Finance structures	Accounts, cost/profit centers and projects support valuation and postings.
3 Business Partners	Suppliers and customers are parents of open documents.
4 Products/materials	Orders, stock and BOMs need valid item identities.
5 Sources/BOM/routing/warehouse data	Planning and execution relationships follow their parent records.
6 Opening inventory	Stock identity, status, lot/serial, quantity and value must reconcile.

Project word	Plain meaning
7 Supported open transactions	Open PO, SO, production, AR/AP and other scope by approved approach.
8 Balances and deltas	Final financial/open-item position and changes since last extract.

SAP migration beginner path

For a new SAP S/4HANA implementation, start by checking whether the SAP Migration Cockpit supports the required migration object for the chosen edition and release. Use a test project and repeated mock loads. Supported scope commonly includes master data, open transactions and balances; closed historical process data usually needs reporting retention or another approved approach.

SAP Activate

Discover → Prepare → Explore → Realize → Deploy → Run

1. Open the Fiori app Migrate Your Data – Migration Cockpit when available for the edition and role.
2. Create a non-production test project; select migration objects and read their dependency notes.
3. Fill staging/file/direct-transfer source using the exact template and code rules.
4. Validate, correct errors, simulate when supported, migrate and retain logs.
5. Reconcile with business owners; repeat until full-volume timing and tolerances are proven.
6. Use a controlled production project and signed cutover sequence only after readiness approval.

Mock cycle	Purpose	Exit
Mock 0	Prove tool/template with small sample	A few records load and can be reversed/repeated safely
Mock 1	Validate mapping and dependencies	Major master objects load with owned errors
Mock 2	Full volume, performance and reconciliation	Counts/quantities/values within tolerance
Dress rehearsal	Execute real sequence and roles	Fits downtime; rollback and evidence proven
Production	Controlled final migration	Signed technical and business reconciliation

Cutover decision	Numeric example
Inventory reconciliation	No unexplained quantity/value difference; approved immaterial rounding only
Critical processes	One complete P2P, O2C and make/ship smoke test per required site
Interfaces	Sent = received + known/recoverable failures; no duplicate postings
Security	Critical roles work; prohibited combinations and emergency access controlled
Rollback	Trigger, latest safe point, restore duration, transaction capture and approver known

Greenfield to closeout execution

Gate	Required evidence	No go trigger
Mobilize	Charter sponsor scope value governance resources	No accountable sponsor or measurable outcome
Discover	Signed baseline process maps site facts data profile pain value	Critical process or data source unknown
Design	Future process controls RACI data architecture signed decisions	Unresolved critical design or compliance issue
Build	Configuration integration reports roles migration training	Core transaction path incomplete
Test	Unit SIT UAT performance security recovery reconciliation	Critical scenario not proven
Readiness	Defect status reconciled data trained roles continuity rollback	Severity one defect or unsigned reconciliation
Go live	Command center controlled cutover opening balances validation	Cutover control fails or rollback point reached

Gate	Required evidence	No go trigger
Hypercare	Severity management KPI review defect burn-down adoption	Service unstable without credible recovery
Stabilize	Sustained tolerances controls ticket aging BAU ownership	Key owner or control still temporary
Close	Benefit owner audit pack lessons residual backlog acceptance	Benefits and BAU obligations unowned

Workstream dependency map

Workstream	Inputs	Outputs	Critical dependency
Process	Strategy volumes constraints	Future flow controls SOP	Operating model and system design
Organization	Process decisions workload	Roles RACI training support	Security and readiness
Data	Source inventory definitions	Clean master migration KPI	Configuration integration reporting
Technology	Requirements architecture	Configured platforms interfaces	Data test cutover
Facilities	Volumes SKU and flow profile	Layout equipment capacity	Process safety go live
Supplier and logistics	Network volumes specs	Contracts capacity lanes	Material and launch
Change	Stakeholders roles impacts	Communication training adoption	UAT and hypercare
Finance and controls	Business case risk policy	Baseline approvals reconciliation	Gate and benefit signoff

Design build and test sequence

1. Translate outcomes into process and control requirements with measurable acceptance criteria.
2. Confirm fit to standard. Log every gap, option, decision, owner, cost, risk and downstream dependency.
3. Freeze solution design by wave. Control changes through impact assessment and authority.
4. Build configuration, integration, reports, roles, migration and training as one end-to-end scenario.
5. Test normal, exception, volume, permission, failure, recovery and reconciliation cases. Retest after fixes.
6. Run at least one mock cutover with real sequence, owners, timing, controls and rollback decision.
7. Operate the command center on customer and transaction outcomes, not ticket counts alone.
8. Exit hypercare only when service, defects, controls, adoption, documentation and BAU ownership are stable.

Test layer	Purpose	Evidence
Unit	Component works as designed	Expected and actual result
SIT	Systems exchange correct complete transactions	Message IDs retry and reconciliation
UAT	User completes business outcome and controls	Signed scenario evidence
Performance	Peak volume and response meet need	Workload result bottleneck and capacity
Security	Access and segregation enforce policy	Role test and audit log
Recovery	Failure restores without loss or duplicate	RTO RPO replay and reconcile
Cutover rehearsal	People sequence data and timing work	Minute plan issue log and improvements

Implementation workbooks inside the book

Copy these column sets into controlled spreadsheets or project tools. Add document ID, version, owner, approval and evidence link to every register.

Workbook	Minimum columns
Charter and value case	Outcome | baseline | target | scope | exclusions | sponsor | owner | cost | benefit | timing | risks
Stakeholder RACI	Process | decision | accountable | responsible | consulted | informed | cadence
RAID	ID | type | statement | impact | probability | response | owner | due | trigger | status

Workbook	Minimum columns
Decision log	ID | question | options | evidence | decision | approver | date | conditions | revisit trigger
Process inventory	Process | trigger | input | steps | output | customer | owner | system | control | KPI
Requirements traceability	Req ID | need | acceptance | design | config | test | result | defect | signoff
Fit gap and design	ID | requirement | standard fit | gap | options | cost | risk | decision | dependency
Data dictionary	Object | field | definition | grain | source | type | rule | owner | sensitivity | lineage
Migration mapping	Source | target | transform | default | reject rule | control total | owner | signoff
KPI dictionary	KPI | decision | formula | grain | inclusions | exclusions | source | owner | frequency | action
Supplier selection	Criterion | weight | evidence | score | weighted score | risk | evaluator | approval
Negotiation matrix	Issue | must | target | fallback | walk away | give | get | authority | final
Test script	Test ID | requirement | precondition | steps | expected | actual | evidence | defect | result
Defect log	ID | severity | scenario | symptom | cause | owner | fix | build | retest | status
Cutover plan	Time | task | predecessor | owner | checker | evidence | status | issue | rollback trigger
Readiness	Area | criterion | evidence | owner | target | status | gap | recovery | approval
Hypercare	Issue | impact | case | owner | action | next update | ETA | workaround | verification
Benefits	Benefit | baseline | method | target | actual | counterfactual | owner | finance signoff | date
BAU handoff	Service | process | owner | SOP | access | support | KPI | control | open item | acceptance

Document and calculation registries

Registry	Required fields
Document	Doc ID | title | process | owner | approver | revision | effective date | storage link | review date | retention
Calculation	Calc ID | purpose | formula | units | grain | source | assumptions | owner | test case | revision
Interface	Interface ID | source | target | object | trigger | volume | SLA | retry | reconcile | owner
Automation	Automation ID | decision | system | identity | action limit | approval | idempotency | rollback | monitor
Model	Model ID | use | version | training period | features | metric | threshold | owner | drift | retirement

Completed workbook examples before the blank row

Workbook	Completed example	Editable next row
Issue log	ISS-014 | Inventory mismatch | owner Lee | contain by 13:00 | open | count evidence linked	ISS-{{ID}} | {{ISSUE}} | {{OWNER}} | {{ACTION / TIME}} | {{STATUS}} | {{EVIDENCE}}
Decision log	DEC-006 | choose split shipment | approver Kim | decided 10:30 | review after delivery	DEC-{{ID}} | {{QUESTION}} | {{CHOICE}} | {{APPROVER}} | {{TIME}} | {{REVIEW}}

Workbook	Completed example	Editable next row
KPI dictionary	KPI-OTIF | order | on time and complete / eligible | TMS+ERP | daily | service owner	KPI-{{ID}} | {{GRAIN}} | {{FORMULA}} | {{SOURCE}} | {{FREQUENCY}} | {{OWNER}}
Migration map	MAP-021 | legacy plant DAL → target plant 1710 | owner Priya | test passed	MAP-{{ID}} | {{SOURCE}} → {{TARGET}} | {{OWNER}} | {{TEST RESULT}}
Test script	TST-032 | receive PO 4500001234 | expect GR document + inventory + accounting | passed	TST-{{ID}} | {{STEPS}} | {{EXPECTED}} | {{ACTUAL}} | {{STATUS}}
Cutover plan	CUT-018 | freeze inventory movement | 22:00 | owner Sam | checker Mia | rollback if variance	CUT-{{ID}} | {{TASK}} | {{TIME}} | {{OWNER}} | {{CHECKER}} | {{ROLLBACK}}
BAU handoff	BAU-009 | daily failed-interface review | owner Data Ops | SLA 08:00 | runbook linked	BAU-{{ID}} | {{SERVICE}} | {{OWNER}} | {{SLA}} | {{RUNBOOK / OPEN ITEM}}

SEE ALSO [SAP-01] SAP, p.38 • [CH27] gates, p.89 • [CH28] migration, p.90 • [CH29] registers, p.95

[CH29] 29 Reusable registers and workbooks

Read first	Plain guidance
In one sentence	Keep decisions, risks, tests and evidence in reusable registers.
Do this first	Use a stable ID, owner, date, status and evidence link for every row.
What changes by company	Tool, naming convention, approval and retention period.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.

Field panel	Execution guidance
Document registry	Doc ID | title | process/chapter | owner | approver | revision | effective date | storage link | review date. Starter records: specification, supplier approval, PO procedure, count procedure, traceability procedure, contract, test plan and cutover plan.
Calculation registry	Calc ID | purpose | formula | input units | data source | assumptions | owner | validation case | revision. Starter records: WAPE (04), reorder point/EOQ (06), TCO (08), labor hours (11), OEE (18), landed cost (20), cash cycle (21).
Issue and action log	ID | problem | priority and impact | containment | root-cause evidence | action | owner | due date | status | verification. Status choices: open, contained, fixing, verifying, closed.
Requirement and test tracker	Requirement ID | business need | acceptance criterion | design reference | test ID | result | defect | signoff. Example: R-014 requires blocked lots to be unpickable; T-014 attempts a pick and must show a block with audit evidence.
Training matrix	Role | process | required competence | trainee | trainer | demonstration date | pass/retrain | refresher due. Attendance alone does not demonstrate competence.
Risk and decision logs	Risk: dependency | failure | impact | early indicator | mitigation | owner | review date. Decision: question | alternatives | evidence | chosen option | approver | date | revisit trigger.
Use in Excel	Copy each register to a separate tab, make it a Table, freeze the header, filter by owner and overdue date, and use controlled status lists. Keep evidence links stable and avoid sensitive data in broadly shared trackers.

[CH30] 30 Quick problem index and escalation

Read first	Plain guidance
In one sentence	Choose the shortest safe path from symptom to verified recovery.
Do this first	Protect, verify, quantify, contain, find cause, correct, prevent and monitor.

Read first	Plain guidance
What changes by company	Severity, response time, owner, escalation and closure evidence.
Proof before closure	Reconcile the affected population, preserve evidence, name the owner and monitor recurrence.
Field panel	**Execution guidance**
Find the right page	**Stockout, excess or forecast:** [CH04] p. 17 • [CH06] p. 19 • [CH17] p. 34 **Wrong quantity, UOM or receipt:** [CH03] p. 16 • [CH12] p. 28 • [CH13] p. 28 **Supplier late or defective:** [CH07] p. 20 • [CH09] p. 21 • [CH19] p. 35 **Contract or SLA:** [CH10] p. 22 • [CH16] p. 31 **Warehouse congestion or short pick:** [CH11] p. 27 • [CH14] p. 29 • [WH-01] p. 32 **Freight cost or border delay:** [CH15] p. 30 • [CH20] p. 36 **Capacity gap:** [CH05] p. 18 • [CH18] p. 34 • [OPT-01] p. 67 **Cash or cost variance:** [CH21] p. 37 • [FIN-01] p. 53 **Excel, SQL or dashboard:** [XL-01] p. 69 • [SQL-01] p. 73 • [PBI-01] p. 79 • [CH25] p. 82 **Outage, launch or migration:** [CH26] p. 88 • [CH27] p. 89 • [CH28] p. 90 • [MIG-01] p. 91 **Tracker or evidence register:** [CH29] p. 95
First 15 minutes	Protect people and contain affected product or transactions. Confirm the actual order, item, quantity, location and deadline. Separate verified facts from assumptions. Assign one owner. Agree the next update time and the evidence needed.
Escalation decision	Safety, regulatory or serious quality concern -> responsible specialist immediately under the company procedure. Customer commitment at risk -> Planning/Customer Service plus the process owner. Material financial or contractual exposure -> Finance/Legal through the authority matrix. System integrity issue -> IT and business owner.
Message template	"Issue: {{WHAT FAILED}}. Impact: {{ORDERS / UNITS / DATE}}. Confirmed facts: {{EVIDENCE}}. Containment: {{ACTION}}. Options: {{COST / TIMING}}. Decision needed: {{SPECIFIC CHOICE}} by {{TIME}}. Owner: {{NAME}}. Next update: {{TIME}}."
Close properly	Confirm affected stock and transactions are reconciled, commitments updated and temporary workarounds removed or controlled. Record the verified cause and prevention. Check recurrence after a suitable operating period.
Boundary	This is a compact general field reference. Apply the company process and industry-specific requirements where product safety, regulated goods, customs or contractual obligations need specialist judgment.

Formula symbol decoder

Symbol or name	Plain meaning
A	Actual observed quantity.
F	Forecast quantity.
D	Annual demand in units.
S	Cost to place one order.
H	Annual holding cost for one unit.
z	Service factor chosen from the approved service target.
sigma d	Variation of demand per day.
L	Lead time in days.
d-bar	Average demand per day.
sigma L	Variation of lead time.
Cu	Incremental shortage cost.

Symbol or name	Plain meaning
Co	Incremental overage cost.
Σ	Add every value in the stated population.
√	Square root.

Read a formula in words before using symbols. Example: reorder point = average daily demand × average lead time + safety stock. Confirm units, population, time basis and assumptions before calculating.

Formula and KPI registry

Metric	Compact formula	Grain	Owner	Guardrail
OTIF	orders on time and in full ÷ eligible orders	order	Service owner	Perfect order and cause
Perfect order	orders time complete damage free document correct ÷ eligible	order	Customer operations	Complaint and margin
WAPE	Σ|actual − forecast| ÷ Σ|actual|	item location horizon	Demand planning	Bias and service
Bias	Σ(forecast − actual) ÷ Σactual	item location horizon	Demand planning	WAPE and inventory
Forecast value add	baseline error − post process error	planning stage	Planning lead	Override volume
Inventory turns	annualized COGS ÷ average inventory at cost	entity period	Supply chain finance	Service and excess
Days inventory	average inventory ÷ annualized COGS × days	entity period	Supply chain finance	Aging and risk
Excess exposure	Σ max(supply − horizon demand − safety stock, 0) × cost	item location	Inventory owner	Expiry and cancelability
Schedule adherence	orders completed in approved window and quantity ÷ due	order	Manufacturing	Output and WIP
OEE	availability × performance × quality	asset shift	Operations	Constraint throughput
Supplier OTD	eligible lines on or before contractual date ÷ due lines	supplier site line	Procurement	Total business OTD
Supplier PPM	defective units ÷ inspected units × 1,000,000	supplier item	Supplier quality	Escape and severity
Cash to cash	DIO + DSO − DPO	entity period	Finance	Service and margin
Expected disruption loss	Σ probability × scenario loss	scenario horizon	Risk owner	TTR TTS and correlation
Resilience gap	max(0 TTR − TTS)	dependency scenario	Continuity owner	Alternate readiness
Revenue at risk	Σ affected revenue × probability × impact fraction	order horizon	Control tower	Margin at risk
Mean time to decide	approved decision time − alert time	exception	Process owner	Outcome and adoption
Exception precision	actionable true alerts ÷ all alerts	model rule	Analytics owner	Recall and missed loss
Decision yield	executed recommendations with verified benefit ÷ executed	recommendation	Product owner	Adoption and regret
Touchless rate	no human touch eligible transactions ÷ eligible	transaction	Automation owner	Escape and override
Data freshness compliance	refreshes inside SLA ÷ scheduled refreshes	data product	Data owner	Reconciliation

Metric	Compact formula	Grain	Owner	Guardrail
Model calibration	outcomes inside stated interval ÷ predictions	model horizon	Model owner	Cost sensitive error
Scenario robustness	scenarios meeting mandatory constraints ÷ tested	decision	Planning owner	Regret and cost
Carbon intensity	tCO2e ÷ chosen activity unit	lane or product	Sustainability	Service and cost

Planning inventory and capacity formulas

Formula	Expression	Worked check and warning
WAPE	Σ|A−F| ÷ Σ|A|	Actual 100 and 120; forecast 110 and 100 → 30÷220=13.64 percent. Undefined when total actual is zero.
Bias	Σ(F−A) ÷ ΣA	Same data → −10÷220=−4.55 percent. State sign convention.
MASE	MAE ÷ seasonal naive MAE	Comparable across scales; denominator fails for perfectly flat naive history.
Safety stock fixed LT	z × σd × √L	z 1.645, σd 20, L 9 → 98.7 units. Assumes independent normal demand and fixed lead time.
Safety stock variable LT	z√(L̄σd² + d̄²σL²)	Units must match; independence assumption can understate correlated risk.
Reorder point	d̄ × L̄ + safety stock	Use inventory position and avoid subtracting demand twice.
EOQ	√(2DS÷H)	D 12000, S 50, H 3 → 632.46. Adjust for packs, MOQ, shelf life and constraints.
Newsvendor ratio	Cu ÷ (Cu + Co)	Translate shortage and overage economics to target quantile; costs must be incremental.
Little law	WIP = throughput × flow time	All values need same stable boundary and time units.
Takt	available production time ÷ customer demand	Do not include planned downtime as available time.
OEE	availability × performance × quality	80 percent × 90 percent × 95 percent = 68.4 percent; validate standards.
Capacity cushion	1 − required hours ÷ available hours	Negative value means overload; available hours must reflect real constraints.

Commercial logistics and cash formulas

Formula	Expression	Use carefully
Total landed cost	price + freight + duty + brokerage + insurance + handling + inventory + quality/risk	Avoid double counting amounts already in Incoterm price.
PPV	(actual unit price − standard) × actual quantity	Separate currency, volume, mix and index effects.
Payment term cash	annual spend × change in days ÷ 365	Cash timing is not automatically P&L savings.
Cost to serve	order + handling + inventory + transport + return + service costs	Reconcile allocation to finance totals.
Cash to cash	DIO + DSO − DPO	Use consistent periods and valuation.
Freight intensity	freight cost ÷ shipment, unit, weight, revenue or margin	Choose denominator that reflects the decision.
Carbon	activity quantity × approved emission factor	Record factor source, scope, geography and version.

[PB-01] Problem → root cause → action → prevention playbook

Use [FIND-01] to choose a route. Use this table after you have one real case. The cause column contains possibilities to test, not conclusions.

Operations and supplier problems

Symptom	Possible causes to test	Ten-minute test	Contain today	Prevent
High inventory and poor availability	Wrong mix/site/status, MOQ inflation, bias, blocked stock	Segment item-location; reconcile usable stock and shortage list	Reallocate/substitute; defer supply	Policy governance and multi-echelon review
Repeated stockouts	Demand censoring, late supply, bad lead time, allocation	Time-phased balance and pegging for affected demand	Expedite/allocate/ resequence	Service-cost inventory policy and supplier recovery
Forecast WAPE worsens	Data break, lifecycle, promotion, regime change	Compare by vintage, horizon, segment and actual completeness	Use approved baseline and exception override	FVA and event-data governance
Negative forecast value add	Biasing overrides, incentives, no expiry	Compare baseline error with each override stage	Remove damaging override	Reason-coded overrides and accountability
MRP creates implausible orders	UOM, BOM, inventory status, lead time, MOQ, duplicate demand	Trace one recommendation through pegging and master data	Hold release and correct approved parameter	Master-data tests and exception thresholds
Schedule changes every day	Unstable plan, poor time fence, constraint not modeled	Measure frozen-window change and source	Freeze critical horizon and protect constraint	Govern time fences and feasibility
Supplier repeatedly late	Overcommitment, material, quality, logistics, payment	Verify WIP, finished stock, daily rate and pickup	Milestone recovery and allocation	Capacity qualification and chronic-failure remedies
Supplier defect spike	Process change, tooling wear, material lot, inspection escape	Contain lots and compare change point	Sort/quarantine and approved alternate	SCAR with effectiveness proof
PO receipt invoice mismatch	Timing, unit, price, tax, freight or duplicate	Reconcile one line across PO receipt return invoice	Hold disputed amount only	Contract/catalog and tolerance controls
Production shortage	Demand change, missing issue, supplier delay, yield loss	Peg shortage to order and earliest feasible recovery	Resequence/substitute/ overtime	Constraint and critical-material control
Low OEE	Availability performance or quality loss	Rebuild OEE from event-level time and count	Protect constraint and attack largest loss	Standard reason codes and reliability plan

Transport, data and implementation problems

Symptom	Possible causes to test	Ten-minute test	Contain today	Prevent
Traceability break	Missing lot event, rework, interface or label error	Stop release and reconcile genealogy population	Widen containment	Scan enforcement and mock recall
Inventory variance	Timing, location, UOM, status, unrecorded movement	Blind recount and last-known-good transaction trace	Controlled adjustment and containment	Cycle count by cause and scan discipline
Warehouse congestion	Peak mismatch, slotting, staging, blocked stock	Map queue by hour and zone	Stagger/re-slot/add controlled staging	Engineered peak design and appointment control
OTIF collapse	Promise logic, shortage, pick, carrier, customs or data	Trace failed orders at order grain	Protect urgent orders and reset promises	Cause-owned leading indicators
Freight cost spike	Rate, fuel, mode, distance, cube, accessorial, expedite	Bridge current to baseline by driver	Consolidate/rebook/dispute invalid charges	Lane strategy and tender discipline
Customs hold	Document, classification, value, origin, license, inspection	Get exact hold reason and owner	Correct evidence or specialist escalation	Pre-clearance and governed trade data
Dashboard totals conflict	Grain, date, status, currency, UOM, mapping, refresh	Reconcile one transaction and control totals	Label or withdraw unreliable view	Certified metric layer and tests

Symptom	Possible causes to test	Ten-minute test	Contain today	Prevent
Pipeline refresh fails	Source, schema, credential, quota, bad data	Locate first failed layer and last successful batch	Quarantine/retry/backfill safely	Contracts, idempotency and observability
Slow or costly analytics	Full scan, fanout, skew, poor model, concurrency	Inspect plan/profile and rows at each step	Filter/aggregate, fix join, right-size	Performance budgets and workload isolation
Go-live reconciliation fails	Cutoff, mapping, duplicate, delta or cost method	Stop release and reconcile count quantity value	Correct/reload or invoke rollback	Mock cutover and signed tolerances
Low adoption after go-live	Workflow, role, training, trust, access or benefit gap	Observe users completing real cases	Remove blockers and coach by role	Adoption measures and process ownership

Incident message and closure

Issue: {{WHAT FAILED}}. Impact: {{ORDERS / UNITS / VALUE / DEADLINE}}. Confirmed facts: {{EVIDENCE / CUTOFF}}. Containment: {{CURRENT ACTION}}. Options: {{COST / TIME / SERVICE / RISK}}. Decision needed: {{SPECIFIC CHOICE}} by {{TIME}}. Owner: {{NAME}}. Next update: {{TIME}}.

Close only after affected stock and transactions reconcile, commitments are updated, temporary workarounds are removed or controlled, the cause is evidenced, prevention is implemented and recurrence is checked through a suitable operating period.

Road map	What to look at	How to look
Order late	Promise, ATP, supply, allocation, pick, carrier, customs	Trace one order by event timestamp and owner
Stockout + excess	Item-site mix, status, lead time, MOQ, bias, blocked stock	Segment and build a time-phased balance
Inventory mismatch	Last receipt/move/pick/count, UOM, lot, interface	Blind recount then transaction genealogy
Cost spike	Rate, volume, mix, currency, fuel, distance, accessorial	Bridge each driver from baseline to actual
Dashboard conflict	Grain, date, status, join, currency, UOM, refresh	Reconcile one row, then control totals
Go-live failure	Cutoff, mapping, duplicate, role, interface, balance	Stop posting; reconcile and invoke tested recovery

SEE ALSO [ROAD-01] universal road map, p.14 • [DATA-01] data words, p.12 • [MIG-01] go-live, p.91

[RES-01] Research-to-practice pathway

Research-to-practice path



Follow the arrows from an observed problem to a repeatable result. Each step must preserve the population, definitions, evidence and limitations needed for another analyst to reproduce the work.

Research word	Plain meaning
Variable	A value that can differ, such as lead time, price or service.
Unit of analysis	The thing represented by one observation: order, line, item-site, supplier or day.
Sample	The observations actually studied from the full population.
Bias	A systematic difference that can push a result in one direction.
Validity	Whether the method measures and supports what the claim says.

Research word	Plain meaning
Robustness	Whether the conclusion survives reasonable alternative assumptions or methods.
Replication	Enough data rules, code and method detail for someone else to reproduce the result.

Stage	Work product	Quality check
Problem	Observed operational gap with population, period and consequence	Evidence before explanation
Question	One answerable relationship, comparison or design question	Variables, unit of analysis and scope are clear
Concepts	Definitions and a causal/process model	Terms mean the same thing across sources
Evidence	Structured literature and practice map	Search, inclusion and source quality are reproducible
Data	Dictionary, lineage, permissions and missing-data plan	Grain, bias and measurement validity are known
Method	Case study, survey, experiment, archival analysis, simulation, optimization or mixed method	Method matches the question
Analysis	Transparent calculations, uncertainty and robustness tests	Alternative explanations tested
Decision	Operational implication, limits and implementation path	Claim does not exceed evidence
Replication	Code/query, data rules, version, parameters and audit trail	Another analyst can reproduce the result

Turn a job problem into a researchable question

Job observation	Weak question	Stronger question
We carry too much stock.	How can inventory be reduced?	How does item-location service segmentation change fill rate and working capital compared with one common safety-stock policy?
Suppliers are late.	Why are suppliers bad?	Which early production and logistics signals predict supplier-line lateness seven days before promise?
Forecast overrides hurt.	Is AI better?	At which product lifecycle stages do human overrides improve or worsen WAPE and bias relative to a frozen baseline?
Warehouse travel is high.	What layout is best?	How does ABC-XYZ slotting change pick distance, replenishment work and congestion during peak mix?
Dashboards do not drive action.	How can BI improve?	Does owner-linked exception routing reduce mean time to decide and service loss compared with passive dashboards?

Literature matrix column	What to record
Citation and setting	Author/year, industry, country, organization type
Question and theory	Claim, mechanism, framework and assumptions
Method and sample	Design, unit, period, data and sample selection
Measures	Operational definitions, reliability and data source
Finding	Effect, uncertainty and boundary conditions
Limitation	Bias, missing variables, generalizability and unresolved question
Use in your study	Support, contrast, method choice or construct definition

Promising emphasis	Possible outcome and method
Forecast value add	Outcome: error/bias; archival forecast vintages + interviews

Job observation	Weak question	Stronger question
Supplier resilience	Outcome: TTR/service loss; network scenarios + case study	
Inventory and service	Outcome: fill rate/cash; simulation or controlled policy comparison	
Digital twin	Outcome: scenario robustness; validation against actual events	
AI exceptions	Outcome: decision speed/yield; back-test + human factors	
Sustainable logistics	Outcome: service/cost/emissions; multi-objective optimization	
Warehouse technology	Outcome: errors/travel/adoption; before-after with process controls	
Last-mile design	Outcome: promise/cost/failed delivery; geospatial or route analysis	

Research discipline: preserve a frozen raw extract, a data dictionary, transformation/query version, analysis parameters, exclusion log, output table and decision memo. State what would change your conclusion.

[INDEX-01] A–Z keyword and problem index

Print: read down the left keyword column, then down the right keyword column; turn to the live page number. Digital copy: click the route or search the exact keyword. If the word is missing, open [FIND-01] and choose the closest symptom.

Keyword	Route and page	Keyword	Route and page
3PL inventory	[CH16] p. 31	missed pickup	[CH15] p. 30
3PL invoice	[CH16] p. 31	MOQ	[CH06] p. 19
3PL service	[CH16] p. 31	movement type	[SAP-MM] p. 41
abbreviations	[ABR-01] p. 104	moving average cost	[QB-BUY] p. 58
ABC inventory	[WH-01] p. 32	MRP	[CH17] p. 34
access denied	[SQL-01] p. 73	negotiation	[NEG-01] p. 23
allocation shortage	[CH14] p. 29	OEE	[CH18] p. 34
Ariba approval	[SAP-ARIBA] p. 49	optimization	[OPT-01] p. 67
Ariba contract	[SAP-ARIBA] p. 49	OTIF	[CH21] p. 37
Ariba guided buying	[SAP-ARIBA] p. 49	packing	[CH14] p. 29
Ariba sourcing	[SAP-ARIBA] p. 49	pallet missing	[WH-01] p. 32
ATP promise	[CH14] p. 29	payment	[FIN-01] p. 53
automation	[CH25] p. 82	permission denied	[SQL-01] p. 73
automation failure	[CH25] p. 82	picking	[CH14] p. 29
backorder	[CH14] p. 29	Power BI	[PBI-01] p. 79
balance sheet	[FIN-01] p. 53	price variance	[CH21] p. 37
bank reconciliation	[FIN-01] p. 53	procurement	[CH07] p. 20
barcode	[WH-01] p. 32	production schedule	[CH18] p. 34
BOM	[CH17] p. 34	profit center	[SAP-FICO] p. 47
business partner	[SAP-01] p. 38	putaway	[CH12] p. 28
capacity	[CH18] p. 34	quality defect	[CH19] p. 35
cash to cash	[CH21] p. 37	QuickBooks	[QB-01] p. 55
company code	[SAP-FICO] p. 47	QuickBooks audit log	[QB-CLOSE] p. 63

Keyword	Route and page	Keyword	Route and page
contract	[CH10] p. 22	QuickBooks bank reconciliation	[QB-CLOSE] p. 63
cost center	[SAP-FICO] p. 47	QuickBooks bills	[QB-BUY] p. 58
cost of goods sold	[FIN-01] p. 53	QuickBooks close books	[QB-CLOSE] p. 63
customs hold	[CH20] p. 36	QuickBooks inventory	[QB-BUY] p. 58
cutover	[CH28] p. 90	QuickBooks migration	[QB-MIG] p. 65
cyber	[CH26] p. 88	QuickBooks sales	[QB-SELL] p. 61
cycle count	[CH13] p. 28	receiving	[CH12] p. 28
damaged shipment	[CH15] p. 30	refresh failure	[CH25] p. 82
dashboard	[CH25] p. 82	rejected migration row	[MIG-01] p. 91
data leak	[CH26] p. 88	reorder point	[CH06] p. 19
Databricks	[DBX-01] p. 83	research	[RES-01] p. 100
DAX	[PBI-01] p. 79	returns	[CH19] p. 35
decision log	[CH29] p. 95	risk and continuity	[CH26] p. 88
demand forecast	[CH04] p. 17	rollback	[CH28] p. 90
dispatch	[CH14] p. 29	S&OP	[CH05] p. 18
duplicate invoice	[FIN-01] p. 53	safety stock	[CH06] p. 19
EIR	[QB-01] p. 55	sales order	[QB-SELL] p. 61
EOQ	[CH06] p. 19	SAP	[SAP-01] p. 38
ERP	[SAP-01] p. 38	SAP Ariba	[SAP-ARIBA] p. 49
Excel	[XL-01] p. 69	SAP CO	[SAP-FICO] p. 47
excess inventory	[CH06] p. 19	SAP EWM	[SAP-EWM] p. 44
expiry	[WH-01] p. 32	SAP FI	[SAP-FICO] p. 47
FI document	[SAP-FICO] p. 47	SAP FICO	[SAP-FICO] p. 47
FIFO	[QB-BUY] p. 58	SAP goods receipt	[SAP-MM] p. 41
finance	[FIN-01] p. 53	SAP invoice block	[SAP-MM] p. 41
forecast bias	[CH04] p. 17	SAP MM	[SAP-MM] p. 41
freight cost	[CH15] p. 30	SAP warehouse task	[SAP-EWM] p. 44
global trade	[CH20] p. 36	screening	[CH20] p. 36
go-live	[CH28] p. 90	SLA	[CH10] p. 22
goods receipt	[CH12] p. 28	slotting	[WH-01] p. 32
GR IR	[SAP-FICO] p. 47	Snowflake	[SNF-01] p. 84
greenfield	[CH27] p. 89	Solver	[OPT-01] p. 67
handling unit	[SAP-EWM] p. 44	SQL	[SQL-01] p. 73

Keyword	Route and page	Keyword	Route and page
interface	[CH25] p. 82	stockout	[CH06] p. 19
inventory costing	[QB-BUY] p. 58	supplier selection	[CH08] p. 21
inventory mismatch	[CH13] p. 28	transport	[CH15] p. 30
invoice block	[SAP-MM] p. 41	Undeposited Funds	[QB-SELL] p. 61
invoice mismatch	[CH07] p. 20	UOM	[CH03] p. 16
item receipt	[QB-BUY] p. 58	user access	[CH27] p. 89
KPI	[CH21] p. 37	vendor credit	[QB-BUY] p. 58
landed cost	[CH20] p. 36	warehouse allocation	[WH-01] p. 32
late order	[CH14] p. 29	warehouse layout	[CH11] p. 27
late supplier	[CH09] p. 21	warehouse monitor	[SAP-EWM] p. 44
lot traceability	[CH19] p. 35	warehouse task	[SAP-EWM] p. 44
master data	[CH03] p. 16	warehouse tracking	[WH-01] p. 32
migration	[MIG-01] p. 91	WMS	[WH-01] p. 32
migration reject	[MIG-01] p. 91	working capital	[FIN-01] p. 53

[ABR-01] Abbreviations and shortcuts in plain language

The first time an abbreviation appears in your own report, write the full form followed by the abbreviation in parentheses. Example: On Time In Full (OTIF). A local company meaning overrides this general dictionary only when it is documented.

Short form	Full form or meaning	What it is for
3PL	Third-Party Logistics Provider	External warehouse or transport operator
ABC	Inventory classes by value or importance	Use with XYZ variability, not alone
AI	Artificial Intelligence	Models that infer, predict or generate
ALM	Application Lifecycle Management	Plan, test, release and support systems
AP	Accounts Payable	Money owed to suppliers
API	Application Programming Interface	Controlled system-to-system request
AR	Accounts Receivable	Money customers owe
ASN	Advance Shipping Notice	Supplier notice before arrival
ATP	Available to Promise	Quantity/date that can be committed
BATNA	Best Alternative to a Negotiated Agreement	Your best option if no deal is signed
BAU	Business As Usual	Stable day-to-day ownership after a project
BOM	Bill of Material	Components required to make a product
CAPA	Corrective and Preventive Action	Fix cause and prevent recurrence
CDC	Change Data Capture	Move only inserted/changed/deleted data
COGS	Cost of Goods Sold	Cost attached to sold products
CO	Controlling	SAP internal cost and management accounting
CSV	Comma-Separated Values	Plain-text tabular file
DAX	Data Analysis Expressions	Power BI calculation language
DDL	Data Definition Language	SQL that creates or changes objects

Short form	Full form or meaning	What it is for
DML	Data Manipulation Language	SQL that inserts, updates, merges or deletes rows
DBMS	Database Management System	Software that stores and queries data
DDMRP	Demand Driven Material Requirements Planning	Buffer-based planning method
DIO	Days Inventory Outstanding	Approximate days inventory is held
DPO	Days Payables Outstanding	Approximate days before suppliers are paid
DSO	Days Sales Outstanding	Approximate days to collect customers
EDI	Electronic Data Interchange	Standard business-document exchange
EIR	Enhanced Inventory Receiving	QuickBooks Desktop separation of item receipt and bill
ELT	Extract, Load, Transform	Load source data before transforming it
EOQ	Economic Order Quantity	Order size balancing order and holding cost
ERP	Enterprise Resource Planning	Integrated transaction and accounting system
ETL	Extract, Transform, Load	Transform data before loading its destination
EWM	Extended Warehouse Management	SAP advanced warehouse system
FEFO	First Expired, First Out	Pick earliest expiry first
FIFO	First In, First Out	Use oldest receipt first
FI	Financial Accounting	SAP external-accounting area
FVA	Forecast Value Add	Error improvement or damage by a forecast step
G/L	General Ledger	Official account balances
GI	Goods Issue	Inventory leaves stock
GR	Goods Receipt	Inventory or service is received
GR/IR	Goods Receipt/Invoice Receipt	Clearing between receipt and supplier invoice
HU	Handling Unit	Identified pallet, carton or tote
IM	Inventory Management	SAP plant/storage-location stock and goods movements
JDBC	Java Database Connectivity	Java standard for database connections
JSON	JavaScript Object Notation	Structured text used by APIs and files
IBP	Integrated Business Planning	Demand, supply and finance decision process
ID	Identifier	Stable code for a record
KPI	Key Performance Indicator	Measure tied to a decision
LAMBDA	Excel custom reusable function	Named function in Microsoft 365
LT	Lead Time	Elapsed time from trigger to availability
MASE	Mean Absolute Scaled Error	Forecast error scaled to a naïve baseline
MAC	Moving Average Cost	Inventory cost recalculated from quantity and value
MES	Manufacturing Execution System	Runs and records shop-floor execution
MOQ	Minimum Order Quantity	Smallest supplier or production quantity
MM	Materials Management	SAP purchasing, inventory and valuation processes
MRP	Material Requirements Planning	Net demand into planned supply
MRB	Material Review Board	Authorized nonconforming-product disposition
MTO	Make to Order	Make after a customer order

Short form	Full form or meaning	What it is for
MTS	Make to Stock	Make before orders using a plan
OEE	Overall Equipment Effectiveness	Availability × performance × quality
O2C	Order to Cash	Customer order through payment collection
OTIF	On Time In Full	Order delivered by promise and complete
P&L	Profit and Loss	Revenue, costs and profit for a period
P2P	Procure to Pay	Need through supplier payment
PBI	Power BI	Microsoft analytics and reporting platform
PGI	Post Goods Issue	SAP dispatch inventory posting
PO	Purchase Order	Authorized supplier commitment
PK	Primary Key	Column or columns that uniquely identify a row
PPM	Parts Per Million	Defects per million inspected units
PPV	Purchase Price Variance	Actual purchase price compared with standard
PR	Purchase Requisition	Internal request to buy
QOH	Quantity on Hand	Recorded stock quantity
QBO	QuickBooks Online	Cloud QuickBooks product
QBD	QuickBooks Desktop	Installed QuickBooks company-file product
RACI	Responsible, Accountable, Consulted, Informed	Role assignment method
RAID	Risks, Assumptions, Issues and Dependencies	Program control register
RFQ	Request for Quotation	Request for price and terms
RLS	Row-Level Security	Restrict report rows by user
RMA	Return Material Authorization	Approved return identity/process
ROP	Reorder Point	Inventory position that triggers replenishment
RPA	Robotic Process Automation	Rule-based user-interface automation
RF	Radio Frequency	Scanner/mobile warehouse execution
RPO	Recovery Point Objective	Maximum acceptable data-loss period
RTO	Recovery Time Objective	Target time to restore service
SAP	Systems, Applications and Products	Enterprise software company/platform
SES	Service Entry Sheet	Evidence that an ordered service was performed/accepted
SLP	Supplier Lifecycle and Performance	SAP Ariba supplier onboarding, qualification and performance
SCD	Slowly Changing Dimension	Method for dimension history
SCAR	Supplier Corrective Action Request	Formal supplier cause and prevention request
SIT	System Integration Testing	End-to-end system connection test
SLA	Service-Level Agreement	Measured service obligation
SKU	Stock-Keeping Unit	Distinct stocked item
SQL	Structured Query Language	Language for relational data
SOP	Standard Operating Procedure	Approved repeatable work instruction
UC	Unity Catalog	Databricks governance catalog
TCO	Total Cost of Ownership	All relevant lifecycle costs
TMS	Transportation Management System	Plans and controls freight
TTR	Time to Recover	Time to restore a dependency

Short form	Full form or meaning	What it is for
TTS	Time to Survive	Time operations can continue without it
UAT	User Acceptance Testing	Business proof that the process works
UOM	Unit of Measure	EA, KG, CS and other quantity unit
UTC	Coordinated Universal Time	Common reference timezone
ODBC	Open Database Connectivity	Standard database connection interface
FK	Foreign Key	Column that points to a key in another table
WAPE	Weighted Absolute Percentage Error	Total absolute error ÷ total actual
WBS	Work Breakdown Structure	Hierarchy of project work or cost objects
WIP	Work in Process	Partially completed production
WMS	Warehouse Management System	Plans and records warehouse work
WO	Warehouse Order	Group of EWM warehouse tasks
WT	Warehouse Task	One EWM warehouse movement
VBA	Visual Basic for Applications	Desktop Microsoft Office automation language
XLOOKUP	Excel lookup function	Returns a matching value from another range
XYZ	Inventory classes by demand variability	X stable, Z highly variable

Keyboard shortcut words

Key	Meaning	How to press it
Ctrl	Control	Hold Ctrl and tap the next key; e.g., Ctrl+T
Alt	Alternate/menu command	Hold Alt and tap the next key
Shift	Upper/alternate key action	Hold Shift with another key
Fn	Laptop function layer	May be needed for F2–F12
F4	Function key 4	Excel repeats/locks references; SAP GUI often opens value help
Esc	Escape	Cancel the current edit or dialog
Enter	Accept/run/go to next cell	Press once after checking input

[SRC-01] Official sources and application notes

This handbook is complete without opening these links. They are version evidence for editors and a final check when software screens, functions, laws or trade requirements change. The practical methods and examples in this handbook are original. Verify current company policy and obtain the required specialist approval before a production posting, regulated decision or contract signature.

Source ID	Reference	Official link
SRC-SAP-01	SAP Fiori spaces and pages	Open at help.sap.com
SRC-SAP-02	SAP S/4HANA product learning	Open at learning.sap.com
SRC-SAP-03	SAP procure-to-pay process	Open at learning.sap.com
SRC-SAP-04	SAP manufacturing learning journey	Open at learning.sap.com
SRC-SAP-05	SAP material valuation and documents	Open at learning.sap.com
SRC-SAP-06	SAP EWM inbound process	Open at learning.sap.com
SRC-SAP-07	SAP Activate methodology	Open at learning.sap.com
SRC-SAP-08	SAP Migration Cockpit process	Open at help.sap.com

Source ID	Reference	Official link
SRC-SAP-09	SAP Integration Suite managed gateway for spend management and SAP Business Network	Open at help.sap.com
SRC-QB-01	QuickBooks purchase orders	Open at quickbooks.intuit.com
SRC-QB-02	QuickBooks inventory assets and COGS	Open at quickbooks.intuit.com
SRC-QB-03	QuickBooks account reconciliation	Open at quickbooks.intuit.com
SRC-QB-04	QuickBooks Online FIFO and moving-average inventory valuation	Open at quickbooks.intuit.com
SRC-QB-05	QuickBooks Desktop Enterprise enhanced inventory receiving	Open at quickbooks.intuit.com
SRC-QB-06	QuickBooks Desktop Enterprise Advanced Inventory	Open at quickbooks.intuit.com
SRC-MS-01	Microsoft Excel support	Open at support.microsoft.com
SRC-MS-01A	Excel structured references	Open at support.microsoft.com
SRC-MS-01B	Power Query best practices	Open at learn.microsoft.com
SRC-MS-02	Microsoft DAX reference	Open at learn.microsoft.com
SRC-MS-03	Power BI guidance	Open at learn.microsoft.com
SRC-MS-04	Power BI star-schema guidance	Open at learn.microsoft.com
SRC-DBX-01	Databricks documentation	Open at docs.databricks.com
SRC-DBX-02	Databricks SQL editor	Open at docs.databricks.com
SRC-DBX-03	Databricks medallion architecture	Open at docs.databricks.com
SRC-SNF-01	Snowflake documentation	Open at docs.snowflake.com
SRC-SNF-02	Snowflake Workspaces	Open at docs.snowflake.com
SRC-SNF-03	Snowflake identifier rules	Open at docs.snowflake.com
SRC-ICC-01	ICC Incoterms resources	Open at iccwbo.org
SRC-US-01	US consolidated trade screening list	Open at www.trade.gov

Ten rules that prevent expensive errors

Rule	Working habit
1 Confirm the grain	Say exactly what one row, one count and one percentage represent.
2 Verify one real case	Trace a known order, item, lot, supplier or payment before trusting totals.
3 Separate fact from assumption	Label the source, cutoff, timezone, unit, currency and status population.
4 Change one controlled thing	Use a version, owner, reason, approval, effective date and rollback.
5 Reconcile count, quantity and value	A matching grand total can still hide wrong items or locations.
6 Protect irreversible decisions	Use qualified authority for payments, contracts, trade, safety and write-offs.
7 Keep the evidence	Store input, output, document number, query/model version and approval together.
8 Design the exception path	Define what happens when the normal process, interface or person fails.
9 Prove prevention	Monitor enough future opportunities to show that the cause did not recur.
10 Improve the handbook	Add a stable ID, filled example, change list, expected result, error path and proof check.